Functional Specification

Customer	**DHL eCommerce**
Project	**Distribution Center of the Future - EWR and LAX**
Revision	**V3.9**
BG Project No.	**617-010013 / 617-000189**
Revision Date	**May 7th, 2024**

Version

Version	Date	Author	Change
1	09/06/17	SaLu	First Draft based on proposal rev 3
1.1	09/15/2017	KrTo	Changes based on meeting with DHL at BC 9/14-15/2017
2.0	10/26/2017	KrTo	Changes based on kick-off meeting with DHL 10/24-25
2.1	1/5/2018	KrTo	Added maximum item dimensions for indirect sack chutes. Updated PC hardware specification. Included new scope and implementation details.
2.2	1/18/2018	KrTo	Details added from meeting with DHL in Atlanta 1/16 and 1/17/2018.
2.3	May 15, 2018	Christopher Reyes	Changes from multiple discussions with DHL over the past few months.
2.4	May 20, 2018	DuPh	Update with conveyor sections based on conveyor DOOs
2.5	August 9, 2018	Christopher Reyes	Updated diagrams and descriptions based on deployment meeting comments
2.6	October 4, 2018	Dunia Pham	Divert arms operation at inbound conveyors
2.7	November 8, 2018	Christopher Reyes & Dunia Pham	Updated descriptions with latest functionality (post-commissioning)
2.8	February 6, 2019	Dunia Pham	Selector switches at inbound and sack commissioning area
2.9	February 8, 2019	Christopher Reyes	Added barcode prioritization details for each barcode type
3.0	May 23, 2019	Dunia Pham	Added Indexing functionality at sack manual sort area
3.1	August 26, 2019	Subha Srikanth	Updated definition of "recirculation reassign" parameter
3.2	May 8, 2020	Subha Srikanth	Updated "Too Many Recirculating Trays" sorting exception to add description of PLC logic to stop inductions
3.3	July 27, 2022	James Kelly	Adding PlusOne Robotics communication and upgraded 5-sided Datalogic scan tunnels information
3.4	Feb 21, 2023	Ramyaa Parthasarathy	Adding auto-clearing feature on fully automatic cross inducts and Slug release

			mode for Inducts with scale
3.5	April 5, 2023	Kristian Toft	Added details on the first robotic induction
3.6	June 12, 2023	Andy Rears	Added details about STC Reject feature
3.7	January 5, 2024	Christopher Reyes	Updated HOST Communication details due to changes to discharge message fields and authentication scheme.
3.8	April 5th, 2024	Kevin Blum	Updated Sort Preparation, Sorting, and Master Data sections to include cuboidal update to sort plan and cuboidal logic
3.9	May 7th, 2024	Kevin Blum	Added Gao Reject to Sorting Exceptions

Table of Contents

1 Introduction

This document provides an overview of the functionality of the BEUMER sorter control system as proposed for the sorters to be installed at the DHL facility in Avenel, New Jersey.

BEUMER has attempted to provide a comprehensive description of the proposed functionality matching the scope provided in our proposal, although some details may be incomplete and will require further clarification before the final system design is complete. From that perspective this will be viewed as a live document that will be updated with relevant implementation details in the design and development process. However, the basic functionality and functional features describes the delivery scope corresponding to the original proposal and subsequent change orders.

We welcome any critique or suggestions to ensure that the final solution meets DHL's unique operational needs and are correctly and comprehensively described in this functional specification.

To the extent this document provides specific quantities or other quantitative measures, it is exclusively provided as support for this functional description. In case of any discrepancy, the proposal documents have priority.

2 System Description

DHL receives and collects parcels from its customers and sorts and transports these parcels to the appropriate destination NDC, SCF or DDU of USPS.

For this DHL receives parcels in outbound mode, evaluates for each parcel the destination SCF or DDU as well as the appropriate transportation route and sorts the parcels according to the DHL DC in which service area the destination NDC, SCF or DDU is located and according to their product.

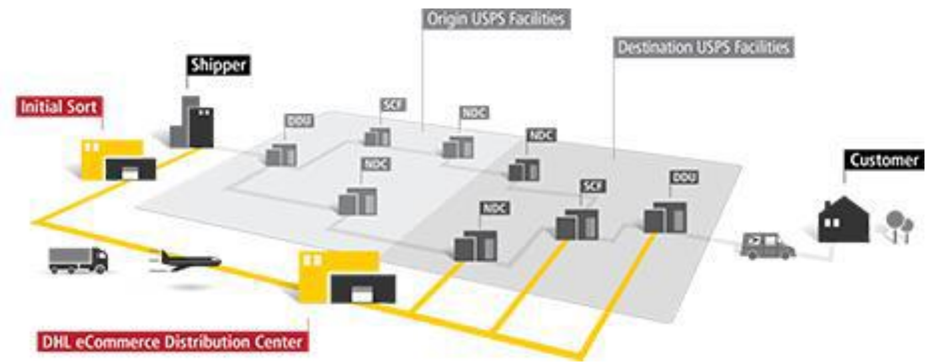


Figure 2-1 - Illustration of the drop shipment process

The facility in New Jersey, nominally termed EWR, but actually located in Avenel, NJ is a part of the DHL infrastructure implementing the mail process shown in the illustration above.

2.1 Design Requirements

The system design is driven by the design criteria provided by DHL as well as several different factors, such as but not limited to, the products to be handled, throughput rates to be achieved, the building structure, the footprint, the project timeline, the costs, etc. Several of these factors that are critical to the design are the focus of this section.

2.1.1 Product Specification

The products to be handled are parcels. To provide a reliable system operation, sortable products are specified in more detail. The product dimensions for parcels are based on the RFP design specifications. If product information was missing, BEUMER made some assumptions for the system design. All assumptions may have changed during the design phase.

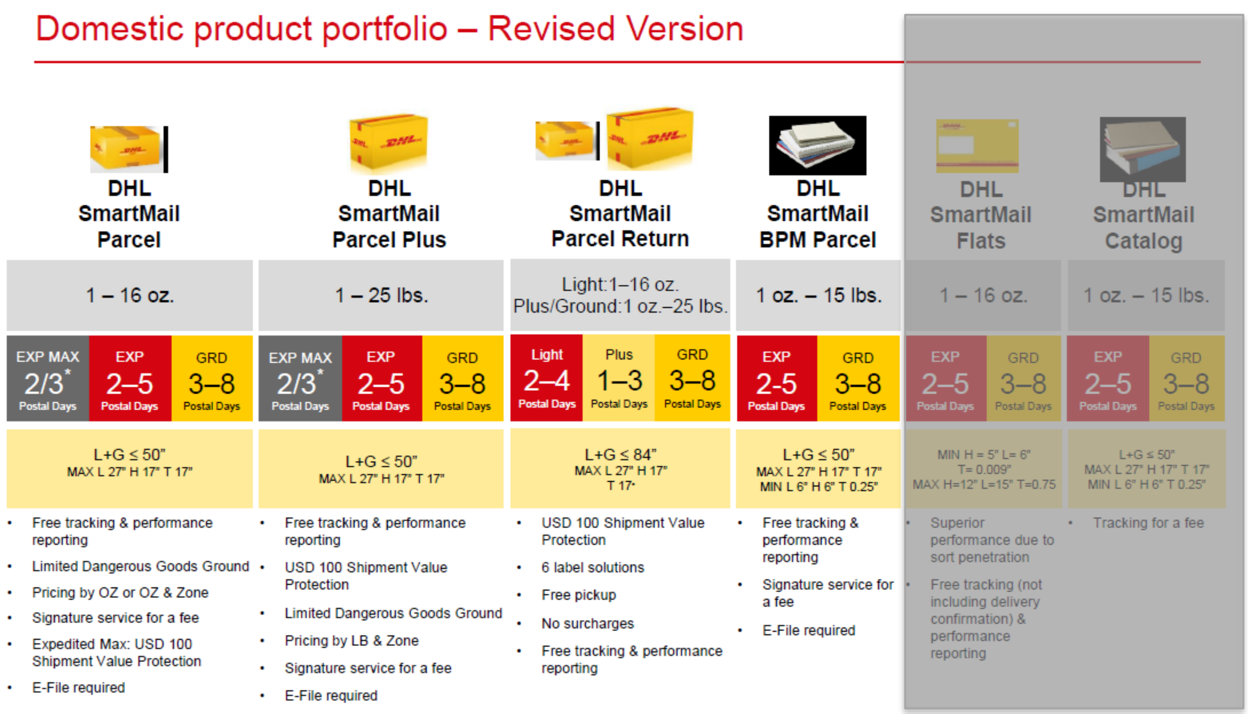


Figure 2-2 RFP Appendix 3 DHL Product Overview Domestic

DHL requested and clarified in the DCoF_QA_V1.1 – list that the sorter system needs to be able to sort the first four parcel categories. Flats and Catalogs are going to be sorted by a Flats Sorter provided by DHL.

The product dimensions that can be handled with the BEUMER system are shown in Table 2-1.

Table 2-1 Product Dimensions

Type	Dim.	Length [in]	Width [in]	Height [in]	Weight
Parcel	Max	27	17	17	16oz
57.7%	Avg.	TBD	TBD	TBD	0.34-0.35 lbs.
	Min	6	4	0.25	1oz
Parcel Plus	Max	27	17	17	25 lbs.
7.1%	Avg.	TBD	TBD	TBD	1.84-2.19 lbs.
	Min	6	4	0.25	1 lbs.
Parcel Return	Max	27	17	17	25 lbs.
TBD	Avg.	TBD	TBD	TBD	TBD
	Min	6	4	0.25	1oz

BPM Parcel	Max	27	17	17	15 lbs.
8.7%	Avg.	TBD	TBD	TBD	1.46-1.49 lbs.
	Min	6	4	0.25	1 oz.

Note:

1. Length > Width > Height
2. The minimum product weight needs to be higher than 1.31oz/ft^2
3. 90% of the product surface needs to be at the minimum height
4. The BS 25 Belt Tray Sorter can handle product up to a weight of 70lbs – However it is not recommended to sort such a heavy weight due to the direct to sack and container sort
5. Tubes within the product dimensions can be sorted if they are positioned and secured on a tray

Table 2-2 Product Dimensions DHL Size Type1 for Direct Sack Destinations

Type	Dim	Length [in]	Width [in]	Height [in]	Weight
Parcel	Max	18.9	15	4.7	11lbs
	Min	6	4	0.25	1oz

Table 2-3 Product Dimensions DHL Size Type1 for In-Direct Sack Destinations

Type	Dim	Length [in]	Width [in]	Height [in]	Weight
Parcel	Max				
	Min				

2.1.1.1 Product characteristics and operational recommendations

The following product and operational characteristics are recommended in order to reach the optimal accuracy:

1. The frictional coefficient must be equal over the complete bottom surface. The maximal allowed deviation of the frictional coefficient is 0.1. The Coefficient of friction is unit-less. The relationship is based on the point at which static friction (no motion) and kinetic friction (motion occurring) and the magnitude in which the opposing forces of friction are overcome to start or stop motion. Generally, the definition is based on "Dry Friction" or Coulomb friction, which categorized as static or kinetic (dynamic).
2. No friction influencing substances or materials are permitted to adhere on the product top or bottom. For the allowed deviation see point 1.
3. No friction influencing substances are permitted from entering the induction zone up to leaving the chute. This is especially important over the areas

where the product will be sliding over (induction to tray, tray to transitions, transitions to chute floor. For the allowed deviation see point 1.

4. The center of gravity must be in the middle of the product relating to the length and width. The maximal allowed deviation is +/-10% of the length. Product center of gravity is imperative to ensure the product tracks correctly on the inductions, remains positioned on the tray without influence of forces in curves, loss of static friction to the tray surface due to forces on the sorter, and/or environment forces such as air resistance to the product surface that may limit the ability of the item to remain in the correct position for discharge. Discharge may be influenced by a center of gravity higher than 10% of the length causing the product to be early or late in discharge to the destination due to loss of friction with the belt tray, or discharge characteristics outside of the normal early late range discharge parabola to the destination.
5. The center of gravity must be in the lower half of the product in relation to the height.
6. Cartons are to be conveyed lengthwise (short side leading) on the feed conveyors to the induction lines. The maximal allowed deviation is +/- 10°.
7. Round, cylindrical items can’t be inducted or sorted without the use of an induction tray.

2.1.1.2 Additional characteristics and operational recommendations for conveyor equipment

1. All products must have the necessary properties to bear the deceleration and acceleration of 4 m/s^2. Products are decelerated against end stops, e.g. at right-angled transfers or at the end of gravity roller conveyors at higher rates and must be able to bear these forces. Besides, the customer is aware that these values are exceeded in case of exceptional operational conditions – e.g. during an emergency stop.
2. Products that can be transported by the sorter and conveyor system meet the following criteria:
   a. Product type: parcels
   b. Bottom is flat and smooth: Curvature max 2%
   c. Product is dry, free of oil and grease and has an even texture/surface
   d. The maximum load eccentricity of the stored/transported material with regard to loading must not exceed the value of ¼ of its respective lateral length.
3. The bottom of the cartons must not bulge towards the inside and must be flat, even and undamaged. We assume a uniform frictional resistance of the parcel (e.g. no sticky contact surface, no magnets). At least 75% of the bottom area must be supported on the conveyor

2.1.1.3 Parcel Package

The parcels must be packed in a way that they won't get damaged by a free fall of 65" (5’ 5”) into the largest container (roller cage).

The label should be printed and applied in a way that bulk transfer does not damage the label or in a way that the barcode is not readable anymore.

Please also refer to the Clarification List in Attachment F of the BEUMER proposal.

2.1.1.4 Product Distribution

The volume was provided for the different parcels types but not for the different sizes and weights and the distribution to the destinations.

The size and weight distribution can influence the number of necessary drives and the number of oversize and regular chutes. It will also impact the labor analysis and the number of operators in the system.

Volume by Product

Q1 2017

Product	%	Average Weight
DHL SM BPM Expedited	0.8%	1.46
DHL SM BPM Ground	7.9%	1.49
DHL SM Flats Expedited	5.4%	0.39
DHL SM Flats Ground	19.8%	0.45
DHL SM Parcel Expedited	31.7%	0.34
DHL SM Parcel Expedited Max	1.0%	0.83
DHL SM Parcel Ground	25.0%	0.35
DHL SM Parcel Plus Expedited	2.6%	1.84
DHL SM Parcel Plus Ground	4.5%	2.19
First Class Parcels	0.1%	0.70
SM Marketing Parcel Expedited	0.4%	0.37
SM Marketing Parcel Ground	0.9%	0.52
	100.0%	

Figure 2-3 - RFP Volume Distribution for Parcel Types

2.1.2 Barcode Specification

2.1.2.1 Domestic Parcels

All domestic parcels will contain a 2D Ticket-to-Ride and/or a 1D USPS eVS barcode.

<u>Ticket-to-Ride barcode</u>:

The Ticket-to-Ride barcode is a DHL-defined barcode type. There are two barcode constructs in use: customer-generated & DHL-generated (shown below). All Ticket-to-Ride barcodes are QR codes, with first characters 'V', of barcode lengths 26, 32, or 36, of size 1.5cm x 1.5cm. **The sorter system identifies a barcode as Ticket-to-Ride if it is of type QR, has first character 'V', and is of length 26, 32, or 36.**

Within the Ticket-to-Ride barcode, the concatenation of the 'DHL Sort Code Version Number', 'DHL Outbound (Destination) Sort Code', and 'DHL Inbound (Final Slot) Sort Code' fields will be used as the *sort code* during sortation. **The sorter system extracts this *sort code* by parsing out characters 1 through 8.**

Within the Ticket-to-Ride barcode, the concatenation of the 'Bin Constraint Parameter 1' and 'Bin Constraint Parameter 2' fields will be used as the *item attribute* during sortation. **The sorter system extracts this *item attribute* by parsing out characters 9 through 10.**

The prioritization scheme among Ticket-to-Ride barcodes will be as follows:

- Priority 1: QR-Code of length 26 containing a V in the first position → Ticket-to-Ride
- Priority 2: QR-Code of length 32 containing a V in the first position → Ticket-to-Ride
- Priority 3: QR-Code of length 36 containing a V in the first position → Ticket-to-Ride

QR Barcode Example:

Figure 2-4 - Example of a QR barcode

DHL-Generated Ticket-to-Ride Barcode Elements:

Position	**Field Description**	**Field Length**
1-2	DHL Sort Code Version Number • Capital "V" + single-digit version number • No spaces	2
3-5	DHL Outbound (Destination) Sort Code • Underscore "_" + 2-digit outbound sort code	3
6-8	DHL Inbound (Final Slot) Sort Code • Single alpha character + 2-digit number	3
9	Bin Constraint Parameter 1 • Single-digit number	1
10	Bin Constraint Parameter 2 • Single-digit number	1
11-26	DHL Internal Identifier Number • 16-digit internal DHL tracking number	16

Example: V6_71A01023072607608496710

Customer-Generated Ticket-to-Ride Barcode Elements:

Character Position	Field Description	Field Length
1-2	DHL Sort Code Version Number • Capital "V"+ single-digit version number • No spaces	2
3-5	DHL Outbound (Destination) Sort Code • Underscore "_" + 2-digit outbound sort code	3
6-8	DHL Inbound (Final Slot) Sort Code • Single alpha character + 2-digit number	3
9	Bin Constraint Parameter 1 • Single-digit number	1
10	Bin Constraint Parameter 2 • Single-digit number	1
11-32 or 11-36	USPS Confirmation Number (PIC) • 22 or 26-digit USPS tracking number	22 or 26

Example: V6_71A0102927489999632731301 6349

USPS eVS Barcode:

The USPS eVS barcode is a USPS-defined barcode type. There are six barcode constructs in use (shown below). All USPS eVS barcodes are GS1-128, with "420" as the Routing Application Identifier field (shown below). **The sorter system identifies a barcode as USPS eVS if it is of type GS1-128 and starts with "420".**

Within the USPS eVS barcode, the 5-digit ZIP Code field will be used as the *sort code* during sortation. The full ZIP Code field can be found between the "420" Routing Application Identifier field and the second FNC1 character, as illustrated below (note: the first FNC1 character is part of the barcode header and will not actually be transmitted by the scanner). **The sorter system extracts the 5-digit *ZIP Code* by parsing out the 5 characters immediately following the leading "420" string.**

Barcode Constructs:

		POSTAL ROUTING CODE Field Length		TRACKING NUMBER Field Length						
Code	Description	Postal Routing AI	Dest ZIP	Channel AI	STC	MID	Serial Number	Check Digit	PIC Length	Total Barcode Length
C01	Commercial Mail - (Nine-digit Mailer ID, 9-digit ZIP Code)	3	9	2	3	9	7	1	22	34
C02	Commercial Mail - (Nine-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	9	11	1	26	34
C03	Commercial Mail - (Nine-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	9	7	1	22	30
C05	Commercial Mail - (Six-digit Mailer ID, 9-digit ZIP Code)	3	9	2	3	6	10	1	22	34
C06	Commercial Mail - (Six-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	6	14	1	26	34
C07	Commercial Mail - (Six-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	6	10	1	22	30

Figure 2-5 - USPS eVS barcode constructs to be encountered on the sorter

Barcode Fields:

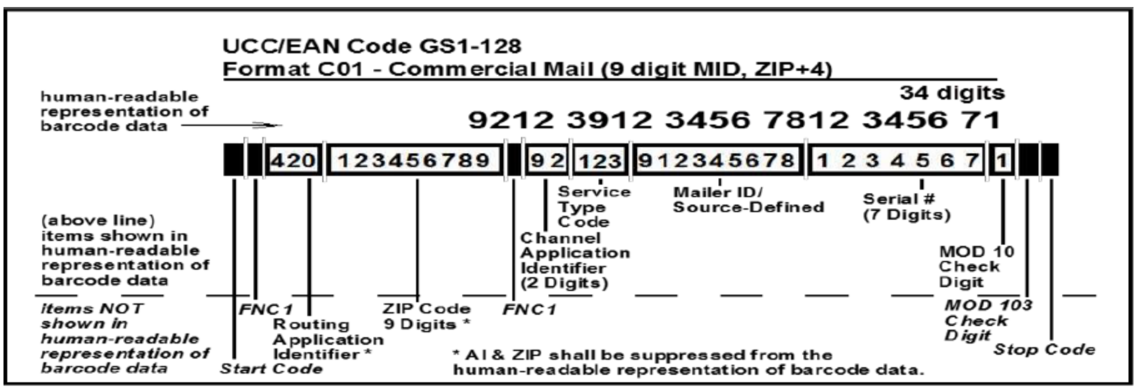


Figure 2-6 - Barcode fields of the USPS eVS barcode

2.1.2.2 Cross-border Parcels

All cross-border parcels will contain either DHL eCommerce barcode or a customer-applied barcode.

<u>DHL eCommerce barcodes</u>:

The DHL eCommerce barcode is a DHL-defined barcode type. All DHL eCommerce barcodes are Code 128 (subset B, ½-inch minimum height, ¾-inch recommended).

The entire DHL eCommerce will be used to lookup its corresponding *sort code* during sortation.

Sample DHL eCommerce barcode:

DHL eCommerce

GM605111234901234564

Figure 2-7 - Example of a DHL eCommerce barcode

<u>Customer-Applied barcodes</u>:

The customer-applied barcodes are customer-defined barcode types. All such customer-applied barcodes to be expected on the sorter will be one encoded as one of the following barcode symbologies:

*QR (2D)
Data Matrix (2D)
PDF417 (2D)
**GS1-128
Code 128
Code 39
IMb (Intelligent Mail barcode)
Code EAN8
Code EAN13

*Mail items with QR barcodes **not** identified as a Ticket-to-Ride barcode (as described in the above Ticket-to-Ride section) shall be considered cross-border packages.

Mail items with GS1-128 barcodes **not identified as a USPS eVS barcode (as described in the above USPS eVS section) shall be considered cross-border packages.

The prioritization scheme among CrossBorder barcodes will be as follows:

- Priority 1: Code128 starting with "GM" → CrossBorder
- Priority 2: GS1-128 starting with "GM" → CrossBorder
- Priority 3: 2/5 Interleaved → CrossBorder
- Priority 4: Code128 not matching the above → CrossBorder
- Priority 5: GS1-128 not matching the above → CrossBorder
- Priority 6: QR-Code not matching the above → CrossBorder
- Priority 7: DataMatrix → CrossBorder
- Priority 8: PDF417 → CrossBorder
- Priority 9: Code39 → CrossBorder

- Priority 10: EAN8 → CrossBorder
- Priority 11: EAN13 → CrossBorder
- Priority 12: IMb → CrossBorder

2.1.3 Container Specification

The design is based on the assumption that on average 250 parcels fit into one container / blue bin and 25 parcels fit into one sack. One bed-loaded trailer can contain up to 18,000 parcels.

Table 2-4 Container Dimensions

Container	Length [in]	Width [in]	Height [in]
Small Gaylord(2)	48	40	35
Medium Gaylord(2)	48	40	46
Large Gaylord(2)	48	40	60
Hamper(1)	48	31.5	38.25
Blue Bin(1)	48	40.75	45.75
DHL Roller Cage	47.25	39.37	74.41

Note:

1. The total weight of parcels inside a container is important to design the Gaylord dumper. Actual values might change the design of the equipment.
2. It is assumed that no parcel sticks out of any of the container types
3. It is assumed that DHL will allow for the drop height that is necessary to dump the items on a collecting conveyor

2.2 Process Throughput

This section gives an overview of the throughput rates and explains the calculation.

2.2.1 Infeed Rates

Depending on the operator throughput, the telescopic conveyor in the inbound area can handle a throughput up to 2,000 pph.

Outbound Receiving

It is assumed that each encoding module provides a throughput of 2,500 pph on one conveyor line. The conveyor lines provided by BEUMER are designed to handle 2,636 pph, each. Please note that the mechanical capacity of the inductions is 3,100 pph.

Inbound Receiving

Depending on the container type and the number of items per container the throughput of each Gaylord dumper might vary. The design plans for six (6) Gaylord dumpers[1] and the one (1) infeed from the telescopic belt to feed two (2) bulk conveyor lines that can handle a throughput of 10,000 pph each. The design also allows for an expansion to add additional Gaylord dumpers in case the product mix or the number of parcels per container might change.

2.2.2 Sorter Rates

Induction Capacity – outbound area

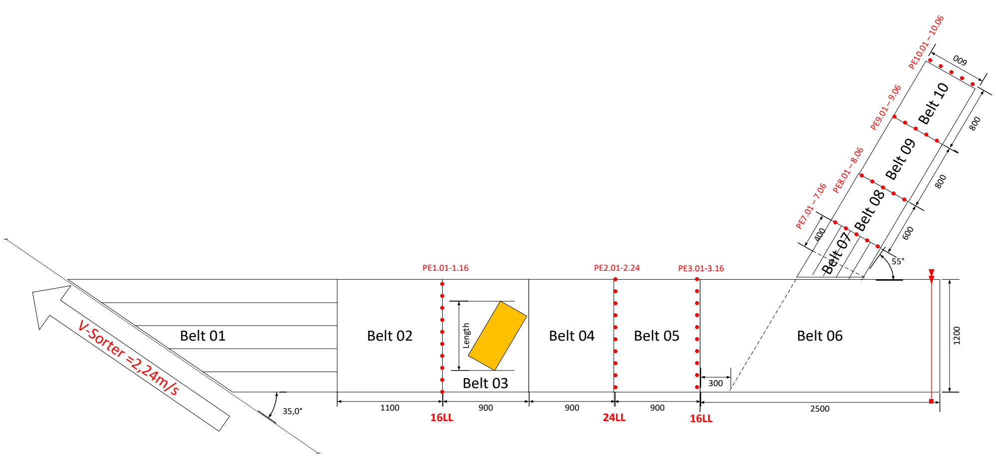


Figure 2-8 - Induction capacity for automatic cross induction unit

The automatic cross induction units are designed for the following capacities:

[1] Only four (4) of the six (6) Gaylord dumpers will be installed day 1

- Design: each fourth tray of the sorter is utilized
- Mechanical capacity: 3,100 [items/h] / 51.68 [items/minute]
- Sustained capacity: 2,636 [items/h] / 43.93 [items/minute]

Besides the product sizes, the operational induction capacity also depends on the infeed capacity of the conveyors and the continuous throughput of the encoding system.

The design features eight (8) encoding lines, feeding two induction areas (one on each sorter), and each area with four automatic cross induction units for a total of eight (8) induction units. The sorting system in the outbound area can theoretically be fed with 20,000 items per hour from the infeed lines, depending on the infeed rates.

Induction Capacity – inbound area

The semi-automatic induction units are designed for the following capacities:

- Design: each fifth tray of the sorter is utilized
- Mechanical capacity: 2,481 [items/h] / 41.35 [items/minute]
- Sustained capacity: 2,109 [items/h] / 35.5 [items/min]

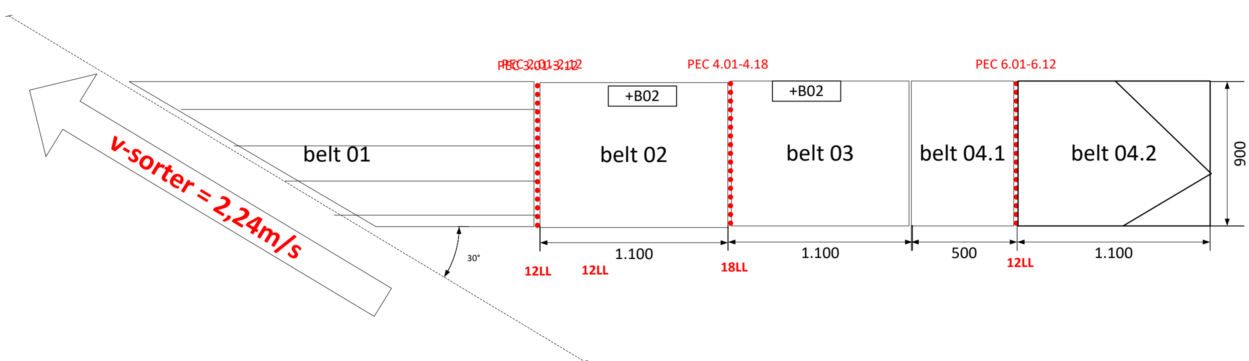


Figure 2-9 - Induction capacity for semi-automatic induction unit

The sustained induction capacity also depends on the infeed capacity of the bulk conveyors and the induct rate of each operator at a semi-automatic induction unit.

Considering five (5) semi-automatic induction units per area (one area on each sorter), ten (10) induction units in total, the sorting system can theoretically be fed with 20,000 items per hour from the Inbound Receiving with Gaylord dumpers, and one (1) feed from the dock door.

The induction units are designed to handle the normal operating throughput rate, as well as any operational peaks during the system operations.

Sorter capacity

To meet the required throughput, the BEUMER solution consists of a stacked system consisting of two (2) belt tray sorters.

The parameters for calculating the mechanical sorter capacity are shown in the following table.

Specification	Value
Number of Sorters	2
Sorter Pitch *s*	650 mm
Sorter Speed *v*	2.24 m/s
Mechanical Capacity	2 x 12,406 trays/hour

The formula used to calculate the mechanical sort capacity is as follows:

$$Mechanical\ Sort\ Capacity\ \lambda = \frac{v}{s} * 3600$$

Several factors that impact the system performance are accounted for in order to calculate the design rate. The factors and resulting design rates are shown in the following table.

Specification	Value
Mechanical Capacity	12,406 trays/hour
Utilization Factor *r*	85%
Performance Factor *f*	2
Design rate per induction area	10,000 pph
Design rate per sorter	20,000 pph

The formula used to calculate the design rate is as follows:

$$Design\ rate\ \lambda_t = \frac{v * r * f}{s} * 3{,}600$$

If several induction areas are used, the sorter capacity will be increased significantly, as the trays can be used several times per cycle. Empty trays and recirculation determine the utilization factor and reduce the sorting capacity. This factor's value is in line with the industry standard. These factors are either determined theoretically or based on empirical values. The capacity resulting from this represents the design rate.

System performance

With two (2) sorter systems a system performance of 40,000 pph is expected.

To achieve a throughput of 20,000 pph in the outbound and the inbound area at the same time, it is assumed that no local-to-local parcels are crossing the inbound induction areas.

When the local-to-local part is around 20-25% the infeed capacity of the inbound induction will be reduced due to occupied items on the sorter. That means that about 20,000 pph of outbound can be sorted, but only 15,000 – 16,000 pph of inbound. If no local-to-local is sorted, or if the outbound operation does not take place at the same time, 20,000 pph can be induction and sorted in the inbound area.

The maximum throughput for cross-border items is 20,000 pph due to the fact that they typically will be inducted at the outbound induction areas

2.2.3 Take away and Sack Commissioning Rates

The take-away conveyor is designed to handle 1,600 sacks per hour (peak).

Currently, the sack commissioning area is designed to provide space for twelve (12) workstations assuming that one (1) operator can weigh and tag 100 to 150 sacks per hour.

2.3 System Layout

The system will be installed in a building in Avenel, NJ.

Site Address: DHL, 191 Blair Rd, Avenel, NJ 07001

2.3.1 Sorter layout

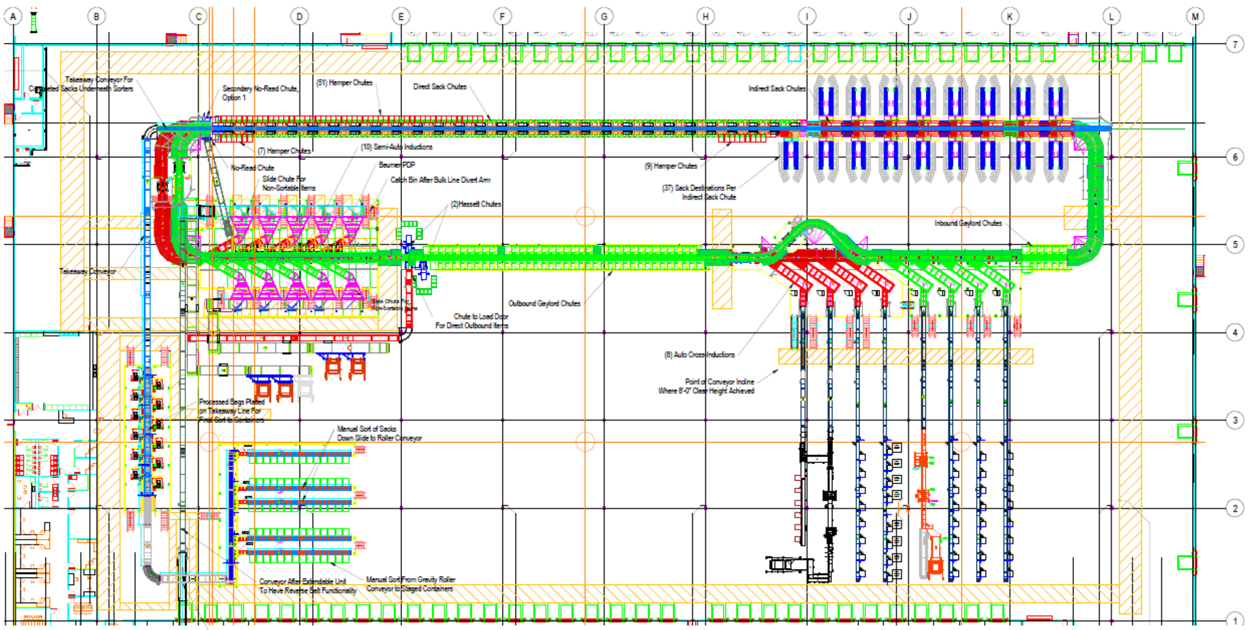


Figure 2-10 - System layout overview

2.3.2 Sorter data

System Specification	
Sorter	
Sorter type:	2 x BS 25 BT
Sorter length Upper:	338.65 m (~1,111 ft.)
Sorter length Lower:	337.35 m (~1,107 ft.)
Carriage (tray) pitch:	650 mm (~25.6 in)
Belt Length:	900 mm (~35.4 in)
Belt width:	450 mm (~17.7 in)
Sorter speed:	2.24 m/s (~441 ft./min)
Mechanical sort capacity per induction area	12,406 pph
De-rated sort capacity (85%) per induction area	10,000 pph
De-rated system sort capacity (85%)	40,000 pph
No. of sorter elements (trays) Upper:	521 trays
No. of sorter elements (trays) Lower:	519 trays
Sorter drives (Linear Motor)	2 x 13 (including one (1) redundant drive)
Upper Elevation Sorter Height @ Upper Tray Level	~ 4,166 m (13'-8")
Upper Elevation Sorter Height @ Lower Tray Level	~ 2,921 m (9'-7")
Lower Elevation Sorter Height @ Upper Tray Level	~ 3,150 m (10'-4")
Lower Elevation Sorter Height @ Lower Tray Level	~ 1,905 m (6'-3")
Curves for Upper Sorter (Horizontal Curves @ 3.5 m radius):	4 x 90° curves
Curves for Lower Sorter (Horizontal Curves @ 3.5 m radius):	5 x 90° curves 2 x 45° curves
Scanner	
Camera scanner	(2) Datalogic Camera Scan Tunnel (each with 1x AV7000 camera for omni-directional top reading)
Camera scanner	(2) 5-Sided camera Tunnel: DM3610
Dimensioning unit	(2) Datalogic dimensioning units

Destinations

There are a total of 339 chutes.

System Specification	
Destinations	
Simple Direct Sack Chutes	(140) destinations, each with (1) single entry per sorter level directly into a designated sack location
Hybrid Direct Sack Chutes	(76) destinations, each with (1) single entry per sorter level directly into a designated sack or hamper location
Indirect Sack Chutes	(34) destinations, each with (1) single entry per sorter level, transferred onto a roller table at pack-out level, with product to be sorted manually into (1) of multiple unique sack destinations
Outbound Gaylord Chutes	(72) destinations, each with (1) single entry chute, into a single designated Gaylord location at the pack-out level
Max Chutes	(2) destinations, each with (1) single entry per sorter level, into a Max chute, and then manually sorted into (1) of multiple unique Hassett boxes at the pack-out level
Inbound Gaylord Chutes	(12) destinations, each with (1) single entry chute, into a single designated Gaylord location at the pack-out level
No Read Conveyor Chutes	(1) destination with (1) single entry per sorter level, transferred to a conveyor to the semi-auto induction area
Jackpot Container Chutes	(1) destination with (1) single entry chute, opposite to the No-read discharge, into a single designated container location
Direct Outbound Conveyor Chutes	(1) destination with (1) single entry per sorter level, transferred to a conveyor to the outbound load door

Non-Conveyable Chutes	(10) destinations, per semi-automatic induction, with no entry per sorter level, into a single designated container location below the induction platform *These chutes do not count towards the 339 chutes, as it is not connected to the sorter for discharge and is solely for manual discharge.

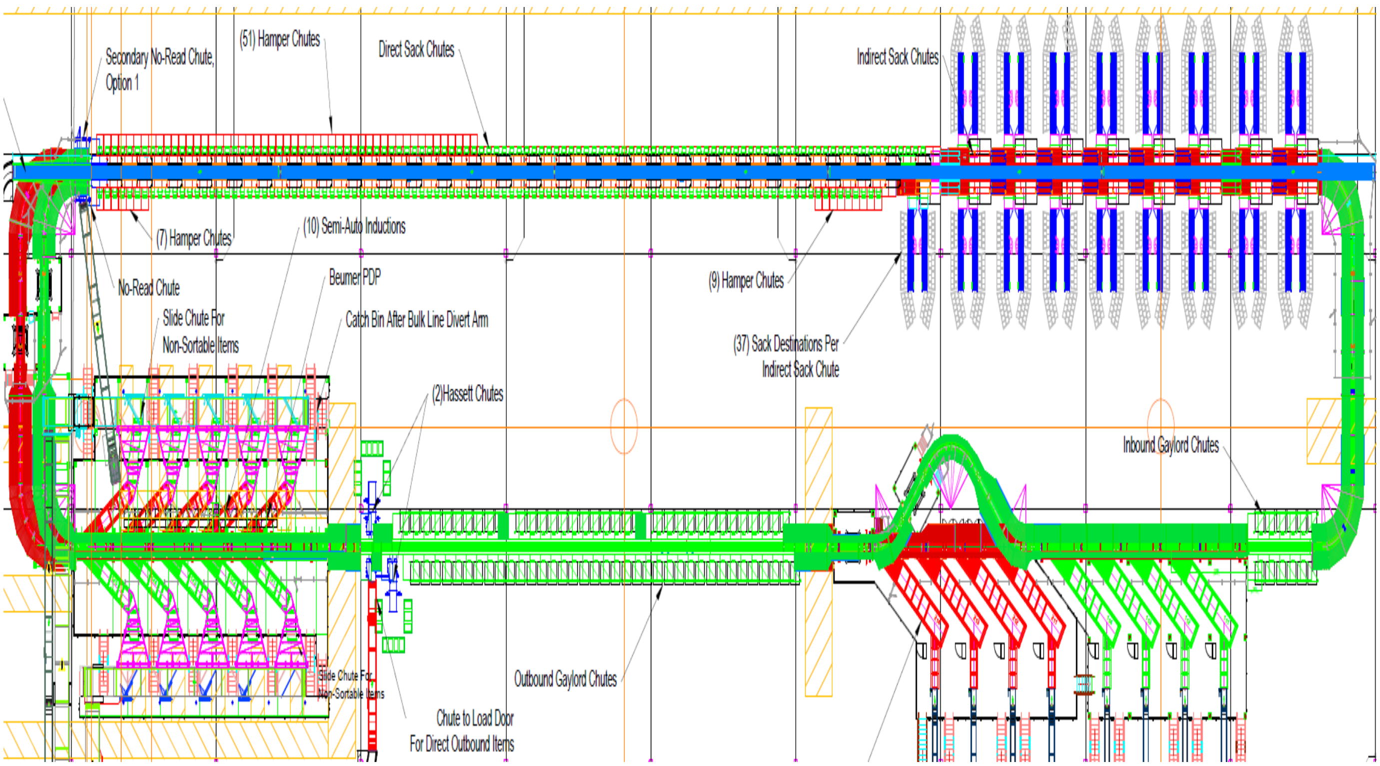


Figure 2-11 – Destination locations

3 Description of Operation

The BEUMER solution is designed to accommodate three different types of processes:

- Domestic Outbound
- Domestic Inbound
- Cross-border

The domestic outbound process handles parcels that have been collected locally (within the area of this facility) and are bound for other facilities in the country or abroad.

The domestic inbound process handles parcels from other facilities that are bound for the local area covered by this facility.

It may be the case that some parcels sorted via the domestic outbound process are bound to the local area, so they will be sorted directly to an inbound sack destination. Such parcels will be termed 'locally sorted items' throughout this document.

The cross-border process handles parcels that are bound for international locations.

All three processes can take place in parallel and are therefore described by DC building area in the following sub-sections.

3.1 Receiving and Infeed

The process is designed to handle all parcels described within the specifications above. Processes to handle non-conveyables before entering the system or on an exceptional basis after they enter the system are not included nor described.

3.1.1 Domestic Outbound

Parcels arrive at the receiving dock doors in bed-loaded trucks, containers, on pallets, or in sacks. The bed-loaded parcels are manually placed on a telescopic conveyor (extendable) provided and integrated by BEUMER but powered by DHL. Consequently, the extendable will be part of the encoding module e-stop zone.[2]

All domestic outbound parcels enter the encoding area provided by DHL. In this area, manual or automatic labeling systems will scan, weigh, and dimension each parcel and apply a DHL label. Then, the parcel leaves the encoding area <u>singulated and spaced with the label barcode face up</u> towards one of the conveyors feeding the outbound automatic inductions.

The transport conveyors, supplied by BEUMER, receive the parcels from the encoding and labeling area. It is assumed that all parcels leaving the encoding and labeling area have a readable shipping label placed on the top of the parcel, the short side leading and the longest axis in parallel to the direction of travel. Please refer to section 2.2 for the expected throughput rates.

[2] The outbound telescopic conveyor will not be installed day 1.

The transport conveyors hand over the parcels to the automatic cross inductions. The induction units work as a fully automatic process and no operator is necessary. It is assumed that only parcels within the product specifications will arrive to this area. A dimensioning system is incorporated in the automatic encoding and labeling area and therefore is not included in the BEUMER scope of supply.

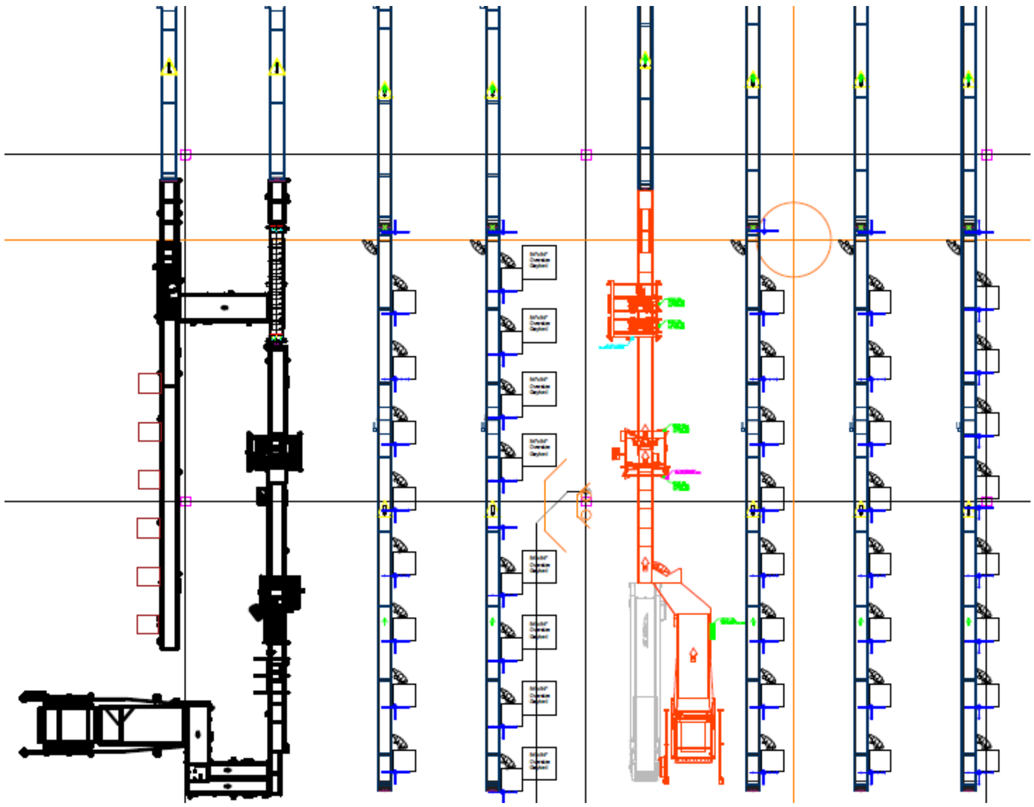


Figure 3-1 - Domestic Outbound to Automatic Cross-inductions

The encoding area feeds 8 automatic inductions in two induction areas, each with four inductions. One induction area is feeding the lower sorter, the other the upper sorter.

The interface between the encoding conveyors and the induction feed lines provided by BEUMER will be based on communication via digital signals.

☝	The encoding module must be designed to handle the inevitable start and stop situations (die back), where passing occupied trays on the sorter will cause the inductions and hence the infeed conveyors to temporarily stop.

3.1.1.1 Manual Encoding Lines

Operationally, the manual encoding process can be described as follows:

Packages arrive on pallets to the manual encoding station.

1. The operator grabs a package from the pallet
2. The operator scans the package barcode
3. The label printer prints either a 2D domestic mail barcode or a DSP cross-border barcode
4. The print is initiated by the workstation based on data from CONNECT or NewOps.
5. The operator applies the label to the package
6. The operator places the item barcode up on an empty space on the conveyor

Note: The operator must assure that items are singulated and that no items are placed side by side or on top of each other. To meet the desired rate, packages must be oriented by length in the direction of travel, aligned to the left in the direction of travel, and with an approximate pitch.

The encoding workstations must accommodate the following controls equipment, provided by others:

- Hand scanner
- Static scale
- Label printer
- PC workstation
- PC keyboard
- 24" flat screen monitor

The hand scanner is connected to the workstation that in turn is connected to the CONNECT host.

The manual encoding workstations, including all hardware and software, are provided by DHL.

Each single conveyor manual encoding line, along with the feeding line and induction, is controlled by the SIMATIC S7-300 PLC which controls the cross-induction. There is a 60-foot long belted conveyor where the operators will place the product with the barcode facing up in a singulated manner. Each conveyor is equipped with a sensing photo eye.

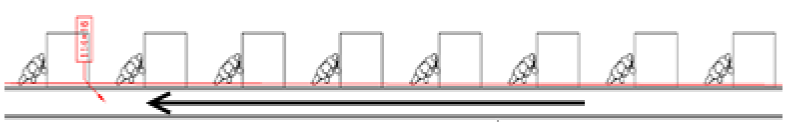
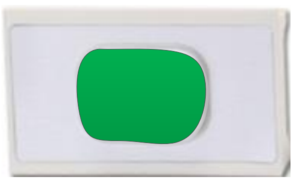
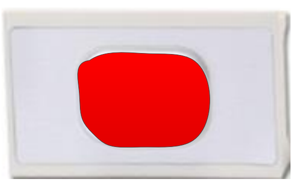
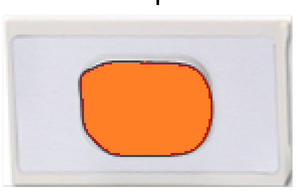


Figure 3-2 Manual encoding workstations line

Figure 3-3 Control Box

Each manual encoding line is started by pressing the start button located at the control box. If the sorter is not running or the induction and feeding lines are not ready to receive product due to a fault or dieback, the manual encoding line stops. Once the fault or dieback situation cleared, the line will start automatically.

The status of the line is signaled by the tower lamp located at the tail end of the operator belt. The green lamp is steady illuminated when the line is running. The green lamp is flashing when the line is about to start after the "Start" button was pressed. The red lamp is flashing if the emergency stop pull cord or push button at the manual line is active and it will be steady illuminated if the line is not in emergency stop condition anymore waiting for reset. To reset the emergency stop, the operator must press the reset button at the induction cabinet.

Element	Status	Description
Green Lamp 	OFF	Line stopped
	Flashing GREEN	Line is about to start after the "Start" button was pressed
	Solid GREEN	Line running
Red Lamp 	OFF	No emergency stop present
	Flashing RED	Emergency stop activated
	Solid RED	Emergency stop not active, waiting for reset
Amber Lamp 	OFF	No fault present
	Flashing AMBER	Line is faulted
	Solid AMBER	N/A

The items need to be released at the required pitch, with minimum gap, and aligned to the long side of the induction unit. **It is imperative for the operators to ensure that the product is properly singulated, aligned, and pitched. Failure to do so will cause multiple-reads or mis-sorts.**

Packages will be checked for overlength and overheight using a set of 2 photoelectric sensors located between the manual conveyor and the beginning of the incline conveyor. If an item is detected as being longer or taller than the specified maximum length or height, it will be stopped, and the amber lamp will start flashing. The operator must remove the out-of-specification item and press the "Reset" push button located at the tail end of the manual encoder conveyor.

The line will automatically start and resume normal operation. Failure to remove the oversize package before pressing the "Reset" button could damage the Beumer induction machines.

Note: The operators should perform a visual size inspection for each item before placing it on the conveyor to minimize stopping the manual line due to oversize items.

This single-conveyor version of the manual encoder has no gapping belts and no alignment section; therefore, there is no way for errors in gap, pitch, or alignment to be corrected automatically.

Due to these factors, it must be the operator's responsibility to place products with a pitch that approximates the required rate. Unfortunately, this will be difficult with no markers on the belt, and rate will decline if the pitch between products is either too large or too small. Specifically, if the pitch is too large then the lines will not supply enough products to the induction. If the pitch is too small, then the lines will force the induction to die back causing unnecessary stops and starts.

3.1.2 Domestic Inbound

Parcels arrive at the receiving dock doors in bed-loaded trucks or containers. The containers are transported to the dump area.

The containers are emptied at one of four (4) Gaylord dumpers (provided by BEUMER), two (2) feeding the upper sorter and two (2) feeding the lower sorter. Initially, only these total four (4) Gaylord dumpers are installed, with space provisioned for two (2) additional Gaylord dumpers, one (1) per sorter level.

The operator must perform the following tasks at the dumpers:

1. Take out the empty container if present with a pallet jack or manually
2. Insert the full container with a pallet jack
3. Activate the tipping process by pushing a button
4. Move on to the next dumper

The containers are emptied onto a bulk conveyor. To prevent jams on the bulk conveyor, the bulk conveyor must be running to enable the dump process. Additionally, some timing constrains in the control system will attempt to prevent large mounds of product.

Each dumper is equipped with a local control system and control panel but is tied to the central conveyor system provided by BEUMER. This allows a coordination of the dumper and the bulk conveyor.

The dumper entrance is protected by a light curtain that is connected to the local control system.

The first bulk conveyor in front of the dumpers runs continuously, with a fixed speed or stopped. The subsequent bulk conveyors leading to the induction area can be run in either continuous or indexing mode. This is controlled individually by conveyor line at two control boxes placed near the Gaylord dumpers.

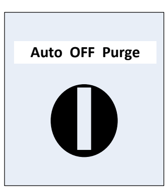


Figure 3-4 Selector Switch at Inbound Bulk Conveyors Area at Floor Level

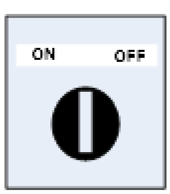


Figure 3-5 Selector Switch at Inbound Bulk Conveyors Area on the Platform

When the switch is in **Off** position the entire line from the Gaylord dumpers to the induction chutes is stopped. An additional selector switch is on the platform with

On and **Off** position. If either the switch on the platform or the one at floor level is in **Off** position will switch off the entire line. The **On** position of the switch on the platform will allow the line to run in the mode indicated by the position of the switch at the floor level.

When the switch is in the **Auto** position, the bulk conveyors upstream from the induction area will be indexing items. In the drawing below the yellow marked conveyors are indexing for the line feeding the lower sorter and the blue marked ones are indexing for the line feeding the upper sorter.

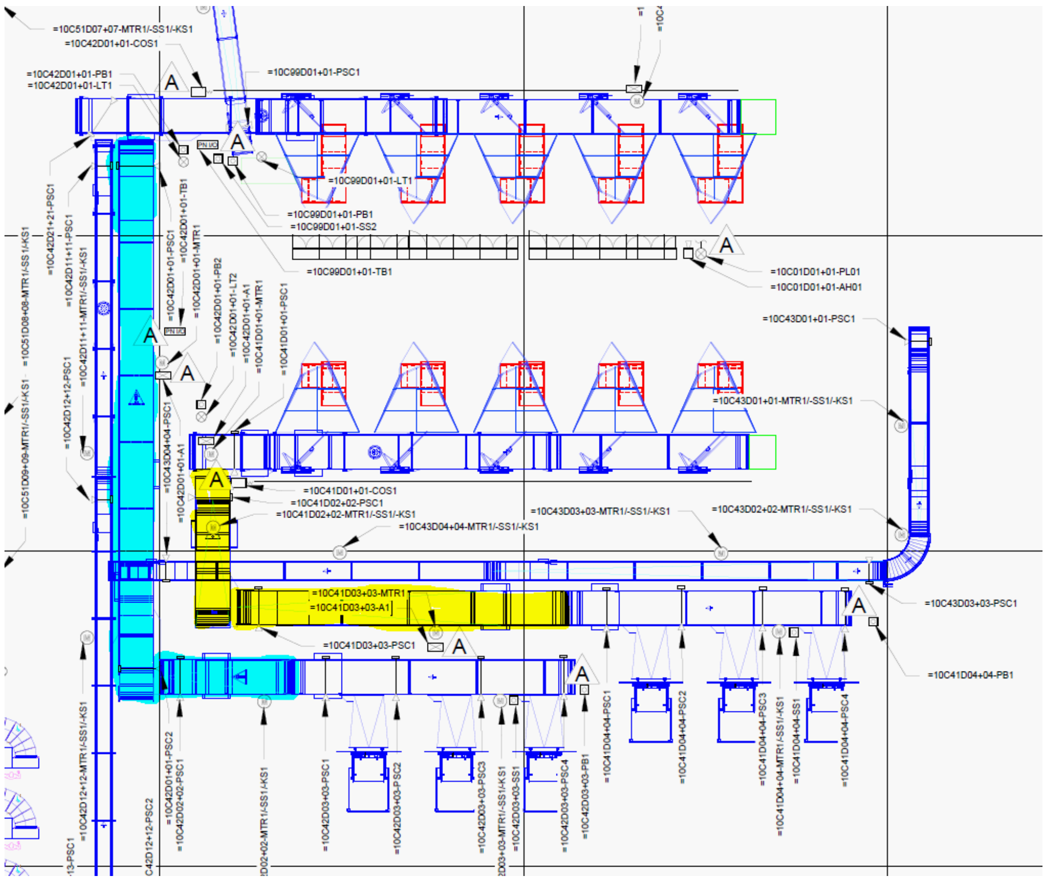


Figure 3-6 Inbound Bulk Conveyors Area

When the switch is in the **Purge** position, the bulk conveyors will run without indexing.

The conveyors equipped with divert arms will stopped if the selector switch for the line is in **Off** position or if there are no available chutes at the semi- inductions to discharge the items.

Operators can manually unload product directly to the collection belt through a table located at the beginning of one of the collection belts. Both, sacks and Hassett Boxes or other smaller boxes can be manually unloaded in this area.

The parcels are then transported in bulk to the semi-automatic inductions. Prior to arriving at the semi-automatic inductions, the parcels from the bedload unload merge onto one of the bulk conveyors.

This direct inbound conveyor originates at the extendable and may operate in two modes, direct in-bound and direct outbound. The direct outbound mode will be described later. In the inbound mode, the conveyor runs continuously until a blocking/full signal is received from the merge conveyor (the one with the divert arms).

For a visual illustration please refer to the illustration on the next page.

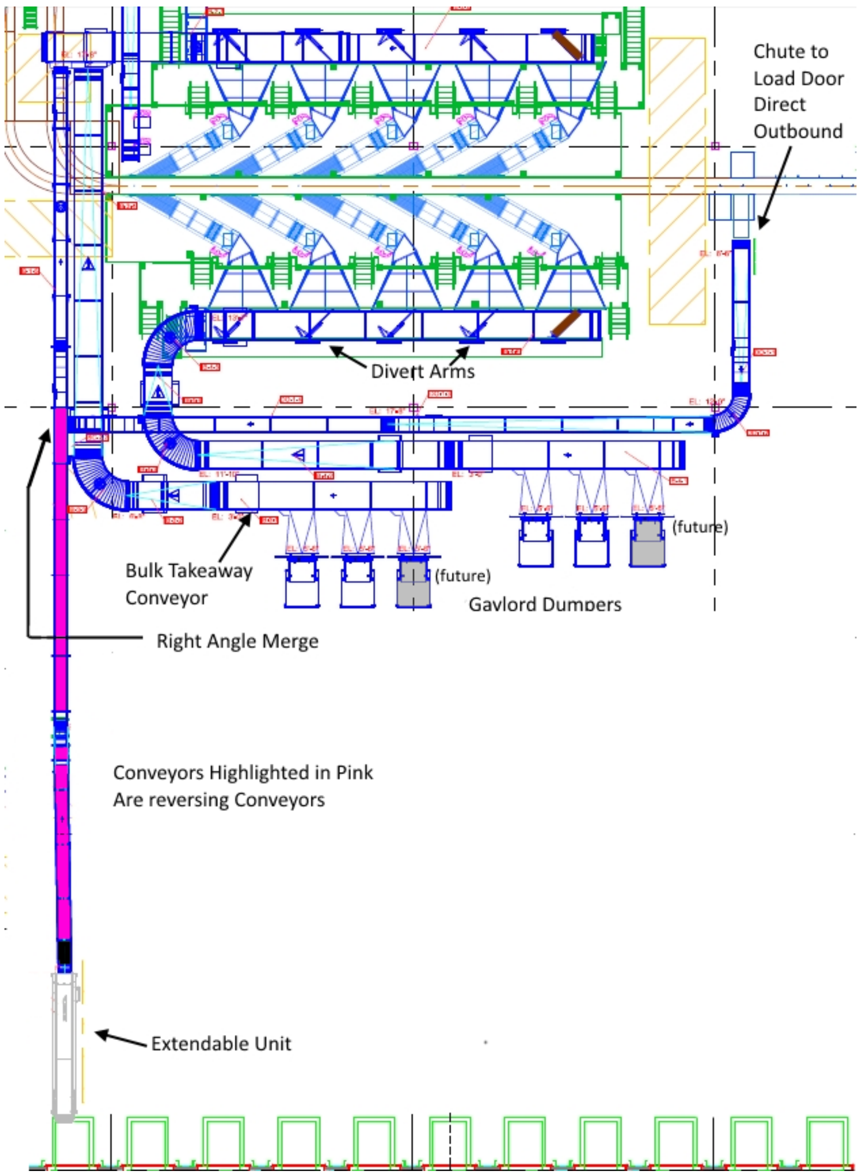


Figure 3-7 - Domestic Inbound to Semi-automatic inductions

A selector switch at each chute feeding a semi-automatic induction will allow the induction operator to activate or de-activate the discharge. A full photo eye placed on each of those chutes determines if the operator can accept or not more items. <u>If all induction stations are not active, full, or the first arm is detected as faulted in extended position, the feed conveyors will not run.</u>

Each induction will be feed using the Round Robin method as long as it is enabled, and the chute is not full. To reduce the amount of overflow items going to the last divert, the chutes will be processed from left to right, then return right to left: chute 1->2->3->4->5->4->3->2->1 etc. where chute 5 is the last chute in direction of travel of the feeding conveyor. The overflow items will be split between the chutes after the one is being processed by extending those divert arms. For example, if chute 2 is currently processed the divert arms for chutes 3 and 4 will also extend even if they are not enabled, distributing the possibly existing overflow items to the remaining chutes.

It is recommended that operations staff the semi-inductions consecutively starting from the tail-end of the feed conveyor.

3.1.3 Cross-border

Cross-border parcels are locally collected parcels that are destined to an international location. They are primarily fed to the sorter through the outbound induction area passing through the DHL CONNECT manual encoding stations, but it is possible that a secondary sort is performed through the inbound induction area. In this case, a different CONNECT sort plan should be used for the secondary sort to be meaningful.

Cross-border parcels will originally contain a CCN (customer-applied) barcode. They may be relabeled with a DSP (Delivery Service Provider) label, containing a DHL eCommerce barcode. In both cases, the barcode to sort code information is to be provided to the sorter by the CONNECT system. Both the customer-applied and DHL eCommerce barcodes will be used for sortation.

DHL CONNECT manual encoding stations will be positioned in the outbound induction area. Here, cross-border parcels will be encoded and labeled with DSP (Delivery Service Provider) labels. These labels will physically cover the original CCN barcode. Parcels passing through the DHL CONNECT manual encoding stations will be fed into the outbound induction lines featuring automatic inductions.

The CONNECT system will communicate with the BeSS server to transmit sort code information linked to either the DHL eCommerce or CCN barcode. This data transmission will be based on RESTful API communication (see section 0 for details). The common sort plan layout is:

Barcode	Sort Code	Additional optional parcel information

BeSS will use this information to translate scanned DHL eCommerce barcodes to sort codes, to be used in the lookup against the active CONNECT sort plan.

Cross-border parcels must conform to the general package dimensions.

3.2 Induction

3.2.1 Domestic Outbound

In the outbound process the conveyor system provides singulated and spaced packages to the automatic cross-induction units from where they are inducted onto passing empty carriages on the sorter.

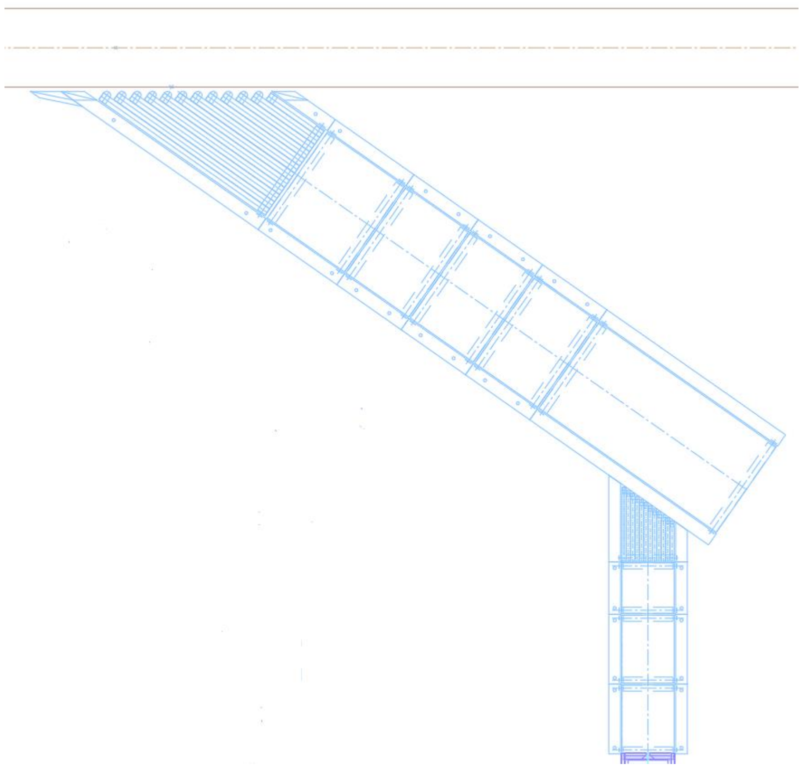


Figure 3-8 - Outbound automatic induction

The infeed conveyors immediately before the automatic cross-inductions and the manual encoding lines are provided and controlled by BEUMER.

3.2.2 Domestic Inbound

In the inbound process packages are bulk fed to the semi-automatic induction units, where operators face and place packages onto the first laydown belt, barcode face-up. A wooden template shows the exact orientation of the product to achieve the best induction result. The induction units then automatically induct the packages onto passing empty carriages.

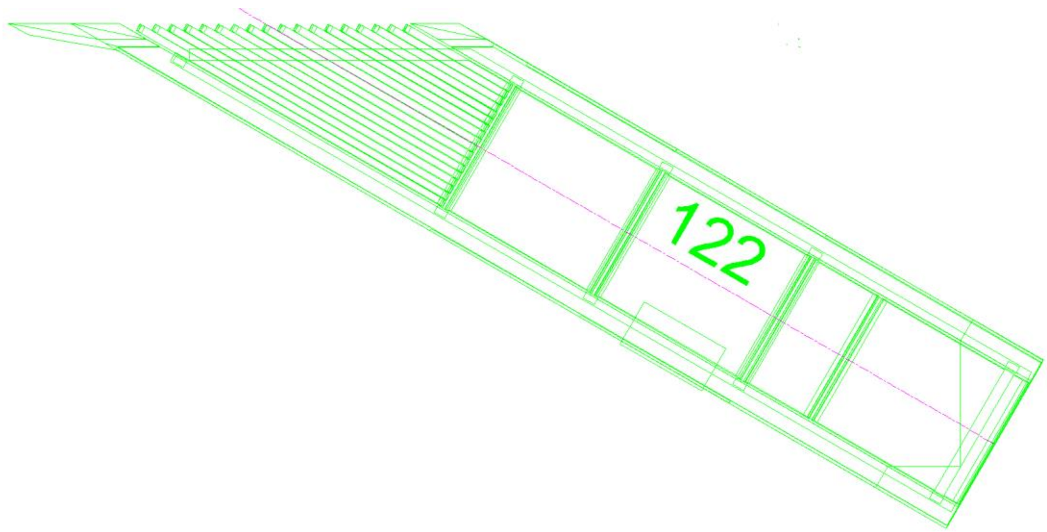


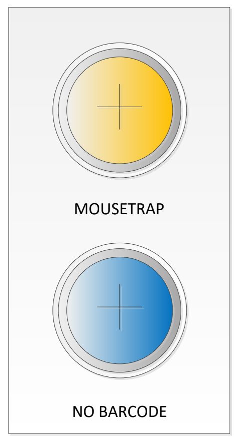
Figure 3-9 - semi-automatic induction at the inbound induction area

One common non-conveyable chute will be installed at each semi-automatic induction. The operators can collect some non-conveyables next to the induction unit and depending on the throughput either they or an additional operator (runner) can place them in the non-conveyable chute.

The no-read conveyor chute feeds adjacent to induction 225 (the last induction with respect to the direction of the sorter, on the upper platform). If configured as such, this no-read conveyor shall accept parcels that were unable to be read by the sorter scanners.

Also at induction 225, a no-read terminal with scanner is mounted at the laydown belt. Here, parcels may be manually scanned before being inducted on the sorter. Such parcels will be sorted using the manually-scanned barcode, without relying on the sorter-scanner to read the parcel's barcode.

Mouse Trap / No Barcode push-button unit at semi-automatic inductions	***Note**: These features are not included, but the pushbuttons are physically present at the inductions. As such, there is no current logic behind these pushbuttons and no support for these features in the sorter control system.* At each semi-automatic induction, a box with two lighted pushbuttons will be added, illustrated below. "Mousetraps" are small reusable trays to be used to hold and support non-inductables, such as small or cylindrical parcels, at the semi-automatic induction units. However, these mouse traps cannot go into direct sack or Gaylord chutes. A "MOUSETRAP" pushbutton may be pushed by an operator prior to inducting a parcel to flag it as a parcel in a mouse trap. In this case, the sorter will not consider direct sack or Gaylord chutes when sorting the parcel.

	For parcels where the operator identifies the barcode is missing or is impossible to be scanned by the sorter scanners, a "NO BARCODE" pushbutton may be pushed prior to inducting a parcel to flag it as a parcel with no barcode. In this case, the sorter will be sent directly to the configured no-read destination. Upon pressing the any of the pushbuttons, it will flash to indicate to the induction operator that the next inducted item will be flagged by the sorter control system as the corresponding status. Then, the next item to be inducted will be flagged as such and the pushbutton light is extinguished. If the pushbutton was pressed by mistake, it may be pushed again to no longer flag the next inducted item with the corresponding status. In this case, too, the pushbutton light is extinguished.  Figure 3-10 – Conceptual illustration of the induction pushbutton unit

3.2.3 Common induction features

In front of each induction area, every carriage is evaluated to check the status, i.e. if the discharge process was successful and the carriage is in proper working condition. The sorter PLC compares the physical information from the occupation sensors with the carriage's logical status saved in the sorter PLC. If a carriage is identified as non-functioning, it is blocked automatically. A blocked tray can only be reactivated in the BeOS.

Before each induction area, if a tray is verified to have discharged, it is set to available for induction, if considered empty and operational.

Note: BEUMER's automatic and semi-automatic inductions have a number of optimization features where poorly loaded packages are corrected on the induction or in extreme cases faulted. However, despite these corrective measures, packages that still end up between two belts will be detected, marked as an exception, and discharged to the corresponding exception destination, as configured via the BeSSView.

3.2.4 Semi-Automatic Induction Scales

The semi-inductions have been upgraded to include scale systems on both sorters. This is handled by OCS.

These scales provide weight measurements of items as they pass from the induction to the sorter. This information is captured by the Induction PLC and is then sent to the BeSS servers through an RR message.

These scales are present on the following inductions.

Lower sorter:

- Induction 113
- Induction 114

Upper sorter:

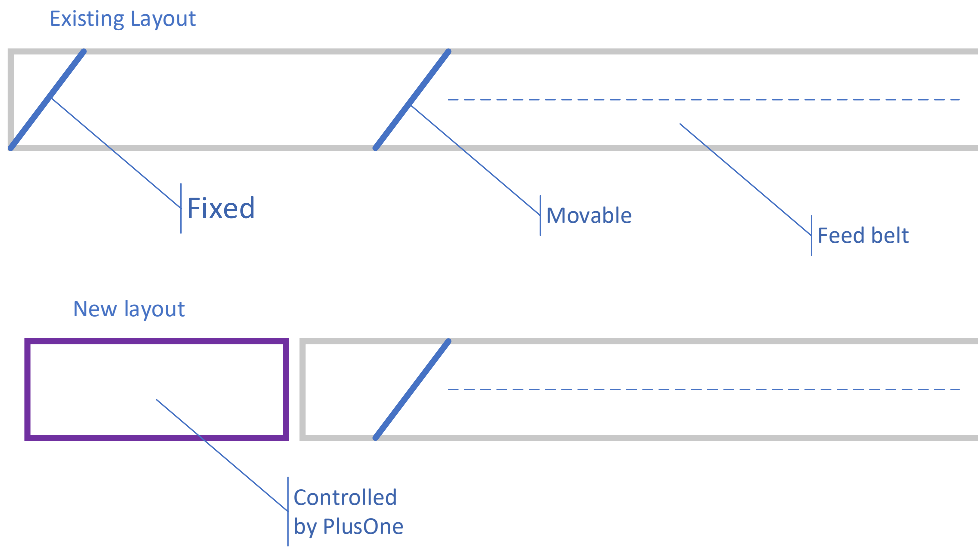
- Induction 211
- Induction 212

3.2.5 PlusOne Robotic Induction

One induction, induction 121 on the lower sorter, has been upgraded with a robotic induction addition.

The company PlusOne provided a robotic arm that automatically selects items from a bulk pile and places these items onto the BEUMER semi-automatic inductions. This bulk pile consists of items transferred via a PlusOne conveyor from the existing bulk distributor provided and controlled by BEUMER.

To establish the control of the product feed to the robotic induction, the existing distribution belt was shortened, and a new belt was installed in place, provided and controlled by PlusOne.



When the robot induction is ready to receive, the diverter arm logic will work as today, so the last diverter arm will at times – when signaled to be ready - be opened to allow the product flow onto the new belt. . When the robot induction is not able to receive product, the (new last) diverter arm will stay in position closed. That is achieved by using the current logic.

The ready signal is provided by the PlusOne control system via a digital opto-coupled signal back to the conveyor PLC (BCC).

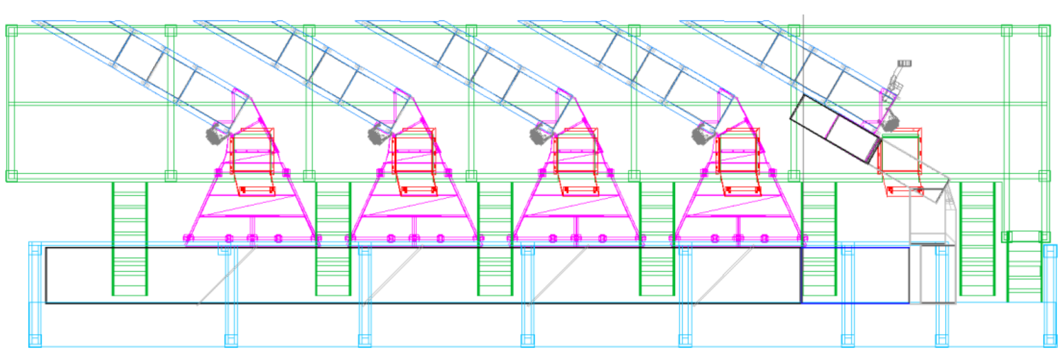


Figure 3-11 - Illustration of feed to the robotic induction

When the robot induction is ready, the induction PLC will indicate to the robotic control system that the laydown belt is clear and ready to receive the item. The robot arm moves the item from the pick to the place location, or the BEUMER induction location.

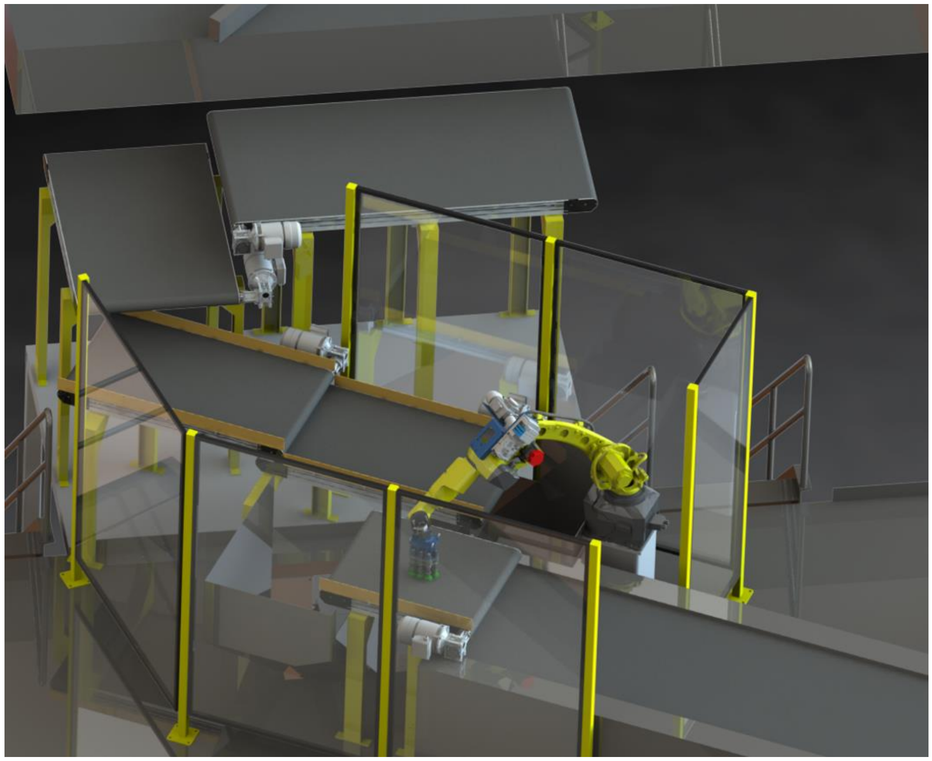
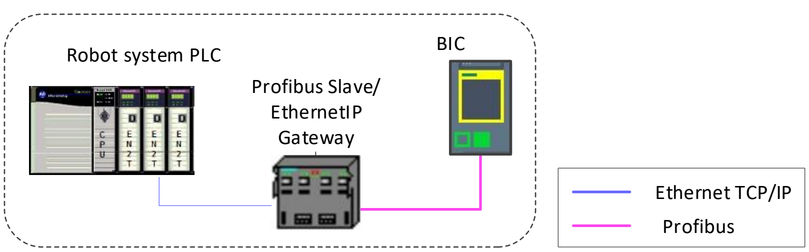


Figure 3-12 - Illustration of the robotic induction addition

When the robot is transferring the item, a bottom scanner positioned in between these two locations will attempt to read a barcode from the side of the item facing downwards. This bottom scanner is part of PlusOne's scope and controls. This scan data will be captured by the induction PLC and is then sent to the BeSS servers through an RR message.

The interface between BEUMER's machine control system (induction PLC) and the PlusOne robot control system is described in:

- 822-010184_Beumer_PlusOne_Robotics_Interface (version 2.2)



As the BEUMER machine control system is based on Siemens technology and hence ProfiBus / ProfiNet, an AnyBus adapter is deployed in the communication link to the Allen-Bradley based robotic machine control system.

The adapter is provided by PlusOne and placed in their cabinet. The wiring between their cabinet and the induction cabinet is done by PlusOne.

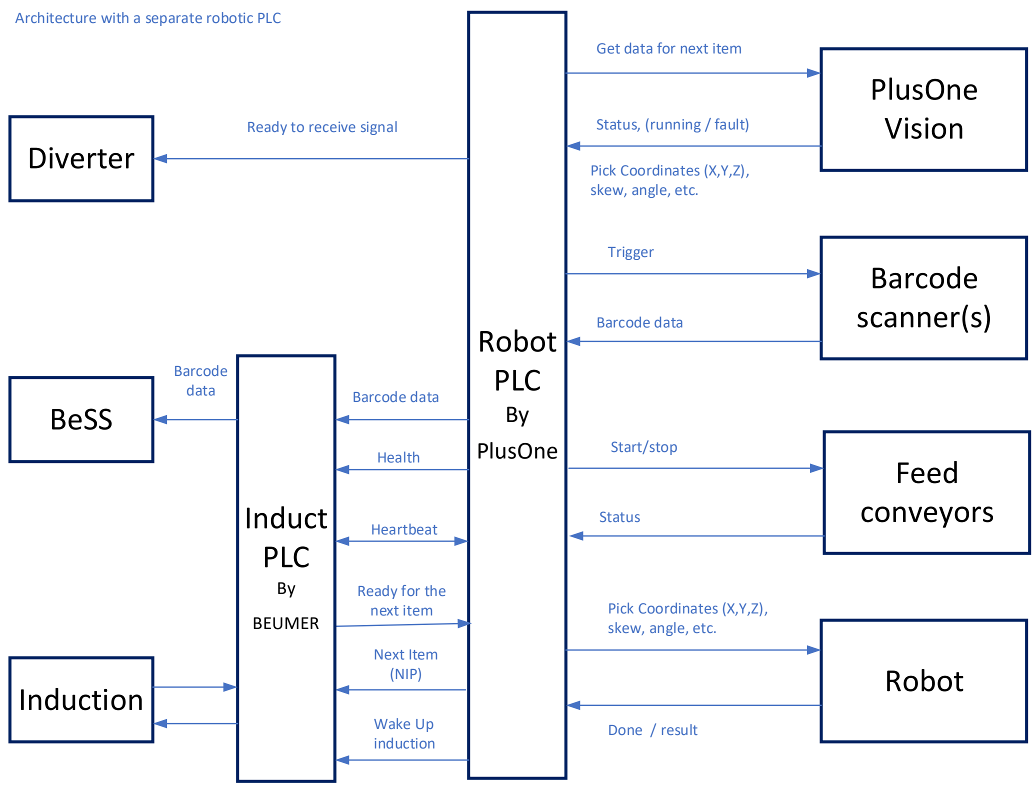


Figure 3-13 - Communication overview

The communication interface from the bottom-read barcode scanner allows up to five barcodes per item. The PlusOne system , however, only ever captures and transmits one barcode.

The barcode data is received by the induction PLC (BIC) via the Adapter. The BIC reads the barcode data when triggered by the NIP signal and the item is detected in the induction photo-eye. It then creates a TAN number and forwards the data to the BeSS using a new RR message via the TCP/IP socket interface.

3.2.6 Cross-Induction Auto-Clearing

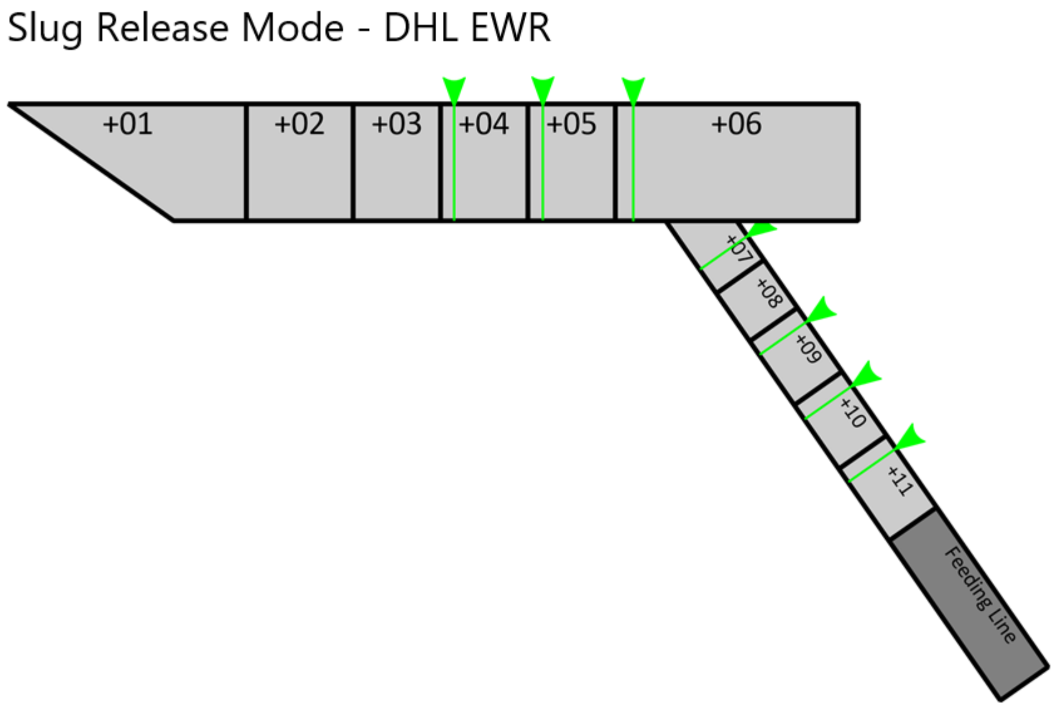
To reduce human intervention due to induction faults, auto clearing feature is introduced in the cross- inductions. The functionality will be active during all induction operational faults such as jam, overlength, etc.

All belts 1-6 on the main will reverse for specific amount of time until the items are discharged into a catch basket at the end of induction belt 6.

A PEC is placed across the catch basket to detect a full condition and alert Operations to clear the products and re-induct.

After certain amount of time taken for the above functions, the induction will run in forward clearing mode until all the PECs are clear, then return to normal operation.

3.2.7 Cross-Induction with Scale- Slug Release Mode



The induction operates with a concept of "stop positions". These are positions on the induction where items will stop if necessary (Item ahead is stopped, or tray is not available on the sorter). On the DHL EWR cross induction, items will stop at stop positions on belts +04, +05, +06, +07, +09, +10 & +11 if trays are not available on the sorter. Presently, to install a scale belt on the cross inducts, the last gapper conveyor belt is replaced with the scale belt. This causes unnecessary starts and stops of the conveyors and induction belt when trays are not available on the sorter. The slug release mode modifies the operation of the induction interface to the feeding line when the number of available trays is low, and the logic is modified to only allow new items onto the induction when there is space for a higher number of items at once instead of just one. This way the feeding line can feed multiple items continuously without stopping instead of feeding one or two at a time.

The way the mode works is when trays are not available on the sorter, the items stop on the main induction belts +04, +05, +06, and all finger induction belts. If a single tray is available, the item on belt 4 will be inducted, but the other items will not advance. Advancing the items is not necessary, because all information required for inducting the item on belt 5 is already known. No throughput is lost by this function. Once a second tray is available, the item on belt 5 will be inducted and all other items on the induction will move forward by at least two belts. In this way an item will never advance by a single belt at a time.

The slug release mode is enabled automatically when available trays on the sorter goes below a parameterized threshold. (Currently: 80%) The tray availability is calculated at a position before the induction bank and is updated every minute.

There is a hysteresis applied to the enable/disable. While slug release mode will be enabled when tray availability goes below 80% percent, it will only be disabled once tray availability goes above 85% percent.

Additionally, there is a lower threshold for "larger slug mode" where the size of the slug is increased (currently 70% percent). When tray availability goes below this value, the size of the slug will be increased by an additional item.

When slug release mode is enabled by tray availability, the function operates as follows:

The feeding line interface operates normally until the finger induction becomes "full of items". When the finger induction becomes full, the feeding line is blocked from feeding additional items until the induction is "sufficiently empty". This guarantees the ability to feed a minimum of 4 items consecutively from the feeding line to the induction.

The induction is considered "full of items" (stopping the feeding line) when either:

• The last item is actively stopping / is stopped at the stop position on belt +10 (second to last induction belt). In other words, an item had to stop on belt +10, but the gap in items behind was large enough that another item is not on the induction already behind it.

• The last item will be stopped at the stop position on belt +11 (last induction belt; scale belt). In other words, there is an item ahead that would stop on belt +10, and this item will stop on belt +11 unless items ahead advance forward. The addition of the first condition is done to minimize the number of items that are stopped on the scale (belt +11).

The induction is considered "sufficiently empty" (reenabling the feeding line) when:

- In "normal slug mode": The stop position on belt +07 is available.
  - Usual situation for feeding line restart: An item on belt +06 was stopped and just started advancing. An item stopped on belt +07 began advancing forward, and the rest of the finger induction is completely empty.
  - A minimum of four items will advance from the feeding line.
- In "larger slug mode": The stop position on belt +06 is available.
  - Usual situation for feeding line restart: An item on belt +05 was stopped and just started moving to be inducted to the sorter, and an item stopped on belt +06 began advancing forward. The finger induction is completely empty.
  - A minimum of five items will advance from the feeding line.

3.3 Sort preparation

Prior to start of sortation, the DHL Master Data server(s) will download any necessary **sort plan** and **chute attribute configuration** files to the sorter control system (often referred to as the BeSS). The sort files are necessary for sorting parcels to pre-determined destinations based on the parcel's extracted sort code. The chute attribute configuration files are necessary for sorting Ticket-to-Ride items only to allowed destinations, based on the parcel's extracted attribute.

Sort plan and chute attribute configuration files may only be created via download to the BeSS server by the DHL Master Data server(s). The download consists of the Master Data server(s) transferring the file over a sFTP site hosted by the BeSS server (see section 3.7.3 for details). The BeSS server will monitor a particular directory for any newly downloaded files. Upon successful validation, the file will be moved to a "pass" subdirectory. Upon encountering an error in the validation, the file will be moved to a "fail" subdirectory along with an error file containing an error description. Both subdirectories are accessible by the Master Data server(s).

3.3.1 Sort Plan Files

The downloaded sort plan files are used during sortation to translate sort codes to chute(s). For all sort plan files, up to three (3) chutes may be specified per sort code. The translation from mail item barcode to sort code is described in section 2.1.2.1, for each mail item type.

There will be three (3) separate sort plan files to be downloaded to the BeSS:

1. Ticket-to-Ride Sort Plan for sortation of domestic mail items containing Ticket-to-Ride barcodes
2. USPS eVS Sort Plan for sortation of domestic mail items containing USPS eVS barcodes
3. CONNECT Sort Plan for sortation of cross-border mail items containing cross-border barcodes

Additionally, all sort plan files may contain chute display information and a list of cuboidal chutes. Chute display information specifies text to be displayed at a chute's display module, where available. The list of cuboidal chutes specifies which chutes will be considered as a cuboidal chute. If a package is marked with the "cuboidal" flag it will be assigned with first priority to the chute that is defined as a cuboidal chute. Only "cuboidal" items may be assigned to a cuboidal chute. The maximum number of chutes that can be added in the cuboidal row is the maximum number of chutes on the sorter.

The common sort plan layout consists of zero or more of the following messages, in any order:

<u>SORT message</u>

Sort code	Primary sort destination	Secondary sort destination	Tertiary sort destination

DISPLAY message

Chute	Text to be displayed

CUBOIDAL message

1st Priority Cuboidal Chute	2nd Priority Cuboidal Chute	3rd Priority Cuboidal Chute	..Nth -1 Priority Cuboidal Chute	Nth Priority Cuboidal Chute

Note: The BeSS server copies successfully-downloaded sort plan files in the "pass" directory for informational purposes only. The actual sort plan files are stored in a separate internal directory, inaccessible by the Master Data server(s) via the sFTP site. Therefore, deleting a file within the "pass" subdirectory will not delete the file from the sorter system. Sort plan files may only be deleted from the system by authorized operators via the BeSSView.

3.3.2 Chute Attribute Configuration Files

The downloaded chute attribute configuration files are used during sortation to determine to which chutes Ticket-to-Ride mail items are permitted to discharge. Ticket-to-Ride barcodes are the only barcodes that contain a two-digit hexadecimal attribute field. Within the chute attribute configuration file, a chute may be configured to allow any combination of 256 different attributes (actual attribute values are '00' – 'FF').

The principle domestic sort table layout consists of zero or more of the following message:

CHUTE message

Chute	List of allowable attributes ('00' – 'FF')

Note: The BeSS server copies successfully-downloaded chute attribute configuration files in the "pass" directory for informational purposes only. The actual chute attribute configuration files are stored separate internal directories, inaccessible by the Master Data server(s) via the sFTP site. Therefore, deleting a file within the "pass" subdirectory will not delete the file from the sorter system. Sort plan files may only be deleted from the system by authorized operators via the BeSSView.

3.3.3 Activation

The sort plan and chute attribute configuration files may be activated, deactivated, and deleted via the BeSSView. At most one (1) of each type of sort plan file may be active at the same time. Similarly, at most one (1) chute attribute configuration file may be active at the same time. All four file selections (three sort plan & one chute attribute configuration) are activated at the same time, via the BeSSView. Upon activation, the system performs basic validation and also ensures that the domestic sort plans (Ticket-to-Ride & USPS eVS) do not share chutes with the CONNECT sort plan. If validation fails, no changes are made to the currently active files and feedback is given to the operator. If the validation fails for one of the following reasons:

- No active sort plans are selected
- A Ticket-to-Ride sort plan is selected but no Chute Attribute configuration is selected
- The domestic and CONNECT sort plans share chutes

, then the user is prompted with a warning message. The user then has the choice to confirm or cancel the activation of the sort plans.

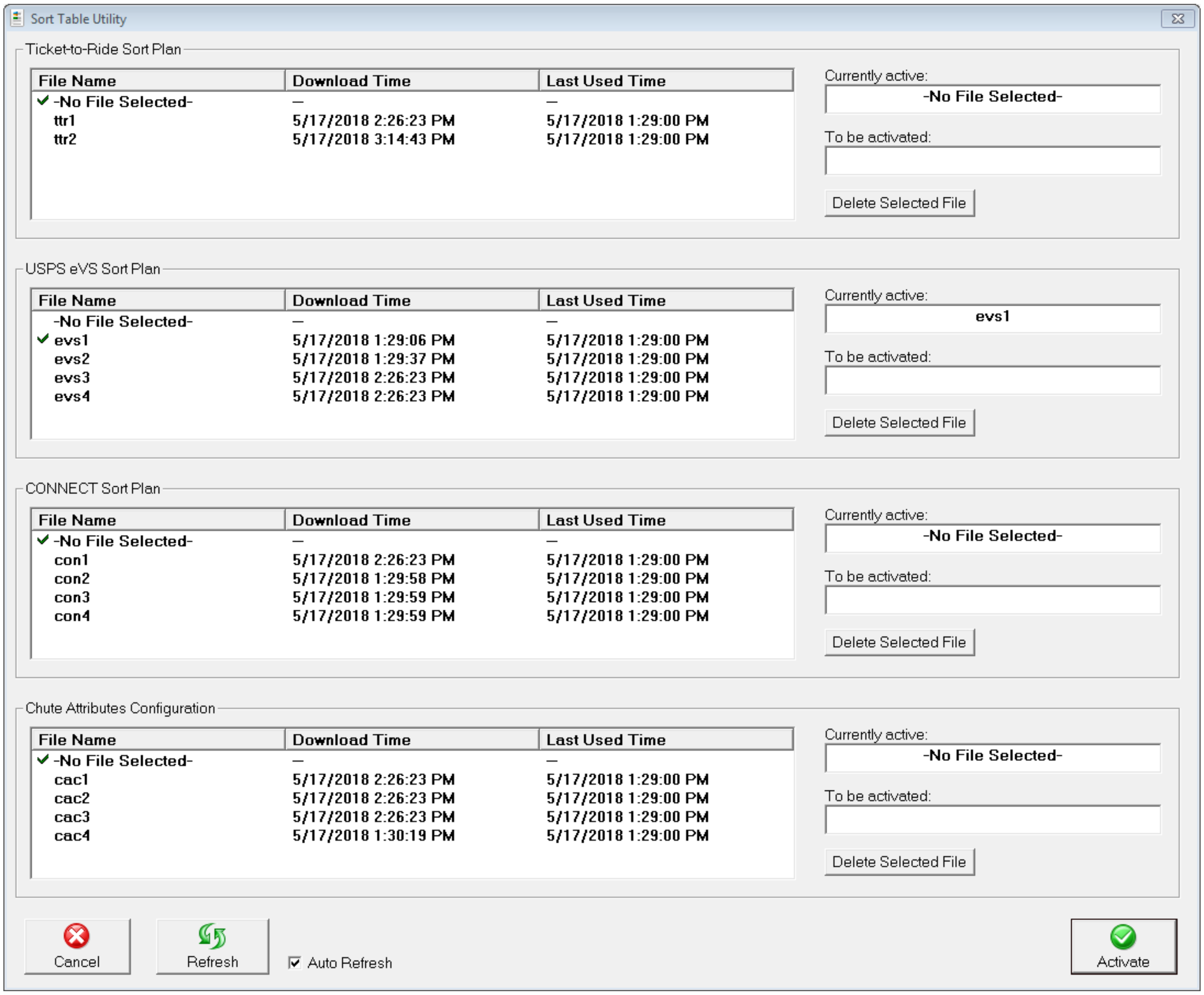


Figure 3-14 - Sort Table Utility window

3.4 Sorting

After the induction area, a centering device locates the package on the belt tray and the sorter PLC (BSC) adjusts the position of the package to be centered to optimize the discharge. The tray then passes the scanning tunnel that reads the barcodes on the parcel and determines if the item is cuboidal or non-cuboidal. This information is then sent to the BeSS. Soon after the scanning tunnel at a pre-defined request point, the BSC sends the BeSS a request for sort destinations. Based on the barcode(s), the parcel is identified as a Ticket-to-Ride, USPS eVS, or Cross-border parcel (as described in section 3.4.4). After the parcel type is identified, the extracted sort code and cuboidal flag on the item is used in the destination lookup against the corresponding sort plan and replies to the BSC.

3.4.1 Scan Tunnel

Included in the BEUMER scope is the supply, mechanical installation, electrical installation, and commissioning of four (4) Datalogic Camera Scan Tunnels. Two (2) are top-reading only, and two (2) are upgraded to be 5-sided scan tunnels.

3.4.2 Top Reading Scanner Tunnels

These camera systems will be installed after two induction areas.

Each of the two (2) DATALOGIC scan tunnel systems is configured as follows:

- (1) AV7000 camera configured for top reading; mounted in Skew/Counter-Skew orientation with an angle of app.+/- 10°
- (1) AI7000-1100 Medium Lighting System
- (1) Power Supply PG-600
- (1) Camera Support
- System Cables and Connectors
- Installation and Commissioning
- Communication with the BSC via Profibus
- Communication with the BeSS via TCP/IP
- WebSentinel maintenance software

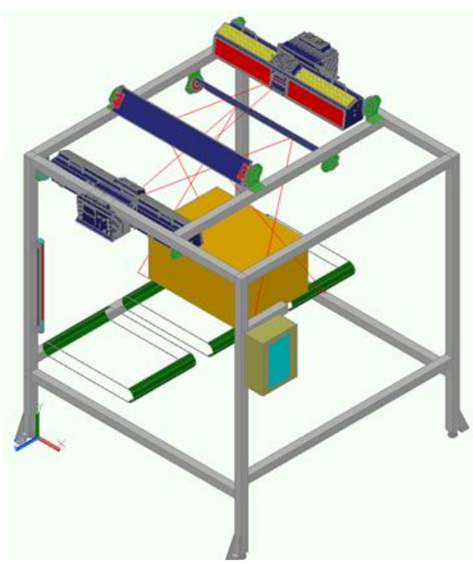


Figure 3-15 - Conceptual illustration of the scanner system and scanner frame

If a scan tunnel is unable to obtain a barcode read, the system can be configured to attempt multiple times based on a configurable *max-scan* parameter (configurable via the BeSSView). In most cases based on practical experience this is limited to a maximum of scan attempts of 3.

3.4.3 5-Side Reading Scanner Tunnels

These scanner tunnels were upgraded from the top-reading scanner tunnels to be 5-sided scanner tunnels with LFT dimensioning. Scanning materials from the top-reading stations are reused for the upgraded 5-side stations. Each of these two (2) DATALOGIC scan tunnel systems is configured as follows:

- (1) AV7000 camera configured for top reading; mounted in Skew/Counter-Skew orientation with an angle of app.+/- 10°
- (4) AV7000 cameras configured for left/front, right/front, left/back, and right/back reading.
- (5) AI7000-800 Lighting System
- (5) EMK-NVS-800 Mirrors
- (2) DM3610-3100 Multi-Head LFT NTEP Dimensioners
- (1) DC3000-2100 Multi-Head LFT NTEP Controller
- (1) 24V READ NOW SIGNAL (BEUMER supplied)
- (1) PGD-100 Contactless Tachometer
- (5) 600W Power Supply (AV7000)
- (1) 240W Power Supply (DM3610/DC3000)
- (1) Power Distribution Panel (PDP)
  (1) Camera Support
- System Cables and Connectors
- Installation and Commissioning
- Communication with the BSC via Profibus
- Communication with the BeSS via TCP/IP
- WebSentinel maintenance software

3.4.4 Item Identification

The barcode specifications are described in section 2.1.2. Based on the barcode(s), the parcel is identified as a Ticket-to-Ride, USPS eVS, or Cross-border parcel. The parcel type determines how to translate from barcode to sort code, and which corresponding sort plan to use for sortation of the parcel.

3.4.4.1 Domestic Parcels

If a parcel contains just a 2D Ticket-to-Ride barcode, or both a 2D Ticket-to-Ride barcode and a 1D USPS eVS barcode, then it is treated as a Ticket-to-Ride parcel. If a parcel contains just a 1D USPS eVS barcode, then it is treated as a USPS eVS parcel. See section 2.1.2.1 for details on how Domestic (Ticket-to-Ride & USPS eVS) barcodes are identified.

All domestic parcels will contain a 2D Ticket-to-Ride and/or a 1D USPS eVS barcode. A maximum of one (1) domestic barcode will be used for sortation. If a domestic parcel contains multiple barcodes but only one is identified as a domestic barcode, then it will be sorted as a domestic parcel using the sole domestic barcode. If Ticket-to-Ride parcel contains both a 2D Ticket-to-Ride barcode and a 1D USPS eVS barcode, then only the 2D Ticket-to-Ride barcode will be used for sortation. If a domestic parcel contains either multiple 2D Ticket-to-Ride barcodes or multiple 1D USPS eVS barcodes, then it is treated as a multiple-read exception item and sent to the corresponding exception destination (see section 3.4.9 for further details regarding sorting exceptions).

For domestic items with a valid barcode, the sort code is parsed directly from the barcode. See section 2.1.2.1 for details on how sort codes are parsed from these barcodes. Depending on the identified parcel type, the extracted sort code is to be used to look up chutes within the active corresponding Ticket-to-Ride or USPS eVS sort plan.

Additionally, for Ticket-to-Ride barcodes, the attribute is parsed directly from the barcode. See section 2.1.2.1 for details on how attributes are parsed from these barcodes. The extracted attribute is to be matched against the active chute attributes configuration to sort only to chutes that allow that attribute.

3.4.4.2 Cross-border Parcels

If a parcel contains only cross-border barcodes, then it is treated as a cross-border parcel. See section 2.1.2.2 for details on how cross-border barcodes are identified.

A maximum of three (3) cross-border barcodes will be used for sortation. For each of these barcodes, the sorter control system will attempt to translate the barcode to a sort code using parcel information previously-transmitted to the sorter during its manual encoding process. See section 3.1.3 for details on how cross-border parcel information is transmitted to the sorter. Each extracted sort code is to be used to look up chutes within the active CONNECT sort plan. Only the first sort code to yield a chute assignment will be used for sortation.

3.4.5 Mousetrap Items

Note*: This feature is not included, but the related induction hardware is physically present. As such, there is no current logic behind the hardware and no support for this feature in the sorter control system.*

Parcels that are too small, cylindrical, or otherwise unsafe to induct and sort may be placed in a special DHL-provided "mousetrap" tray. As this tray must remain in the facility and should not be part of the final sorted parcel, mousetrap items should not be sorted into direct sack or Gaylord chutes. Therefore, mousetrap items shall only be allowed to sort to indirect sack or Max chutes, where operators will have an opportunity to manually handle the item before placing into its next container.

The induction operator must press the dedicated lighted "MOUSETRAP" pushbutton at the semi-automatic induction indicating that the next parcel is in a mousetrap and therefore can only be discharged at an indirect sack chute or a Max chute.

Please refer to section 3.2.2 for further details regarding the illuminated pushbutton unit at the semi-automatic inductions.

3.4.6 Items with Unreadable Barcodes

***Note**: This feature is not included, but the related induction hardware is physically present. As such, there is no current logic behind the hardware and no support for this feature in the sorter control system.*

Parcels that are deemed to have missing or unreadable barcodes may be flagged as such by the induction operator to the sorter control system. All semi-automatic inductions would be equipped with a lighted pushbutton with which the operator may press to that the next parcel did not have a readable barcode and should be sent directly to a pre-configured destination. This feature is not included in the final scope. The lighted pushbutton will be present at the inductions.

Please refer to section 3.2.2 for further details regarding the illuminated pushbutton unit at the semi-automatic inductions.

3.4.7 Chute Assignment

If the item is an exception item, it will be sent to the corresponding exception destination (see section 3.4.9 for further details regarding sorting exceptions).

Once the sorter control system has scanned a parcel for barcodes, identified the parcel type (e.g. Ticket-to-Ride, USPS eVS, or cross-border), extracted its sort code (and attribute, if applicable), and identified if the package is cuboidal or non-cuboidal, chutes can be fetched from the parcel's corresponding sort plan to be considered for sortation.

For a USPS eVS and cross-border parcel, the corresponding sort plan is searched for an entry whose sort code field that exactly matches the parcel's extracted sort code. An entry may contain up to three (3) chutes to be considered for sortation (primary, secondary, & tertiary).

For a Ticket-to-Ride parcel, the Ticket-to-Ride sort plan is searched for all entries whose sort code fields are left-substring matches of the parcel's extracted sort code. Each entry may contain up to three (3) chutes to be considered for sortation (primary, secondary, & tertiary). Entries whose sort code matches more of the parcel's extracted sort code are prioritized for chute assignment. Only the two (2) best matches are used in sortation, for a maximum of six (6) chutes to be considered for sortation. Furthermore, only chutes whose chute attribute configuration contains the parcel's extracted attribute will be considered for sortation.

If no chutes can be considered for sortation, the item is deemed a no-data exception item and sent to its corresponding exception destination (see section 3.4.9 for further details regarding sorting exceptions).

For all chutes being considered for sortation, the chute assignment order (e.g. order of discharge) is determined by the following:

i. Matching sort plan entries whose sort code matches more of the extracted sort code (applies only to Ticket-to-Ride parcels)
ii. Primary chutes will be considered for assignment over secondary chutes, for the same entry

iii. Secondary chutes will be considered for assignment over tertiary chutes, for the same entry

After the list of chutes are determined that match that parcels sort code, BeSS will consider if the item is a cuboidal or non-cuboidal item, and the list of cuboidal chutes in the active sort plans. If an item is determined to be cuboidal by the scanner, BeSS will look at the list of chutes that match the item's sort code and check if any of these chutes are in the list of cuboidal chutes in the parcel's corresponding sort plan. If any of the chutes match the chutes in the list of cuboidal chutes in the sort plan, BeSS will rearrange this list giving the highest priority to the cuboidal chutes.

If the item is determined to be a non-cuboidal item, BeSS will look at each chute labeled as a cuboidal chute across all active sort plans and remove them from the item's list of possible chutes to be assigned to if any of these chutes match. This is because only cuboidal items are allowed to be assigned to a cuboidal chute.

Once the final list of chutes for an item is created, only the highest-priority chute will be used for sortation.

3.4.8 Redirection Rules

If a parcel could not be sorted at its initial assignment, defined rules determine the subsequent behavior.

If a parcel fails to discharge to its initial chute assignment before a configurable number of attempts has been reached (*Recirculation Reassignment* parameter, configurable via BeSSView), the parcel will then simultaneously consider all eligible chutes for discharge. From this point on, a discharge attempt will be made at any eligible chute as the parcel passes the chute.

If a parcel recirculates a configurable maximum number of times (*Max # of Recirculations* parameter, configurable via BeSSView), it will be sent to the corresponding exception destination (see section 3.4.9 for further details regarding sorting exceptions).

3.4.9 Sorting Exceptions

Parcels may experience various sorting exceptions, in which case they will be discharged the corresponding configured exception destination, unless otherwise stated. Exception destinations can be configured via the BeSSView.

- No Read
    - The system can be configured to attempt multiple times based on a configurable *max-scan* parameter (configurable via the BeSSView). If a parcel is detected and the scan tunnel returns a scan result containing no barcode, after the configurable maximum number of attempts, the parcel is marked as "No Read" and will be discharged to the configured "No Read" exception destination.

- Multiple Read
  - If a parcel has been detected and a valid scan result is obtained from the scan tunnel, then if the parcel contains multiple of the same domestic barcode (e.g. two Ticket-to-Ride, or two USPS eVS) barcodes, then it is marked as a "Multiple Read" and sent to the configured "Multiple Read" exception destination.
- No Data
  - If a parcel is detected and a valid scan result is obtained from the scan tunnel, then the parcel may be marked as "No Data" if any of the following occur during the chute assignment process:
    - The cross-border barcode was unable to be translated to a sort code since it did not match any of the received cross-border parcel information (applicable only to cross-border parcels). See section 3.1.3 for details on how cross-border parcel information is transmitted to the sorter.
    - The extracted sort code does not match an entry in the corresponding sort plan.
    - The extracted sort code matches at least one entry in the corresponding sort plan, but the parcel is not eligible to discharge to any of the chutes due to mismatching attributes (applicable only to Ticket-to-Ride parcels).

    If the parcel is marked as "No Data", it will be discharged to the configured "No Data" exception destination.
- Discharge Fault
  - If a parcel that should have been discharged to a destination is still detected on the belt tray, then it is marked as "Discharge Fault" and will be discharged to the configured "Discharge Fault" exception destination. Occupation sensors that detect whether a belt tray is occupied are located at the end of each chute bank, per sorter.
- Stray Item
  - Prior to the scan tunnel, if the centering device unexpectedly detects a parcel on a belt tray (e.g. the parcel was not assigned to that tray at the induction), then it is marked as a "Stray Item" and will be discharged to the configured "Stray Item" exception destination.
- Too Many Recirculating Trays
  - If the total number of recirculating trays exceeds a configurable percentage (*Recirculating Tray %* parameter, configurable via BeSSView), then recirculating parcels are marked with the "Too Many Recirculating Trays" exception and will be discharged to the configured "Too Many Recirculating Trays" exception destination.

Once the total number of recirculating trays drops below the configurable percentage, recirculating trays are no longer subject to this exception.

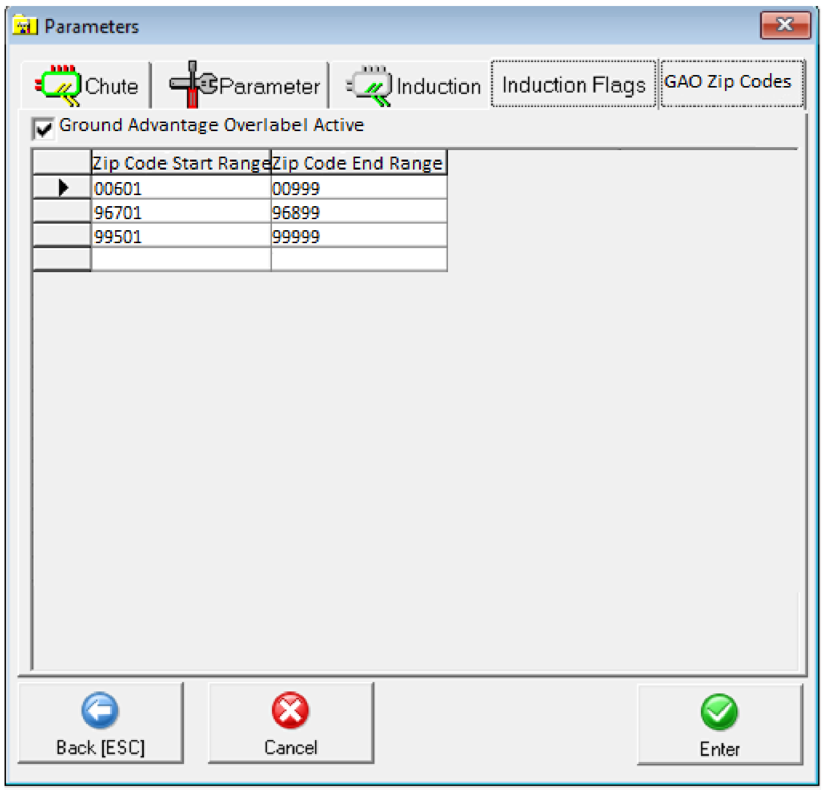
  - There is also a "Recirculating Tray %" parameter in the PLC, which is used to stop the inductions from inducting items onto the sorter if the total number of recirculating trays exceeds the parameter value. The inductions will resume inducting items onto the sorter once the total number of recirculating trays drops below the parameter value.
  - **Note**: If the total number of recirculating trays exceeds the configurable percentage, parcels that have not yet recirculated will not be marked with this exception; only upon completing at least one recirculation will the parcel be subject to this exception.

- Tray Spacing
  - Due to necessary delays when identifying chutes as FULL, consecutive parcel discharges to certain chute types may cause undesirable consequences (e.g. sorter jams, overfilling sacks, parcels reaching sorter track, etc.). As such, the sorter control system may need to limit consecutive discharges to particular chute types to a certain number of trays, and then wait to resume consecutive discharges after skipping a certain number of trays. This feature is known as tray spacing. The actual number of trays that may discharge consecutively and the actual number of trays that are skipped before further consecutive discharges may resume are determined by physical tuning during on-site commissioning.
  - If a parcel cannot discharge to a chute due to tray spacing, it will follow the redirection rules, as defined in section 3.4.8.
- Overlabel/STC Reject
  - Inductions can be assigned the STC Reject flag inside of BessView; this flag gives each item that passes through the induction an additional set of requirements to pass before it can be sorted normally.
  - Based on the STC code present in the EVS barcode, it will be sorted based on the below table.

STC Code	Logic
021, 419, 422	Do Not Reject
001, 703, 704, 748, 835	Reject if item weight is greater than or equal to 1lb
055, 108, 612, 615	Reject if item weight is less than 1 lb
All other STC codes	Reject All

- If the item is rejected, it will go to the Overlabel special destination.
- If an item is inducted from an STC flagged induction, and it has an unknown weight OR fits any other exception category (no read, multi-read), it will be redirected to the overlabel special destination.

- Ground Advantage Overlabel (GAO)
    - Parcels within specific 5D Zip Code ranges, configurable via a BeSSView window, will be rejected and sent to the Ground Overlabel exception destination. BeSSView Window example shown below:



The entire feature can be toggled on/off via the new parameter window.

    - items rejected due to Ground Advantage Overlabel reason will also provide the reject reason "GAO_Reject" to the Reject API that is sent upon discharge confirmation
    - GAO reject reason will always have the highest priority and the items will be diverted to this exception destination despite any other reject reasons the results in a barcode read. It also has priority over the STC reject rule.

   - An Encode & Sort API call should not be made to the DHL Host system for items being rejected due to the new GAO reason.

3.4.10 Operational Exceptions

- Disable / enable tray
   - It is possible for an occupied tray to be manually disabled by an operator via the BeOS application.

     Normally, trays will be disabled after a mechanical defect, so it is not possible to discharge the item and it is assumed that the item will be manually removed.

     The operator must remove the item from the disabled tray and induct it again.
- Disabled destination
   - It is possible for a destination to be manually disabled by an operator via the BeOS application. A disabled destination means that the BSC assumes the destination is electrically disconnected. The lights, valves etc. will not be activated from the control.

     In this scenario, BeSS will not attempt to discharge parcels to disabled destinations.

3.5 Discharge

Domestic parcels are sorted to one of the following destination types:

a) direct to sack chute

b) direct to hybrid sack/hamper chute

c) direct to Gaylord chute

d) to an indirect sack chute

e) to a Max chute

f) to the direct outbound conveyor chute

g) to the No Read conveyor chute

h) to the Jackpot Container chute

When an item discharge is confirmed by the sorter control system (e.g. an item is discharged, AND the tray does not subsequently trigger the occupation sensor), BeSS will prepare a message to send to the NewOps system. If the parcel discharged to a destination as instructed via the sort plan, it is communicated as a successfully discharged parcel. Otherwise (i.e. the item discharged to an exception destination), it is communicated as a rejected discharged parcel. These messages are transmitted via RESTful API communication (see section 3.7.1 for details).

3.5.1 Destination-Operator Interface

At all direct sack and Gaylord destinations, a display on the chute front may inform the operator about the destination and sack status. This will reflect the logical and physical status of the destination.

The display is a 12-character 16-segment display, one per destination.

Figure 3-16 - AT70C 12 digit PTL display unit

Besides the display, each direct sack destination has two lighted push-buttons (with 6 different colors).

Figure 3-17 - Display and pushbuttons at direct sack chutes

The display will have priority to show any chute exception information (e.g. BLOCKED, FULL, etc.). Otherwise, the display will show any destination text/name as dictated by the active sort plan file (see section 3.3.1 for details).

3.5.2 Stack Lights

Around the sorter, stack lights may inform the operator about the logical and physical status of one or more destinations within each stack light's immediate area. The stack lights are placed high on poles to increase visibility from a greater distance than the destination-operator interface. Some destinations share a stack light, so a stack light's state may not reflect the status of all chutes in its area.

Per destination type, there may be amber, red, or stacked amber/red stack lights in position.

Element	Amber Stack Light	Red Stack Light
Chute Full	STEADY	OFF
Sorter E-Stopped	OFF	STEADY **Note**: Whenever possible, the red stack light containing the device that triggered the e-stop (e.g. pushbutton at destination) will be FLASHING.
Conveyor Chute Not-Ready-To-Receive	FLASHING	OFF
Nothing	OFF	OFF

3.5.3 Simple Direct Sack Chutes

Simple direct sack chutes are designed in a similar manner to the container chutes with some modifications. The chute itself has a smaller destinations pitch and is not as wide as the container chute. A sack holder is located at the bottom of the chute which funnels the product into the sack.

Small parcels (refer to section 2.1) can directly be sorted into sacks.

Once a sack is deemed full by photo-eye, the operator gets the signal by an illuminated pushbutton. The operator either rearranges the items to free the phot-eye or removes the sack following the sack close process described in section 3.5.10.

For the simple direct sack chutes, the system does NOT automatically detect if a sack is in place or not.

The sack is closed to ensure a safer transportation on the takeaway conveyor located underneath the destinations as described in section 3.5.12. The takeaway conveyor brings the closed sacks to the sack commissioning area. A new sack is put onto the bag holder underneath the destination and the destination will be activated again by pressing a pushbutton once more.

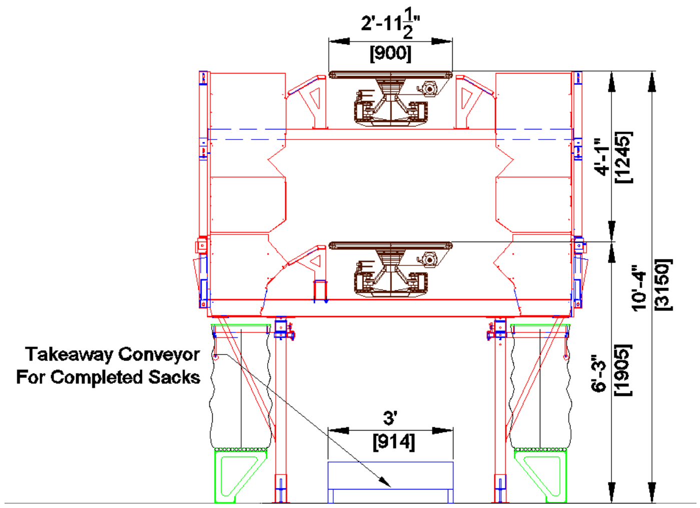


Figure 3-18 - Elevation view of the direct sack chute

Chute equipment

- 1x FULL sensor
- 1x ATOP Alphanumeric Display with 12 characters
- 2x ATOP Pushbutton units
- 1x amber stack light for every four to eight bag chutes
- 1x amber/red stack light only at those bag chutes with E-Stop pushbutton

3.5.4 Hybrid Direct Sack Chutes

Hybrid direct sack chutes are designed in a similar manner to the simple direct sack chutes with one modification: the bag stands that allow the proper mounting of sacks at sack chutes will be removable, thus allowing the use of a hamper container instead of a sack. The logic behind properly sorting parcels to these destinations is driven purely by properly configured sort plans, as BEUMER will have no way of knowing whether the direct sack chute contains a sack or hamper container.

There are future plans to convert the entire direct sack chute area to be able to handle either a hamper container or sack, physically configurable by the operators.

Once a sack/hamper is deemed full by photo-eye, the operator gets the signal by an illuminated pushbutton. The operator either rearranges the items to free the phot-eye or removes the sack/hamper following the sack close process described in section 3.5.10.

For the hybrid direct sack chutes, the system does NOT automatically detect if either a sack or hamper is in place or not.

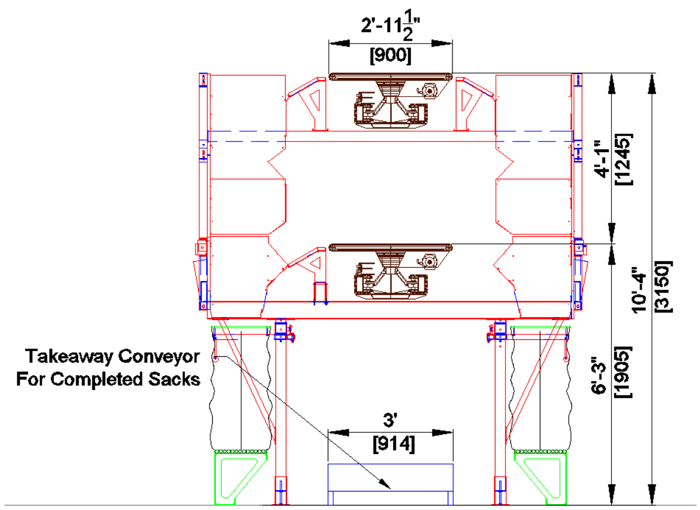


Figure 3-19 - Elevation view of the direct sack chute

Chute equipment

- 1x FULL sensor
- 1x ATOP Alphanumeric Display with 12 characters
- 2x ATOP Pushbutton units
- 1x amber stack light for every four to eight bag chutes
- 1x amber/red stack light only at those bag chutes with E-Stop pushbutton

3.5.5 Gaylord Chutes

Gaylord chutes are designed to accommodate the discharge from both sorter levels to a common destination by utilizing a cascading slide chute configuration. The cascading slide runs in the opposite direction to the sorter flow. This reversed flow helps in slowing down discharge items while traveling down the slide. The interior of the chute is open to prevent any blockage of product, with the side panels being removable to allow the breakup of any blockage if necessary.

A container is placed at the bottom of the chute so that the slide drops product directly into them. The chute is designed to accommodate the varying heights of containers that are placed under the chute. A set of photo-eyes to detect when the containers are full are installed onto a bracket that can be manually adjusted with ease based on the containers height. The container chute and associated material selection are as follows:

- Transition slide, creates a transition between sorter and destination
- Floors that are angled stacked accordingly to allow proper transfer to floor level

- Common support structure that is designed to hold up both the sorter and the chute, utilizing structural channels, tubing and steel brackets
- Wooden sidewalls that are designed to be removed for access to the sorters for maintenance

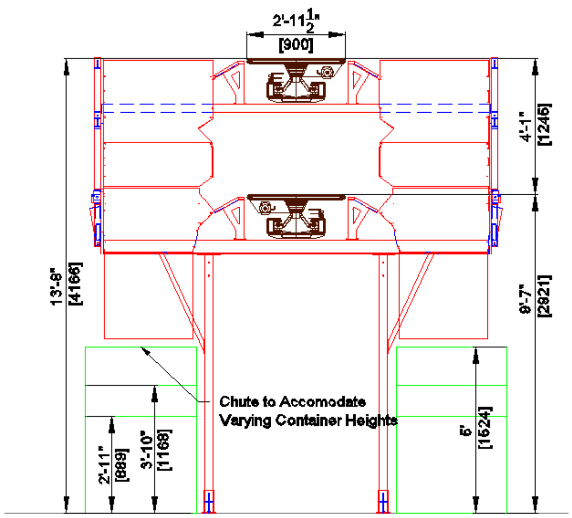


Figure 3-20 - Elevation view of the Gaylord (Container) chute

3.5.5.1 Domestic Outbound – Gaylord Chute

The majority of parcels in the outbound process are sorted to Gaylord chutes. The chute is designed to be mechanically adjusted to handle a variety of container, roller cage and blue bin heights as specified in the RFP.

For all Gaylord chutes the system automatically detects if a container is in place or not.

Additionally, a photo-eye detects if a container or blue bin is full and a light at the destination switches on. After a new container is positioned at the destination, the sort process can continue.

See section 3.5.10 for details on the container close process at Gaylord chutes.

The full containers are transported to the staging area before they leave the DC.

Chute equipment

- 1x FULL sensor
- 1x Container-in-position sensor
- 1x ATOP Alphanumeric Display with 12 characters
- 1x operator-unit (1 FULL/DISABLED light, 1 illuminating CONTINUE button, 1 illuminating BLOCK/REPLACE button)
- 1x amber stack light for every four (4) bag chutes
- 1x amber/red stack light only at those bag chutes with E-Stop pushbutton

3.5.5.2 Domestic Inbound – Gaylord Chute for NDC Volumes

Inbound parcels that are going to NDC containers are sent to the Gaylord Chutes in the inbound area. The design and the process are identical to the design and process described for the outbound Gaylord Chutes.

3.5.5.3 Jackpot Container Chute

See section 3.5.9.2 for details on the Jackpot Container exception chute.

3.5.6 Indirect Sack Chutes

A pre-sort to a specific indirect sack area is done by the sorter; the final sort is done manually by an operator. The indirect sack chute operation after the package discharge is performed by DHL, so the following description is for general information only.

The operator is comparing the number on the label to the number on the final sack destinations and puts the parcel into the specific sack. Due to the fact that the volume inside the sacks is not critical, the operators can toss the parcels into the sacks instead of placing them in there. As mentioned above, there is no additional verification provided if the parcel was sorted to the correct sack destination.

The sacks are placed on sack holders connected to a cart (provided by DHL). When the sacks on one cart are full the operator closes the sacks temporarily for the transport and puts the sacks onto the takeaway conveyor. Afterwards, the operator can attach new sacks to the cart.

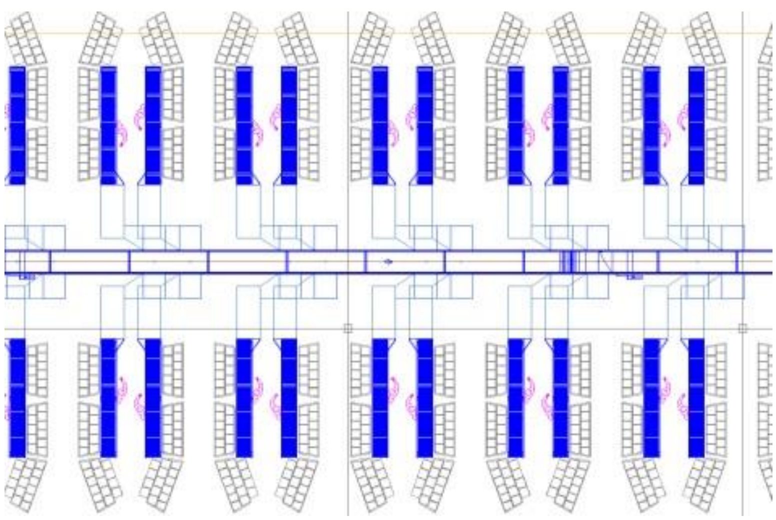


Figure 3-21 - Top view of the indirect sack chutes

Indirect sack chutes are designed in a similar manner to the Max chutes with some modifications. Product is brought to the floor level at an ergonomic height for operators to perform a secondary fine sorter using the additional slide section which runs perpendicular to the sorter. At the bottom of the slide, a long manual roller section allows operators to spread out the product. Behind the roller section are multiple carts, each with multiple sack positions. Operators then manually place product into their respective sack position. This configuration allows the system to provide the mandatory number of sack positions while providing the most minimal footprint.

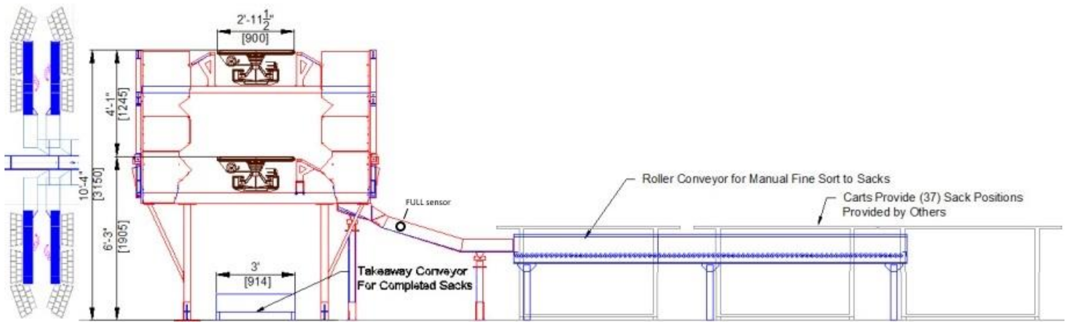


Figure 3-22 - Side view of the indirect sack chute

Chute equipment

- 1x FULL sensor
- 1x amber/red stack light per 2x indirect bag chutes FULL / E-Stop
- 1x E-Stop pushbutton per 2x indirect bag chutes

There are no displays at the indirect sack chutes.

3.5.7 Max Chutes (domestic outbound)

Parcels that are transported via air freight are packed into Hassett boxes. To ensure a condense packing of the parcels, the pack process includes an operator and is not done via a direct sort. Parcels are discharged to a chute and stop at a table at the end of the chute. The operator can work at this workstation continuously, but a full photo-eye is also provided at this destination to signal to the operator that they must unpack in order to receive new parcels at the destination. After picking up the parcel, the operator identifies it by looking at the shipping label and sorts it to the box with the same location as shown on the label. There is no automated confirmation process whether the parcel was sorted into the correct Hassett box or not. The operator must also decide if the Hassett box is full. If the box is full, the operator closes the box and puts it on a pallet. Afterwards the operator erects a new box, puts a label on it, and places it in the position of the old box.

Max chutes are designed in a similar manner to the container chutes with some modifications. At the end of the slide chute, an additional slide chute section perpendicular to the sorter track is installed. This slide chute section brings product to a worktable set at an ergonomic height. An operator stationed at the worktable manually fine sorts the product to one of multiple Hassett box positions staged around the table. To ensure the Hassett boxes are at an ergonomic height for the packing process, additional tables are provided to place the boxes on.

Printers or any other special equipment is not included by BEUMER in this proposal.

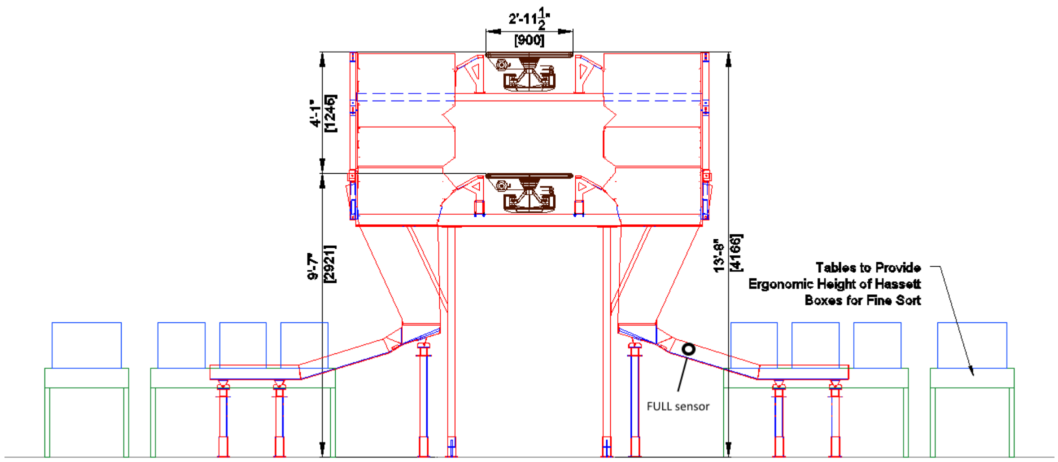


Figure 3-23 - Elevation view of the Max chute

Chute equipment

- 1x FULL sensors
- 1x amber/red stack light FULL / E-stop
- 1x E-Stop pushbutton

There are no displays at the Max chutes.

3.5.8 Direct Outbound – Conveyor to Fluid Load at Dock Doors

Outbound parcels that need to be loaded with a fluid load line are discharged to a chute that is connected to a conveyor. The conveyor will be connected with the infeed conveyor of the inbound area. This infeed conveyor and the telescope (see pink colored conveyors in figure below) can run in a reverse mode to also load the parcels instead of unloading them.

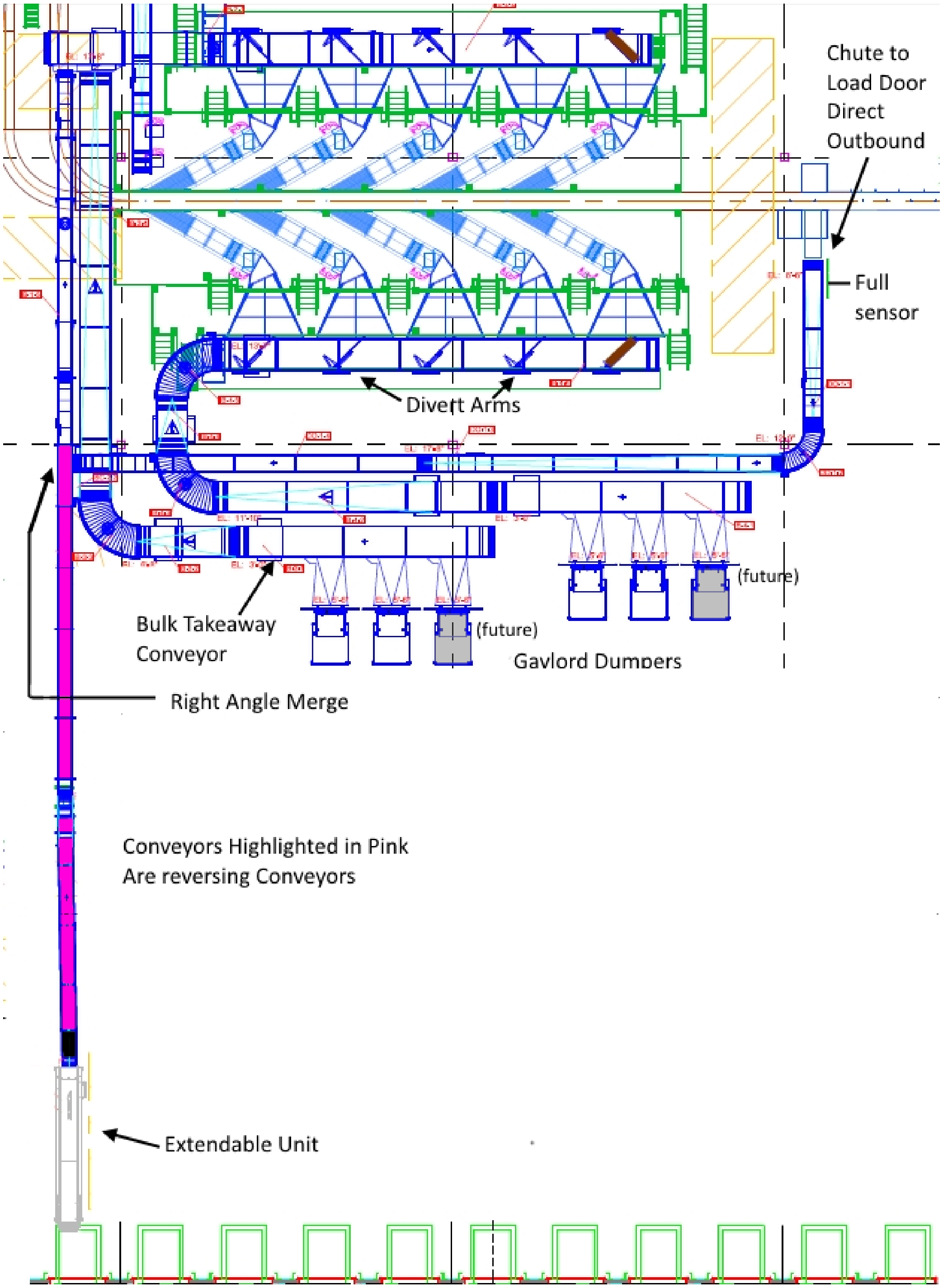


Figure 3-24 Direct Outbound and Inbound Receiving

The conveyor line, connecting the sorter and the inbound conveyor, can buffer some parcels via inch and store in case the inbound conveyor runs in inbound mode. The mode can be manually switched via selector switch at the dock door.

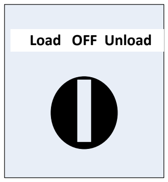


Figure 3-25 Selector Switch at Direct to Outbound Area

When the switch is in the **Load** position the outbound conveyor will run items towards the dock door without indexing and the inbound conveyors will run in reverse.

If the line is not used it can be stopped by placing the switch in the **Off** position. When the switch is in the **Unload** position then the outbound conveyor will index items until this line is full, while the inbound conveyors are running the items towards the upper sorter semi-inductions chutes.

Chute equipment

- 1x FULL sensor
- 1x amber stack light at direct Outbound chute FULL

There is no display at the Direct Outbound Chute.

3.5.9 Exception Chutes

There are two (2) dedicated exception chutes on the sorter: a "Jackpot container chute" which leads to a container on the floor, and a "No-Read chute" which transfers items to induction 225 on the upper induction platform for re-induction.

Please note that other destinations can be used as exception chutes – this is freely configurable – but care should be taken that these destinations are not included in any active sort scheme, so the preference should be to use the dedicated exception chutes.

See section 3.4.9 for details on the correlation between sorting exceptions and exception destinations configured via the BeSSView.

There is also a non-conveyable chute at each semi-automatic induction. The sorter does not discharge into these chutes. Instead of inducting items onto the semi-automatic inductions, induction operators may manually discharge non-conveyables into these chutes which lead to containers on the floor below the induction platform.

There are no displays at the exception chutes.

3.5.9.1 No-Read Conveyor Chute

Parcels that could not be read on the sorter should be discharged to the No-Read conveyor chute. These items are transported on a belt conveyor to near semi-automatic induction 225 of the upper sorter. Here, the operator may attempt to manually scan the barcode using the overhead mounted Cognex scanner or the handheld scanner (both provided by DHL) at the no-read terminal before placing the package on the semi-automatic induction laydown belt. The conveyor is equipped with a selector “Auto/Purge Mode” switch. When the operator places the switch in “Purge Mode” the No-Read conveyor will run continuously bringing discharged items to the No-Read station. When the operator places the switch in "Auto Mode" the conveyor will run in inch-and-store mode, buffering discharged items and filling up the conveyor starting from the start of the conveyor belt. A photo eye placed at the end of the conveyor belt, near the operator, is used to determine when the conveyor (not the entire chute) is full, in which case the No-Read conveyor chute will stop moving, regardless of the mode, but still allow items to discharge to the chute. Once the full photo-eye by the chute portion of the No-Read conveyor chute is blocked, no more discharging to the No-Read conveyor chute will be allowed. Discharging to this conveyor chute is only allowed if the conveyor system is started, there are no technical faults experienced by No-Read conveyor, and the No-Read conveyor chute portion is not full.

At the chute level by the entrance of the No-Read conveyor chute and at the operator by the tail of the No-Read conveyor, there is an amber stack light that indicates the status of the No-Read conveyor chute. If the amber stack light is flashing, then the conveyor is not receiving items (e.g. conveyor system not started, or conveyor faulted). If the amber stack light is steady on, then the No-Read conveyor chute is full (full photo eye at the chute level is blocked).

The run status of the No-Read conveyor line is indicated by the green stack light near the conveyor. If this light is off, then the conveyor system is not started. This light is flashing green when the conveyor system is about to start, and it stays on when the conveyor system is running or enabled to run. The fault status of the No-Read conveyor line is indicated by the red stack near the conveyor (next to the green stack light). This light is flashing red when the Emergency Stop push-button at the No-Read conveyor is activated and is steady on when the system is in emergency stop (not due to the No-Read conveyor).

Note: DHL may choose to place a re-labelling station on the induction platform on the vicinity of the induction equipped with a hand-scanner. This station is not connected to the BEUMER sorter control system.

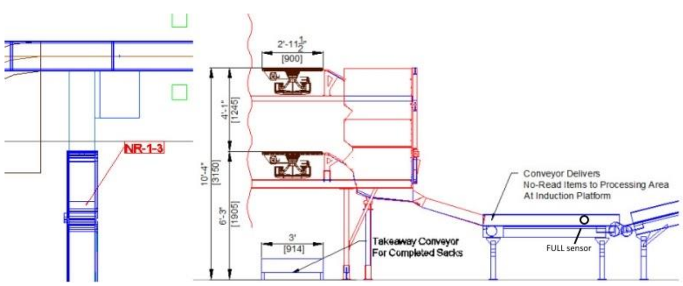


Figure 3-26 - Side view of the no-read chute

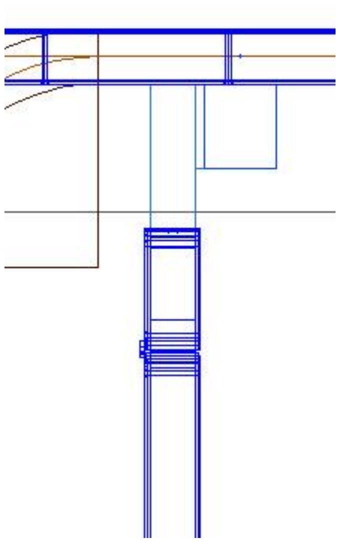


Figure 3-27 - Top view of the No-Read chute

Chute equipment

- 1x FULL sensor
- 1x amber stack light at No-Read chute FULL

Conveyor equipment

- 2x photo eyes (1x head of conveyor, 1x tail of conveyor)
- 1x “Auto/Purge Mode” selector switch
- 1x red and green stack light at No-Read conveyor

3.5.9.2 Jackpot Container Chute

Parcels that are deemed logically un-sortable (i.e. missing data, need special attention beyond re-labeling or re-induction, etc.) may be discharged at the Jackpot container chute. This chute feeds items to a container on the floor level, after the semi-automatic induction platform. To replace the container, the operator must place the 'Block Chute' selector switch in the ON position. No items will be discharged at this chute if the container is full, the container is not in position, or the chute is blocked from the selector switch. The amber stack light will illuminate if the discharge is not possible. The red stack light will illuminate if the emergency stop push button is pressed.

Chute equipment

- 1x FULL sensor
- 1x Container-in-position sensor
- 1x 'Block Chute' selector switch
- 1x E-Stop push button
- 1x amber/red stack light

3.5.9.3 Non-Conveyable Chutes

One common non-conveyable chute will be installed at each semi-automatic induction. The operators can collect some non-conveyables next to the induction unit and depending on the throughput either they or an additional operator (runner) can place them in the non-conveyable chute.

The sorter does not discharge into these chutes.

3.5.10 Sack Close Process

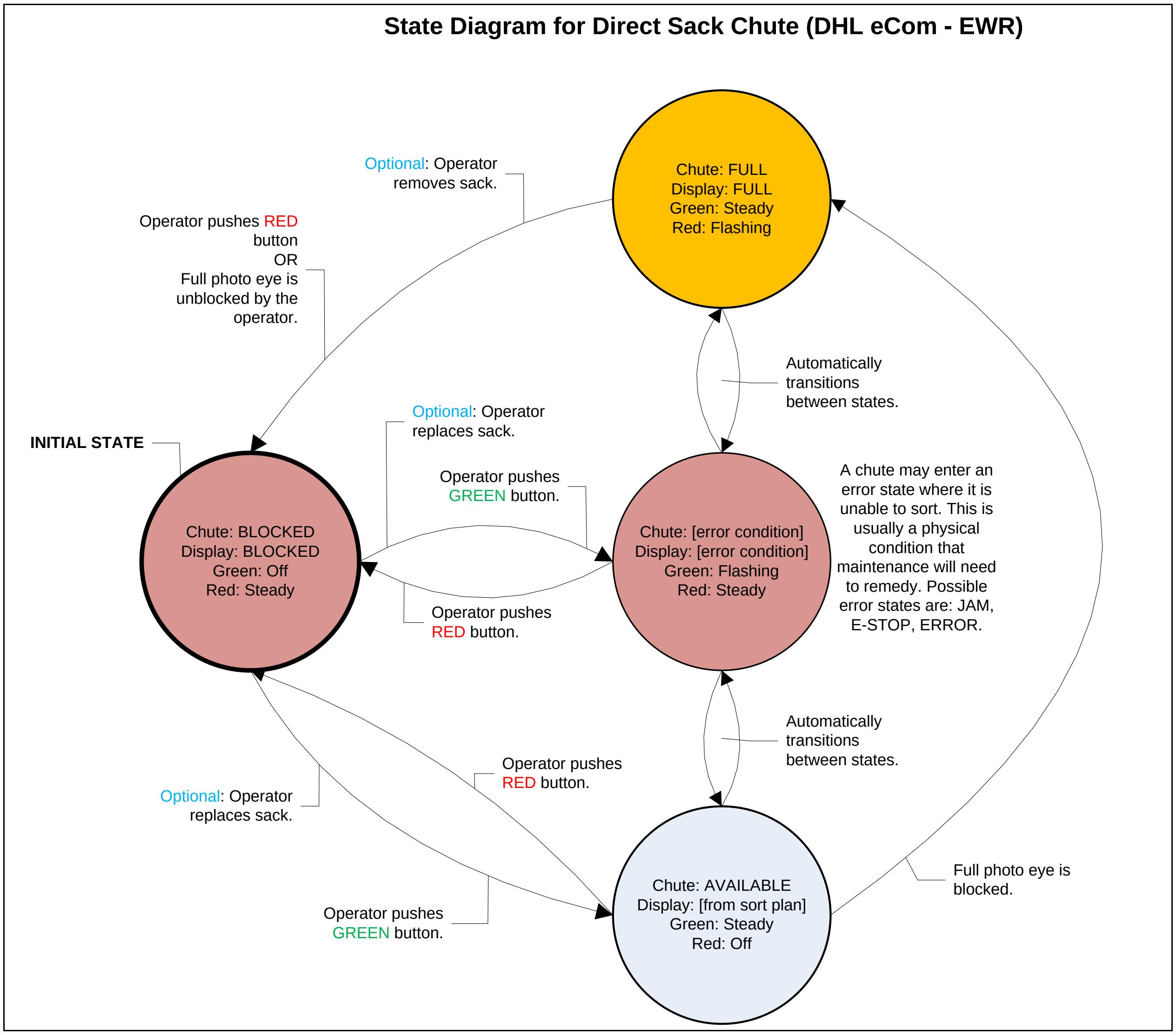


Figure 3-28 - State Diagram for the Direct Sack Chutes

This section covers the handling of sacks at the direct sack chutes and does not apply to any other chute type.

The equipment at each direct sack chute consists of a 12-digit alphanumeric display with two lighted push-buttons (red-lit button on the left, green-lit button on the right).

Figure 3-29 - Display and pushbuttons at direct sack chutes

Please refer to the above state diagram for a detailed description of a direct sack chutes state transitions, including the corresponding ATOP display information.

Direct sack chutes shall undergo the following general workflow:

a) Initially, the sack chute is BLOCKED; meaning items are blocked from discharging to the chute. In this state, the red light is on, and the green light is off.
    - Note: The sack chute will enter this initial state if the BeSS software or server is restarted.
b) The operator should first place a sack under the direct sack chute and then press the green button to set the chute to AVAILABLE. In this state, the red light is off and green light is on and items are now able to discharge to the chute
    - Note: the system does NOT automatically detect if a sack is in place or not before entering the AVAILABLE state.
c) When the sack is FULL (e.g. FULL photo-eye is blocked), the red light is flashing, and the green light is on, and items are blocked from discharging to the chute.
d) If the operator determines the sack is not actually full, the operator may shuffle the items within the sack to unblock the FULL photo-eye. Once the FULL photo-eye is unblocked, the chute will enter the BLOCKED state. After handling the items at the chute and unblocked the FULL photo-eye, the operator should press the GREEN button to set the chute to AVAILABLE.
e) If the bag is full, or to commence the packout operation while the chute is in the AVAILABLE state, the operator may push the red pushbutton to set the chute to BLOCKED.
    - Note: the packout operation is not handled by BEUMER.
f) When the operator is finished with the packout operation, the operator may keep the chute BLOCKED, replace the sack, and then push the green button to set the chute to AVAILABLE.
g) At any point, the chute may enter an error state. In this case, the green light is flashing, and the red light is on. Once the chute no longer

experiences the error condition, the operator may push the red button to return it to its previous state.

Closed sacks may be placed on the sack takeaway conveyor, as described in section 3.5.12.

In the sack commissioning area, a couple of parcels will be taken out and scanned to identify the destination and trigger the printout of a USPS bag tag. This procedure is not handled by BEUMER and does not require any equipment at the direct sack chute.

<table>
<tr><th>Element</th><th>Chute State</th><th>Light Status</th></tr>
<tr><td rowspan="4">Right ATOP lighted pushbutton

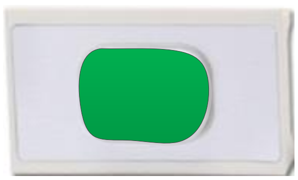



Figure 3-30 - Green pushbutton at direct sack chutes</td><td>BLOCKED</td><td>Off</td></tr>
<tr><td>AVAILABLE</td><td>Steady</td></tr>
<tr><td>FULL</td><td>Steady</td></tr>
<tr><td>ERROR</td><td>Flashing</td></tr>
<tr><td rowspan="4">Left ATOP lighted pushbutton

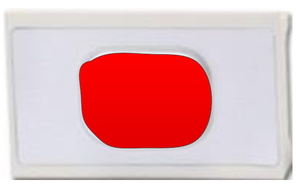



Figure 3-31 - Red pushbutton at direct sack chutes</td><td>BLOCKED</td><td>Steady</td></tr>
<tr><td>AVAILABLE</td><td>Off</td></tr>
<tr><td>FULL</td><td>Flashing</td></tr>
<tr><td>ERROR</td><td>Steady</td></tr>
</table>

3.5.11 Container Close Process

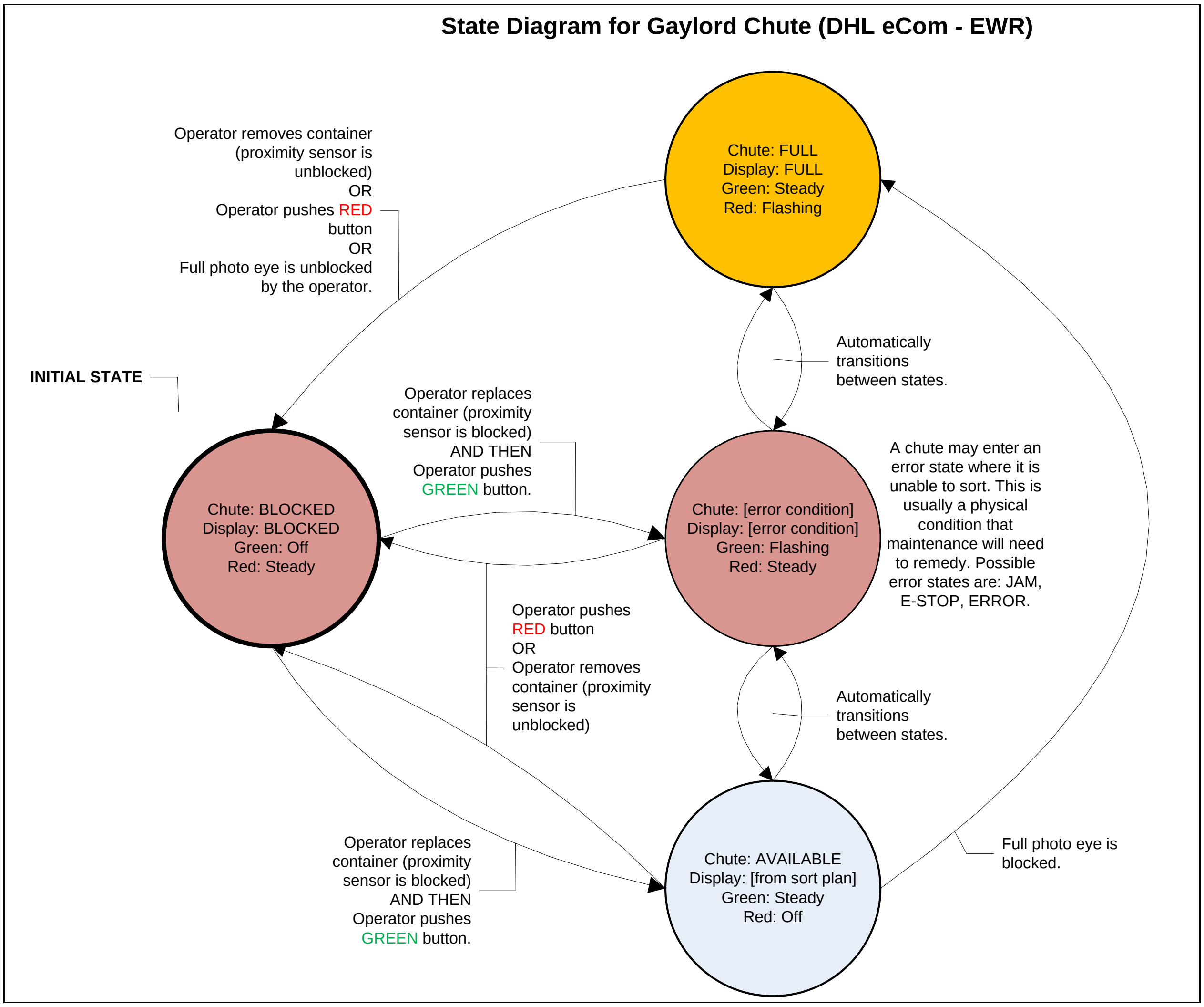


Figure 3-32 - State Diagram for the Outbound/Inbound Gaylord chutes

This section covers the handling of containers at outbound and inbound Gaylord chutes and does not apply to any other chute type.

The equipment at outbound and inbound Gaylord chute consists of a 12-digit alphanumeric display (no push-buttons).

Figure 3-33 - Display at outbound/inbound Gaylord chutes

Since the alphanumeric display is too high and out of reach for the operator, instead of providing pushbuttons as in the direct sack chutes, each outbound and inbound Gaylord chute has a control box at the operator level. Each control box has a FULL/DISABLED yellow light, CONTINUE illuminated green push-button, and BLOCK/REPLACE illuminated red push-button.

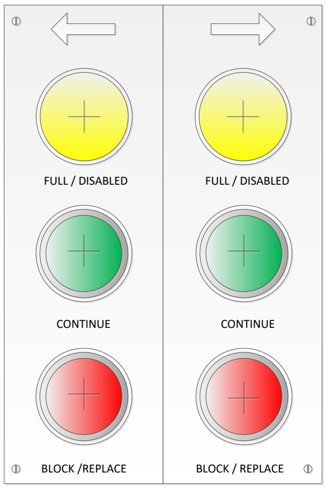


Figure 3-34 - Conceptual illustration of the pushbutton unit at outbound/inbound Gaylord chutes

Please refer to the above state diagram for a detailed description of a direct sack chutes state transitions, including the corresponding ATOP and control box display information.

Gaylord chutes shall undergo the following general workflow:

a) Initially, the Gaylord chute is BLOCKED; meaning items are blocked from discharging to the chute. In this state, the red light is on, and the green light is off.
    - Note: The sack chute will enter this initial state if the BeSS software or server is restarted.
b) The operator must first place a container under the Gaylord chute and then press the green button to set the chute to AVAILABLE. In this state, the red light is off and green light is on and items are now able to discharge to the chute
    - Note: the system DOES automatically detect if a container is in place or not before entering the AVAILABLE state.
c) When the container is FULL (e.g. FULL photo-eye is blocked), the red light is flashing, and the green light is on, and items are blocked from discharging to the chute.
d) If the operator determines the container is not actually full, the operator may shuffle the items within the container to unblock the FULL photo-eye.

Once the FULL photo-eye is unblocked, the chute will enter the BLOCKED state. After handling the items at the chute and unblocked the FULL photo-eye, the operator should press the GREEN button to set the chute to AVAILABLE.

e) If the bag is full, or to commence the packout operation while the chute is in the AVAILABLE state, the operator may push the red pushbutton to set the chute to BLOCKED. Alternatively, if the container is removed (e.g. the container-present photo-eye is unblocked), the chute is automatically set to BLOCKED.

   - Note: the packout operation is not handled by BEUMER.

f) When the operator is finished with the packout operation, the operator may keep the chute BLOCKED, replace the container, and then push the green button to set the chute to AVAILABLE.

   - Note: the system DOES automatically detect if a container is in place or not before entering the AVAILABLE state after packout.

g) At any point, the chute may enter an error state. In this case, the green light is flashing, and the red light is on. Once the chute no longer experiences the error condition, the operator may push the red button to return it to its previous state.

   - Note: the system DOES automatically detect if a container is in place or not before re-entering the AVAILABLE state.

Element	Chute State	Light Status
Control box red lighted pushbutton	BLOCKED	Off
	AVAILABLE	Steady
	FULL	Steady
	ERROR	Flashing
Control box green lighted pushbutton	BLOCKED	Steady
	AVAILABLE	Off
	FULL	Flashing
	ERROR	Steady

3.5.12 Sack Takeaway Conveyor

The takeaway conveyor is located underneath the domestic inbound destination area of the sorter. It transports the temporarily closed sacks to the sack commissioning area, both from the direct and indirect sack chutes.

The takeaway conveyor underneath the sorter is set at an elevation of 1'-6" with 3'-0" belt width to allow ease of transfer of completed bags from both sides of the sorter. Once the conveyor is clear from the sorter system, it inclines to provide clear height underneath. After that, a set amount of space along the takeaway conveyor accommodates the sack commissioning process. All equipment required for the sack commissioning process is not included in the BEUMER scope of supply.

The takeaway conveyor is made up of "belt-on-roller" conveyors. The conveyors under the BEUMER sorter will run as long as there is a demand for sacks on the conveyors downstream. Photo eyes are placed on those conveyors to determine if there is a jam or full condition. The sack takeaway is equipped with green and amber stack lights. When the system is stopped, the green light will be off. The light will flash when the system is about to start, and it will stay on when the system is running or enabled to run. In case of a jam the affected conveyor will stop, and the amber light will be on. The operator must clear the jam (photo-eye must be clear) and reset the fault at the assigned reset push button in order for the system to resume normal operation.

3.6 Sack Commissioning

Sacks are transported to the operator workstations at the sack commissioning area by the sack takeaway conveyors. Afterwards, the sacks will be accumulated using inch-and-store logic prior to the workstations or moved directly to the workstations.

The sack-commissioning platform is equipped with one selector switch from where the operation can be switched between **Auto**, **Off** and **Purge** mode for this line.

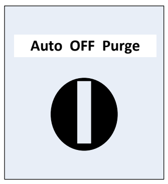


Figure 3-35 Selector Switch at Sack Commissioning Area

When the switch is in the **Auto** mode, the sacks will be inch and stored on the 51-07 through 51-15 conveyors as long as there are no operator requests.
When the switch is in the **Purge** mode, the sacks accumulates up to the last sack commissioning station even if there are no operator requests, filling any empty available conveyors on the sack take-away and sack commissioning line (conveyor line 51).
In the **Auto** and **Purge** mode, line 52 will continually run sending the commissioned sacks to the sack manual sortation platform.
The **Off** position will stop the entire sack take-away, sack commissioning, and manual sack sortation lines (conveyor line 51 and 52).

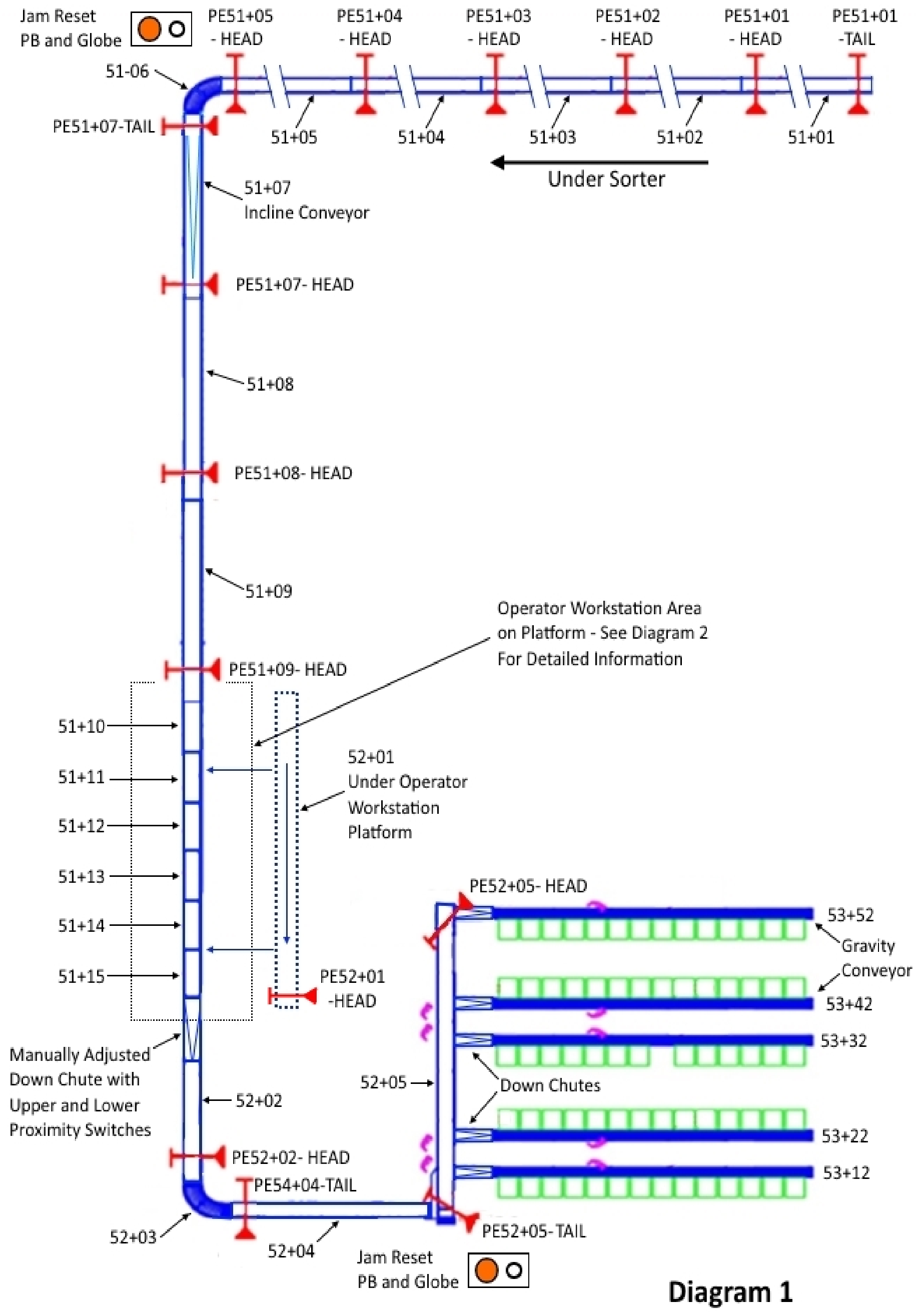


Figure 3-36 Sack Take Away and Sack Commissioning Conveyors

At the workstations, there are six 10.5-feet long indexing conveyors. Each conveyor can be manned with 2 operators, one on the right and one on the left. Each workstation has a photo eye to detect sack presence and a green illuminated push button to request sacks.

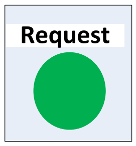


Figure 3-37 Illuminated Request Push Button at Sack Commissioning Area

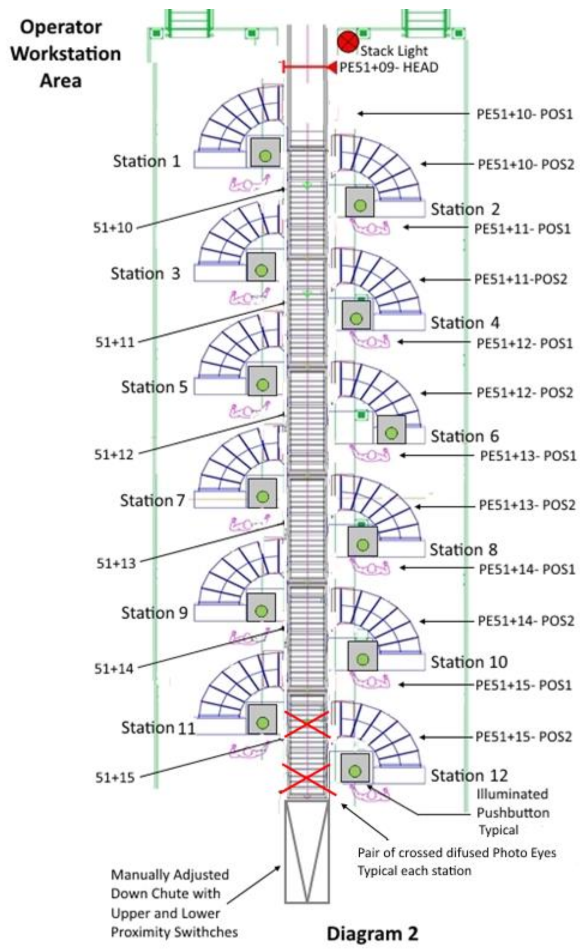


Figure 3-38 Sack Commissioning Operator Stations

The sacks are buffered on the conveyors filling from the back end of the conveyors towards the head up to the operator stations when in **Auto** mode and there are no operator requests. When in buffering mode (also referred as indexing mode) the green pushbuttons at the stations are steady illuminated. Sacks will be filling the stations two at the time until all stations are full. Once operators are

present, they can request sacks by pressing the request push button. The line will exit the indexing mode filling all the available spaces based on head photo eye detection and downstream conveyor running status. During this “accumulation” mode, the green push button will be flashing. If no requests are made in a predefined time (60 seconds) the line enters again the indexing mode filling the conveyors from the tail end towards the head.
In **Purge** mode the line will be constantly operate in “accumulation” mode described above, without timing out if there are no operator requests. The green illuminating lamps at the request push buttons are flashing.

3.6.1 Operator Workstations – Sack Complete

Sacks can be weighed on a scale at the workstation. Apart from the conveyors, this process is supported by equipment provided by DHL.
When a sack's "Weigh & Label" process is complete, the bag will be pushed off to the workstation chute to the takeaway conveyor line underneath that brings the sacks to the final manual sort area. The takeaway conveyor is closed in on the sides such that sacks will stay contained on it even if they fall on top of another sack. An indicating stack light will let the operators know if the takeaway conveyor underneath is running. Sacks should only be pushed onto the chute when the takeaway conveyor below is running. The takeaway conveyor is extra wide to accommodate sacks that come down from the left and right operator at the same time. The status of the system is indicated by the green stack light near the operator workstation. If the light is off, then the system is not started. The light is flashing green when the system is about to start, and it stays on when the system is running. The amber stack light indicates that one or more conveyors are faulted.

It is recommended that when staffing the workstation area, the downstream operator positions are staged first and then the next upstream positions filled in order. This will allow the upstream operator stations to act as transport conveyor when not manned to better supply the operators.

Once the sack commissioning is complete, the completed sacks continue on a belted takeaway conveyor to be manually sorted via a slide chute to one of several secondary lines. The secondary lines consist of non-driven roller conveyor to allow operators to line up the sacks to their respective container location. The operator then pushes the sacks down a slide chute that directly feeds into the container. At sack commissioning platform a selector switch is available for the operators to control the operating mode of the secondary line.

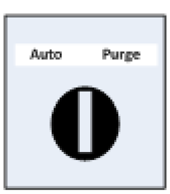


Figure 3-39 Selector Switch at Sack Commissioning Area for Secondary Line

In position **Auto**, the sacks will be stored one at the time until the line leading to the manual sort platform is full. Once full the secondary line will stop. The storing mode is cancelled when the area is in bypass mode and the line will run continuously. In **Purge** position, the sacks will be transported to the platform without buffering. The **Auto** and **Purge** modes are canceled when the line is turned off from the selector switch at sack commissioning area.

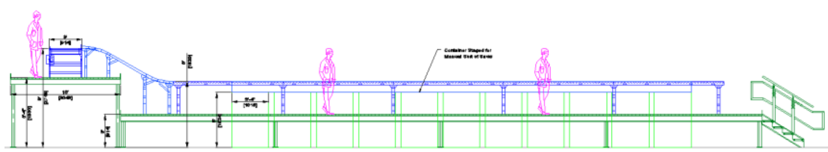


Figure 3-40 - Elevation view of sack sorting - Sack to Gaylord

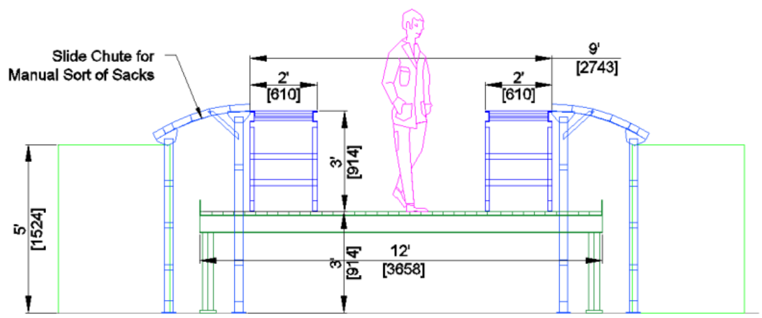


Figure 3-41 - Elevation view of Manual sack sortation

The process of sorting sacks and replacing full containers, when necessary is manual and is not supported by the sorter or conveyor control system. On the manual sack sortation platform, a green and amber stack light are mounted along with a jam reset push button. The status of the system is indicated by the green stack light. If the light is off, then the system is not started. The light is flashing green when the system is about to start, and it stays on when the system is running. In case of a jam the affected conveyor will stop, and the amber light will be on. The operator must clear the jam and reset the fault at the assigned reset push button, system resuming normal operation.

3.6.2 Sack Bypass Mode

The sack commissioning area has a pass-through mode, where sacks by-pass the commissioning process and continue straight to the manual sort area.
The tilt device (tip chute) directs sacks from the upper infeed conveyor to the outbound conveyor.
By-pass mode will be activated by manually lowering the tip-chute. The positioning of the tip-chute is a manual operation. The current status of area (Normal or Bypass) mode will be displayed on the BeOS. The tip-chute will be counterbalanced for ease of movement. Two proximity switches at the chute will be included for position confirmation.

When the tip chute is in the lowered position, the indexing operation on the inbound conveyor will be directly tied to the outbound conveyor to the manual sort area. When the outbound conveyor is stopped, the indexing conveyor will stop.

The belts at the operator stations will be used as transportation conveyor with the operator stations in-operable

3.6.3 Cross-border Sacks and Containers

Cross-border sacks may physically be placed on the take-away conveyor, but they are not processed by the sorter control system. It is the assumption that handling of cross-border mail after the sorter is outside BEUMER's scope.

3.7 HOST Communication

The sorter control system communicates with three separate DHL host systems: NewOps, CONNECT, and Master Data.

The information in this section is an overview of the data and data structures being communicated between the host systems and the sorter control system. Refer to each host system's specific communication protocol document for technical details.

3.7.1 NewOps

Communication between BeSS and the NewOps system is based on HTTP communication using RESTful APIs. Both BeSS and NewOps will host a RESTful API as servers to accept client connections from each other. If there is a loss of connection between BeSS and NewOps, a BeOS alarm will be raised to alert operators to the broken connection.

BeSS will communicate the following information to HTTP endpoints via the NewOps RESTful API:

1. Authentication Request (BeSS → NewOps)
   - This message is sent by the BeSS system to the host Authentication server to get a valid token.
   - Fields to be transmitted via Authentication Request
     - Client ID
     - Client Secret
     - Grant Type
   - A message should be sent by NewOps as an HTTP response to the Authentication Request from the BeSS system. This message provides a valid token requested by the BeSS system.
   - Fields to be transmitted via Authentication Response
     - Access Token
     - Expires In
     - Token Type
   - The access token is then used as a parameter in the "Destination Request" message.
2. Mail Item Sorted (BeSS → NewOps)
   - This message is sent by the BeSS system for each mail item that is successfully sorted. An item is deemed successfully sorted by the BeSS if the item is confirmed to have sorted to any of its intended destinations listed by the sort plan.
   - This message needs an authentication token in the request parameters. If the authorization information is missing or invalid,

the host will not accept the message and respond with HTTP status code 401 (Unauthorized). In that case we will resend the Authentication request to receive a new valid token and then re-send the Destination Request message.

- Fields to be transmitted via Mail Item Sorted message:
    - Scan time
    - Chute name
    - 2D Barcode
    - 1D Barcode
    - Sort plan
    - Induction name

3. Mail Item Rejected (BeSS → NewOps)
    - This message is sent by the BeSS system for each mail item that is not successfully sorted. An item is deemed not successfully sorted by the BeSS if the item is confirmed to have sorted to a destination other than any of its intended destinations listed by the sort plan.
    - This message needs an authentication token in the request parameters. If the authorization information is missing or invalid, the host will not accept the message and respond with HTTP status code 401 (Unauthorized). In that case we will resend the Authentication request to receive a new valid token and then re-send the Destination Request message.
    - Fields to be transmitted via Mail Item Rejected message:
        - Scan time
        - Reject code
        - Reject reason
        - Barcode (2D/1D)
4. Container Closed (BeSS → NewOps)
    - This message is sent by the BeSS system for each chute that is closed so that the associated sack can be packed out by the operator.
    - Fields to be transmitted via Container Closed message:
        - Machine ID
        - Chute ID
        - Associated sack barcode (if any)
5. Container Status Request (BeSS → NewOps)

- This message is sent by the BeSS system cyclically every 10 seconds for a list of chutes with the mobile packout flag enabled to query NewOps for the sack barcode to be associated to its respective chute within the BeSS system. This message is also sent for an individual chute if its green ATOP button is pressed to open the chute so that NewOps can send the associated sack barcode in response.
- Fields to be transmitted via Container Status Request message:
    - Machine ID
    - List of chute IDs
- A Container Status message should be sent by NewOps as an HTTP response to the Container Status Request message. This message provides associated sack and threshold information for the chute requested by the BeSS system.
- Fields to be transmitted via Container Status message:
    - List of container information, where each container information object contains the following:
        1. Chute ID
        2. Associated sack barcode (empty if none)
        3. Threshold information (empty if none)

6. Destination Request (BeSS → NewOps)

- This message is sent by the BeSS system for each mail item scanned on the sorter that has been inducted from an induction with the "Send DRM" induction flag enabled in BeSSView.
- The Destination Request needs an authentication token in the request parameters. If the authorization information is missing or invalid, the host will not accept the message and respond with HTTP status code 401 (Unauthorized). In that case we will resend the Authentication request to receive a new valid token and then re-send the Destination Request message.
- Fields to be transmitted via Destination Request message:
    - Item ID
    - Barcode (2D/1D)
    - Width
    - Length
    - Weight
    - Height
    - DimensionUom

  - weightUnit
  - Cuboidal

- A Destination Assignment message should be sent by NewOps as an HTTP response to the Destination Request from the BeSS system. This message provides any associated destination assignment information for the item requested by the BeSS system.
- If the induction on which the item was inducted has the "wait for destination assignment from NewOps" flag disabled, the BeSS system will use the downloaded sort plan as a default. If NewOps sends a Destination Assignment message, this will overwrite the default sort code extracted from the barcode. BeSS will then look up the newly-provided sort code in the downloaded sort plan to assign destinations to the item.
- Otherwise, if the flag is enabled on the induction, the BeSS system will exclusively use the Destination Assignment message to assign the item to a destination. The item will recirculate until it receives a Destination Assignment message from NewOps or until it exceeds maximum recirculations.
- If for some reason, the hosts is unable to process the message due to a business rule, it will respond with a 422 error response. In that case, BeSS will try to interpret the message body.

  NOTE: If the **errorCode** field in the error response message body is not 'null', then BeSS will treat the errorCode as a sortCode (rejectCode) and will attempt to assign the item to a discharge bin associated with that sortCode in the ASF sort plan.
- Fields to be transmitted via Destination Assignment message:

200 OK	4XX Error Response
AutomationSlot (sortCode)	DetailedUserMessage
Item ID	errorCode
Barcode (2D/1D)	Errors
Width	userMessage
Length	
Weight	
Height	
DimensionUom	
weightUnit	

Cuboidal

Note: BeSS will attempt to preserve sent messages not received by HOST for a certain number of days (configurable via the BeSSView, default: 7 days) before purging these messages from the system. While these are preserved, they will be re-sent per defined retransmission rules (described in the NewOps communication protocol document).

3.7.2 CONNECT

Communication between BeSS and the CONNECT system is based on HTTP communication using RESTful APIs. Specifically, BeSS will act as server and host a RESTful API for CONNECT to consume as the client.

BeSS will communicate the following information to HTTP endpoints via the NewOps RESTful API:

1. Consignment (BeSS → CONNECT)
   - This message is sent by the BeSS system for each mail item scanned on the sorter in accompaniment to the Mail Item message. The consignment number will always be hardcoded to 999999 and the customer name and customer number will always be blank.
   - Fields to be transmitted via this message:
     - Consignment number
     - Customer name
     - Customer number
2. Mail Item (BeSS → CONNECT)
   - This message is sent by the BeSS system for each mail item scanned on the sorter upon receiving the first good-read scan for the item. If the item does not have an assigned destination when it reaches a request point, this message will be sent again by the BeSS system.
   - Fields to be transmitted via this message:
     - Machine ID
     - List of item barcodes
     - Consignment number
     - Width
     - Length
     - Weight
     - Height
     - Cuboidal

CONNECT will communicate the following information to HTTP endpoints via the BeSS RESTful API:

1. Mail Item Data (CONNECT → BeSS)

    - This message is sent by CONNECT for each encoded mail item that is to be sent to the sorter. The message contents shall contain the mail item's barcode and sort code pairing to be used by the sorter when scanning and sorting the mail item.
    - Fields to be transmitted via this message:
        - Barcode
        - Sort code

3.7.3 Master Data

Communication between BeSS and the Master Data server(s) is based on file transfer over sFTP. Specifically, BeSS will act as server and host a sFTP site for the Master Data server(s) to transfer files for processing.

The Master Data server(s) will transfer the following file types to the BeSS server:

1. Domestic Ticket-to-Ride Sort Plan File

    - Contains sort code and destination information for sortation of domestic mail items containing Ticket-to-Ride barcodes. Optionally these files contain display information or Cuboidal destination information. Display information is text to be displayed at chutes with chute displays. Cuboidal Destinations are a list of at least one destination or at most the number of destinations on the sorter. These destinations become priority for cuboidal items.
    - Data to be transmitted via this file type:
        - SORT message, containing the following fields:
            - Sort Code
            - Primary Destination
            - Secondary Destination
            - Tertiary Destination
        - DISPLAY message, containing the following fields:
            - Chute
            - Display Text
        - CUBOIDAL message, containing the following fields:
            - Primary Cuboidal Destination
            - Secondary Cuboidal Destination

- …Nth – 1 Cuboidal Destination
- Nth Cuboidal Destination

2. Domestic eVS Sort Plan File

   ➢ Contains sort code and destination information for sortation of domestic mail items containing USPS eVS barcodes. Optionally these files contain display information or Cuboidal destination information. Display information is text to be displayed at chutes with chute displays. Cuboidal Destinations are a list of at least one destination or at most the number of destinations on the sorter. These destinations become priority for cuboidal items.

   ➢ Data to be transmitted via this file type:

     - SORT message, containing the following fields:
       - Sort Code
       - Primary Destination
       - Secondary Destination
       - Tertiary Destination
     - DISPLAY message, containing the following fields:
       - Chute
       - Display Text
     - CUBOIDAL message, containing the following fields:
       - Primary Cuboidal Destination
       - Secondary Cuboidal Destination
       - …Nth – 1 Cuboidal Destination
       - Nth Cuboidal Destination

3. CONNECT Sort Plan File

   ➢ Contains sort code and destination information for sortation of cross-border mail items. Optionally these files contain display information or Cuboidal destination information. Display information is text to be displayed at chutes with chute displays. Cuboidal Destinations are a list of at least one destination or at most the number of destinations on the sorter. These destinations become priority for cuboidal items.

   ➢ Data to be transmitted via this file type:

     - SORT message, containing the following fields:
       - Sort Code
       - Primary Chute

- Secondary Chute
- Tertiary Chute

○ DISPLAY message, containing the following fields:

- Chute
- Display Text

○ CUBOIDAL message, containing the following fields:

- Primary Cuboidal Destination
- Secondary Cuboidal Destination
- …Nth – 1 Cuboidal Destination
- Nth Cuboidal Destination

4. ASF Sort Plan File

➢ Contains sort code and destination information for sortation of items that BeSS sent a DRM message to host. Optionally these files contain display information or Cuboidal destination information. Display information is text to be displayed at chutes with chute displays. Cuboidal Destinations are a list of at least one destination or at most the number of destinations on the sorter. These destinations become priority for cuboidal items.

➢ Data to be transmitted via this file type:

○ SORT message, containing the following fields:

- Sort code
- Primary destination
- Secondary destination
- Tertiary destination

○ CHUTE INFO message, containing the following fields:

- Chute
- Display text

○ CUBOIDAL message, containing the following fields:

- Primary Cuboidal Destination
- Secondary Cuboidal Destination
- …Nth– 1 Cuboidal Destination
- Nth Cuboidal Destination

5. Chute Attributes Configuration File
    - ➢ Contains attributes per chute to match with Ticket-to-Ride mail item attributes.
    - ➢ Data to be transmitted via this file type:
        - o CHUTE message, containing the following fields:
            - Chute
            - Attribute 1
            - Attribute 2
            - …
            - Attribute 256

When a new file is transferred and process by BeSS, it is stored internally. This will not affect any currently active sort plan or chute attribute configuration. These downloaded sort plans and chute attribute configurations may be set to active via the BeSSView by an operator.

3.8 Production Statistics and Reporting

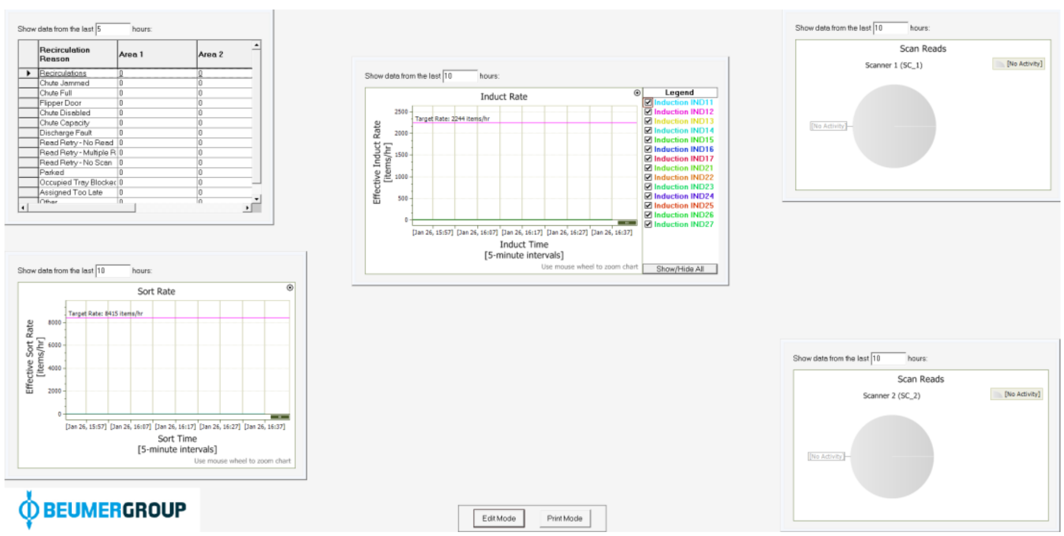
The BEUMER sorter control system provides a rich set of production statistics, most of which will be applicable to all projects. Some, however, will only apply to certain project types, like a packing sorter.

The following screens provide different statistics for the installation parts.

- Statistic data is typically kept for **30 days** within the system.
- The data is saved and available in 5-minute interval.
- To reduce the number of records in the database the data is compressed according to the following rules:
  - After 5 days the data will be compressed to a 1-hour interval.
  - After 14 days the data will be compressed to one record per day.
- Values are available in form of a list and some values are available in a graphical form as a chart.

3.8.1 Performance Overview

The performance overview comprises of several "widgets" providing real-time insight into the performance of various components of the sorter. The screen is fully customizable by the operator, allowing widgets to be moved, resized, and hidden. Additionally, all widgets have printing capabilities for reporting purposes.

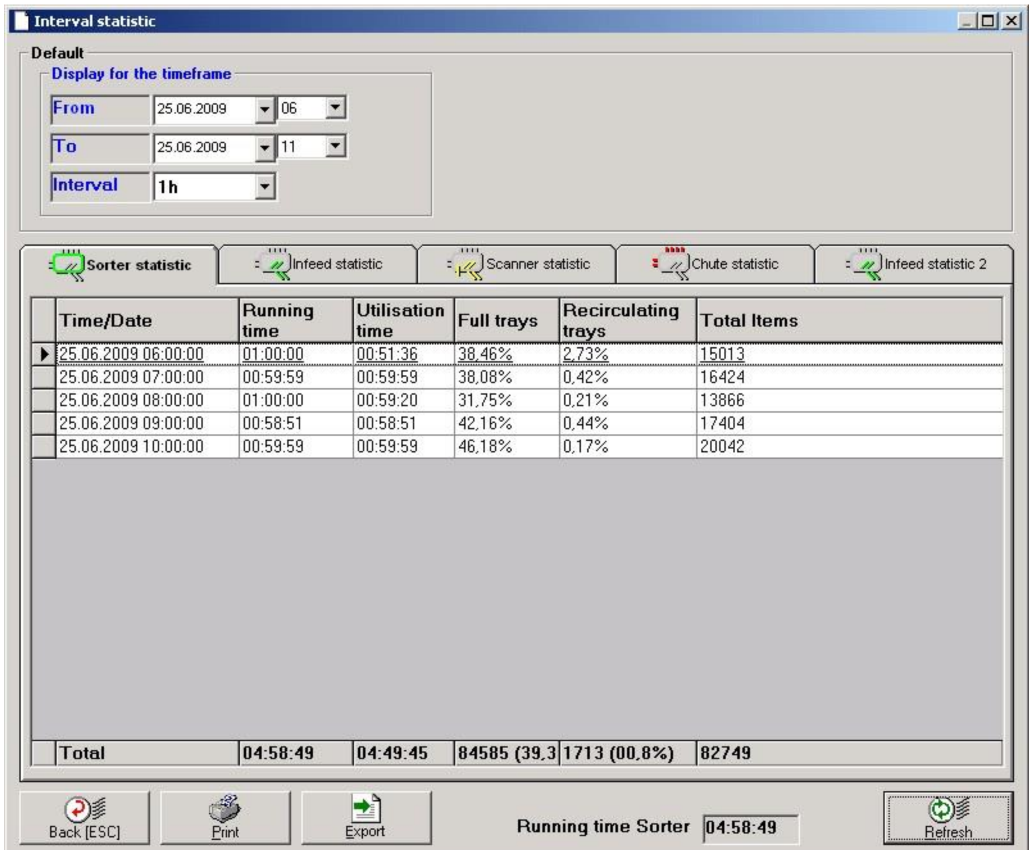


May 7th, 2024

3.8.2 Sorter Statistics

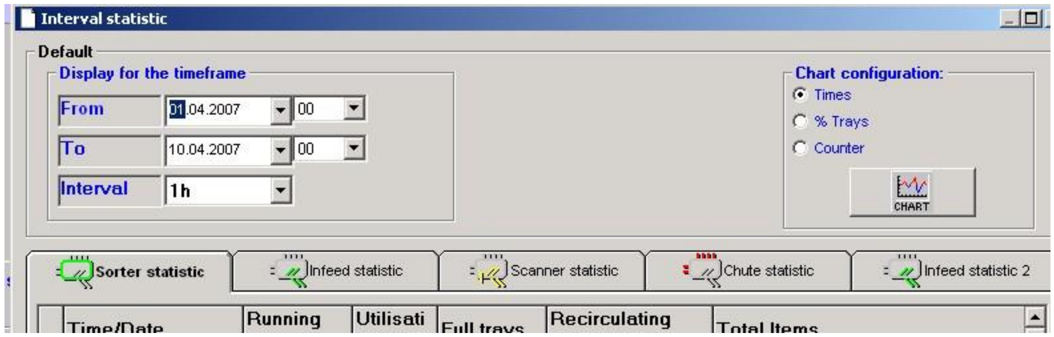
The sorter statistics indicate the utilization of the sorter in a certain time range in a grid.

The time range and the display interval can be selected. The list can be printed, and the data can also be exported into a file.



Column	Description
Running time	Total running time of the sorter within the selected range
Utilization time	Time with at least one tray occupied with item
Full trays	Percentage of trays with item of all trays within the running time
Recirculating trays in %	Percentage of re-circulating trays of the trays with item within the running time
Total items	Number of trays with item within the running time

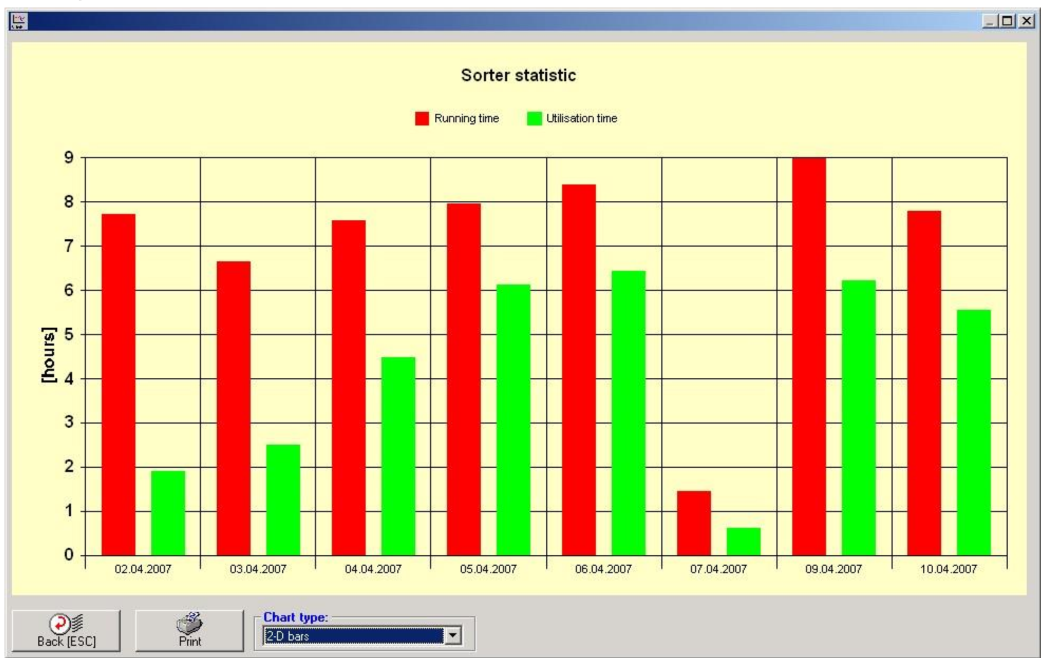
As an alternative to the list view, the values can be displayed in a graphical form as a chart.



Because of different values types within the list (counters, times...), the type must be selected. After pressing the CHART button the current values are shown in a graph.

Within the chart dialog 2-D Bars and 2-D Lines can be selected as the chart type.

Example:



3.8.3 Infeed Statistics

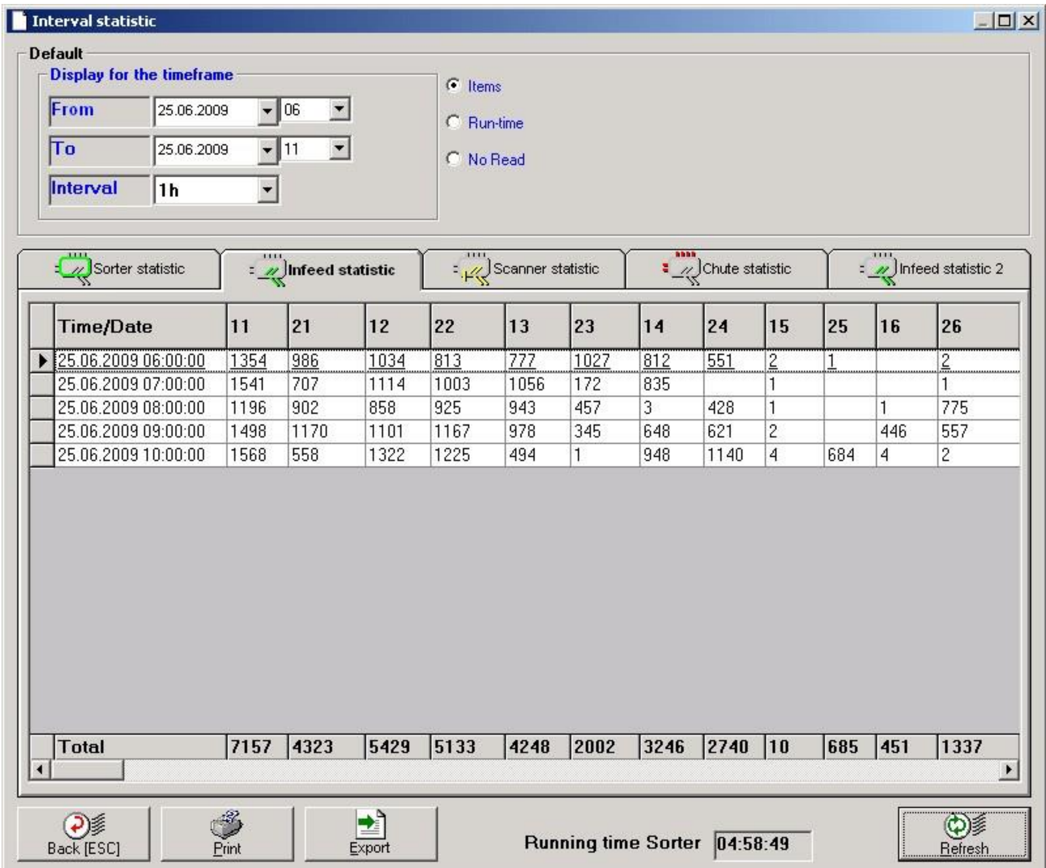
The infeed statistics indicate the utilization of the infeeds on a certain time range in a grid.

There are two statistic types available:

- Statistics arranged by time
- Statistics arranged by infeed

Statistic arranged by time:

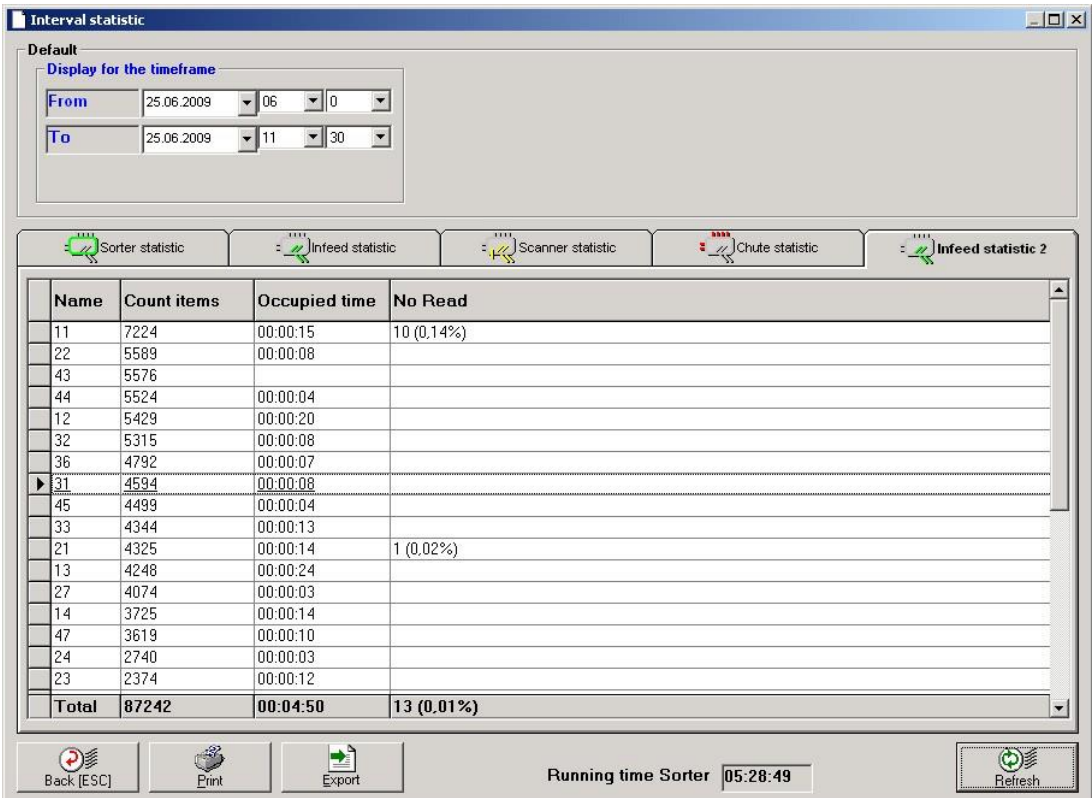
Based on the selected time and interval settings the data is displayed arranged by interval.



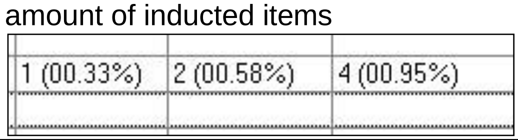
Time/Date	11	21	12	22	13	23	14	24	15	25	16	26
25.06.2009 06:00:00	1354	986	1034	813	777	1027	812	551	2	1		2
25.06.2009 07:00:00	1541	707	1114	1003	1056	172	835		1			1
25.06.2009 08:00:00	1196	902	858	925	943	457	3	428	1		1	775
25.06.2009 09:00:00	1498	1170	1101	1167	978	345	648	621	2		446	557
25.06.2009 10:00:00	1568	558	1322	1225	494	1	948	1140	4	684	4	2
Total	7157	4323	5429	5133	4248	2002	3246	2740	10	685	451	1337

Statistic arranged by infeed:

Based on the selected time range the data is displayed arranged by infeed here.

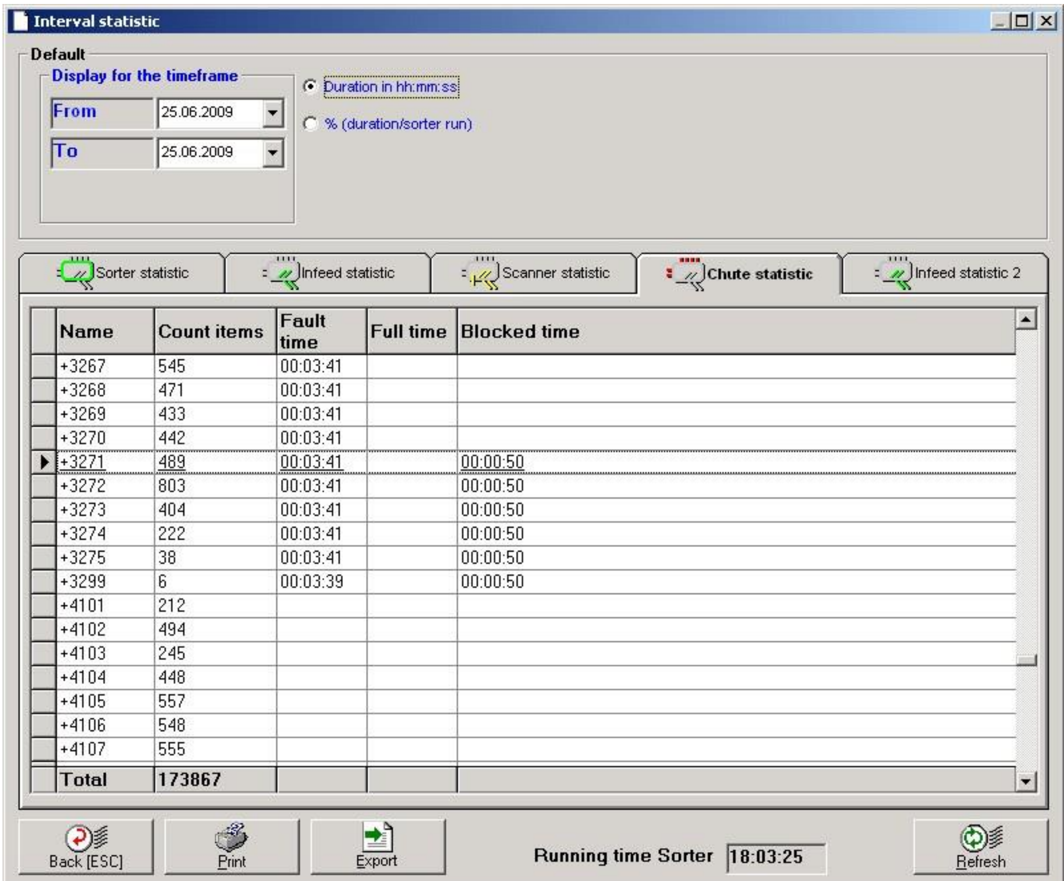


The following statistic values are available:

Column	Description
Items	The total amount of inducted items will be displayed
Occupied time / Run-time	Running time of the induct will be displayed
No Read	The total amount of inducted items which were **discharged** at a special chute due to no reads. The No-read is also shown as a percentage to the total amount of inducted items 

3.8.4 Chute Statistics

The chute statistics indicate the utilization of the chutes on a certain time range in a grid.



Name	Count items	Fault time	Full time	Blocked time
+3267	545	00:03:41		
+3268	471	00:03:41		
+3269	433	00:03:41		
+3270	442	00:03:41		
+3271	489	00:03:41		00:00:50
+3272	803	00:03:41		00:00:50
+3273	404	00:03:41		00:00:50
+3274	222	00:03:41		00:00:50
+3275	38	00:03:41		00:00:50
+3299	6	00:03:39		00:00:50
+4101	212			
+4102	494			
+4103	245			
+4104	448			
+4105	557			
+4106	548			
+4107	555			
Total	173867			

There are two display options available. These options will **only** affect the displayed value for the "time columns" (Fault time, Full time...):

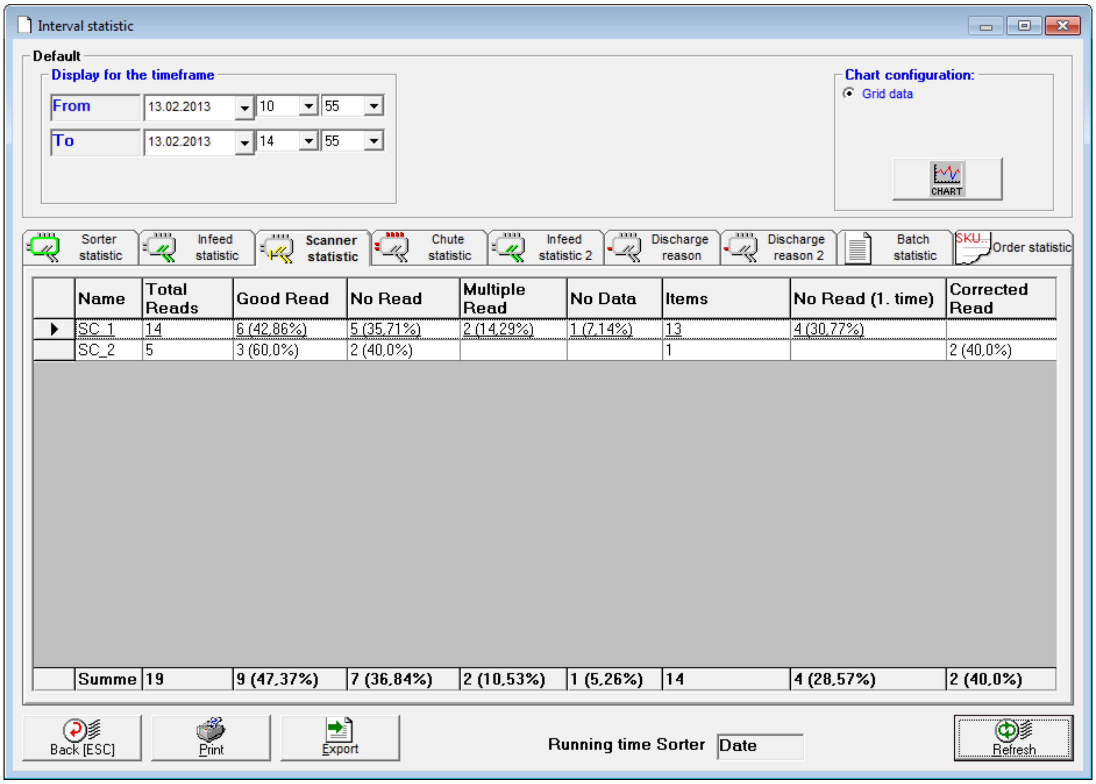
- Duration in hh:mm:ss
  The absolute time value will be displayed
- % (duration/sorter run)
  The percentage relative to the running time will be displayed

The following statistic values are available:

Column	Description
Count items	The total amount of discharged items will be displayed
Fault time	Duration of fault (missing air pressure...)
Full time	Duration of full condition
Blocked time	Duration of blocked condition

3.8.5 Scanner Statistics

These statistics displays various scanner related statistics for a selected time range.



The following statistic values are available:

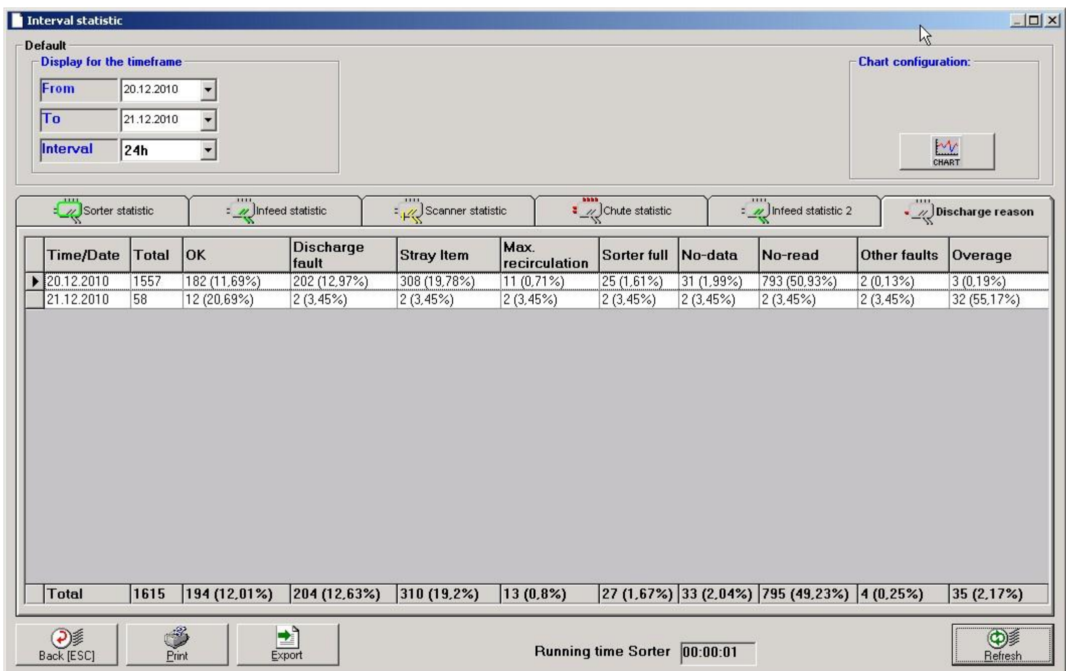
Column	Description
Total	The total amount of scans (not related to number of items)
Good Read	Total “Good reads” (Scanner delivered barcode) The percentage is based on the column Total Reads.
No Read	Total “No reads” (No barcode found from scanner) The percentage is based on the column Total Reads.
Multiple Read	Total “Multiple reads” (Multiple barcodes of same type found by scanner) The percentage is based on the column Total Reads.
No Data	Total “No data” (Scanner did not answer in expected time) The percentage is based on the column Total Reads.
Items	The total amount of items which have been scanned by this scanner the first time.
No Read (1. Time)	This counter shows the number of “No reads” which have been detected by this scanner when the item has been scanned the first time. The percentage is based on the column "Items".

Column	Description
Corrected Read	This counter shows the number of corrected reads. Example: If an item has been a No read the first time and becomes a good read afterwards (second scanner or read retry of the same scanner) then this counter will be increased The percentage value is based on the number of items which could have been 'corrected' by another read (previous No Read or No Data)

3.8.6 Discharge Reason Statistics

These statistics displays the discharges grouped by discharge reason.

The time range and the display interval can be selected. The list can be printed, and the data can also be exported into a file.



The following statistic values are available:

Column	Description
Total	The total amount of discharged items This is 100% basis for all the other percentages
OK	Items which have been successful delivered (not to an exception chute)
Following columns	These statistic counters are based on the configuration of exception chutes. For every possible exception discharge a statistic counter is available.

The discharge reason statistic 2 simply displays the statistic in a different way (not per time slices).

4 Sorter Controls

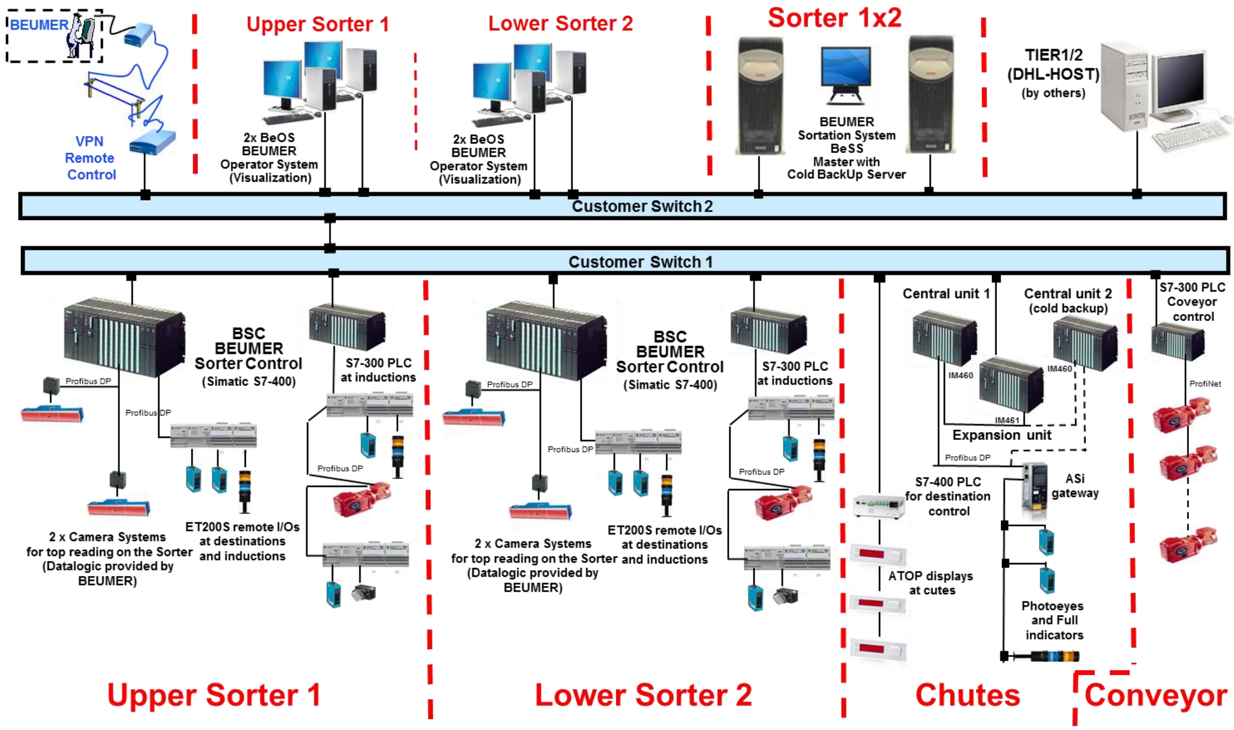


Figure 4-1 - Sorter Controls Layout

4.1 Servers and Terminals

All PC hardware will be provided by DHL. BEUMER will provide minimum hardware specifications in a separate document.

The sorter control system server (BeSS) will be installed in a virtual environment provided and maintained by DHL. BEUMER will provide minimum requirements in a separate document.

Two server instances are planned, one acting as master, the other as a warm backup. This implies that data (e.g. database) is automatically synchronized between the master and backup server instances, but the switch between master/backup is a manual process.

There are two types of sorter control system workstations. The first, the BeSSView/BeOS workstation, hosts two client applications, the BeSSView provides the operational and BeOS the maintenance interface to the sorter control system. These workstations may be placed where power is available and a network connection that can reach the BeSS server. The second, the BeTerm workstation, is part of the induction control and provides the ability to scan or key in a barcode before or when the item is placed on a semi-automatic induction.

For details on the entire machine control level, please refer to the PC hardware description (separate document).

5 BeSS Description

5.1 General BeSS description

The BeSS consist of different systems and software packages. Communication between every part of the system is done with socket communication based on Ethernet TCP/IP protocol.

Overview Standard BeSS

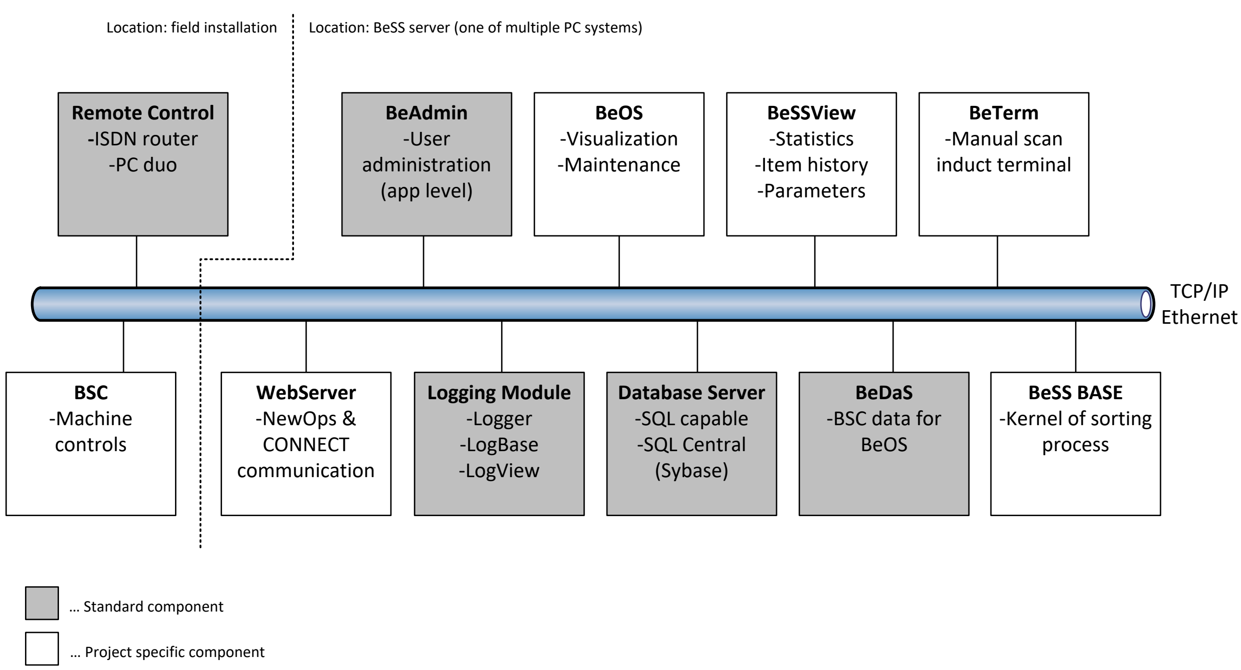


Figure 5-1 - Standard BeSS overview

5.1.1 Logging Module

The logging module is used to trace, display and analyze all events on the system reported by the different BeSS systems and modules.
The logging system creates log files with a size of 1MB. The current will be stored in the folder ..\logger and old log files in the folder ..\save_log. Depending on the hard disk space they will be deleted after 14 days.

5.1.2 BeAdmin

BeAdmin is the user administration system of the BeSS. It is working on application level independent from the user administration of the operating system.

5.1.3 BeSSView

BeSSView is a tool that shows sortation related data like

- Statistics
- System Parameters

- Real-Time information of what is on the sorter (e.g. items, chutes, etc.)
- Total number of items sorted per day and hour
- Total number of items per induction per day and hour
- Discharged items per destination
- It also holds some simple mapping tables to allow for assignments of inductions to destinations for commissioning or maintenance purposes.

5.1.4 BeOS

BeOS or **BE**UMER **O**perator **S**tation is the local visualization of the sorter and is a window into the BSC (Beumer Sorter Controls) system. It includes a graphical display of the system. Different statuses will be shown in different colors.
Error handling will be done at the BeOS. Error messages will be shown and acknowledged at the BeOS.

5.1.5 WebServer

WebServer is a software module responsible for communication to the NewOps and CONNECT systems.

5.1.6 Database Server

All relevant information and system parameter are stored in a local database on the BeSS server system. The database is based on Sybase SQL Central.

5.1.7 BeDaS

The BeDaS is the **BE**UMER **Da**ta **S**erver. It is a software module to provide status and error information to the different visualization systems like BeOS.

5.1.8 BeSS Base

BeSS Base is a software module responsible for the sortation of items in the system. With this sort-plan-driven system it will be primarily responsible for assigning and managing destination assignments based on the sort plans and relaying item information to and from the BSC, NewOps, and CONNECT systems.

5.1.9 BSC

The BSC is the **BE**UMER **S**orter **C**ontrol system. It is a PLC based system that is responsible for the machine level control functionality of the sorter.
Functionality of the BSC is as follows:

- Communications with BeSS Base for controlling of sortation via Ethernet
- Communications with the scanners for barcode information via Profibus
- Communications with the conveyor PLC at infeed and destinations via Ethernet
- Exchange of status information and commands with the BeDaS via Ethernet
- Control of the sensors and actuators in the field via field bus Profibus

- Control power supply controller for the sorter system via field bus Profibus

5.2 Operation

The sorting system can be operated and configured with the following elements:

BeSSView

This program has the following main functions:

- Parameter setting
- Statistics Reports
- Sorter data display (what is on the sorter, and what was discharged from the sorter)

BeOS

The BeOS software serves to operate the sorter. The main functions are:

- Start / stop sorter
- Display of fault messages
- Display of tray data within the PLC

Further Details will be outlined in the operator manuals for both of these applications.

5.3 BeSS application handling

The sorter is equipped with a BeSS Master server and a BeSS Backup server. All processes are running as a service.

On each machine two batch files are available from the start-menu:

- Start BeSS
- Stop BeSS

The function “Start BeSS” performs the steps as follows:

- Try to set the virtual IP address of BeSS. Therefore, BeSS makes a ping to that virtual address. In case of no reply it sets the IP address and continues.
- Start the database service
- Start all BEUMER processes

The function “Stop BeSS” performs the steps as follows:

- Stop all BEUMER processes
- Stop the database service

- Remove the virtual IP address

With this principle it is possible to start BeSS on the BeSS Master or on the BeSS Backup. The other systems like terminals and host will always communicate with the server via the virtual IP address.

5.4 Backup and Recovery

The BeSS server is a virtual machine that is provided by DHL. The two VM servers are in a warm backup set up. Data redundancy between the two BeSS servers is handled by the BEUMER warm backup automated procedures. The switching between master/backup servers, however, remains a manual process to be handled by the customer when needed.

5.5 Remote VPN control

The principles of the remote control access are displayed in the following graphic.

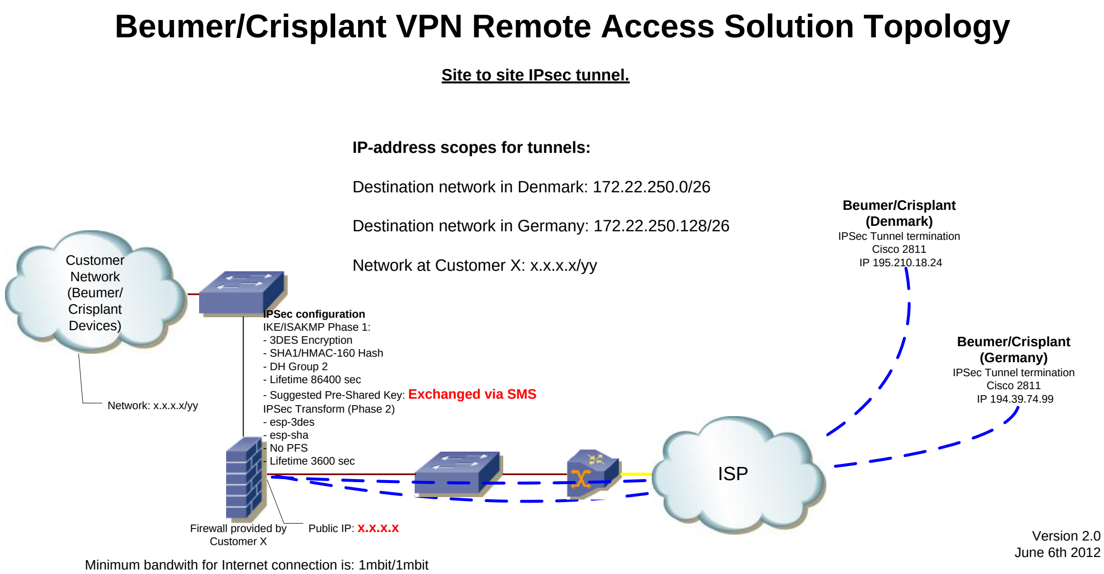


Figure 5-2 - Remote access scheme

Remote control access is a standard part of the system. Since a small problem may have a big impact on the system it is mandatory to be able to give quick and efficient remote support.

This is done by the standard remote support software ‘NetSupportManager’, NSM for the PC systems. All components are connected to a local Ethernet network. This network can be reached remotely from the BEUMER office via a VPN LAN to LAN connection.

6 Appendices

6.1 Mail flow charts

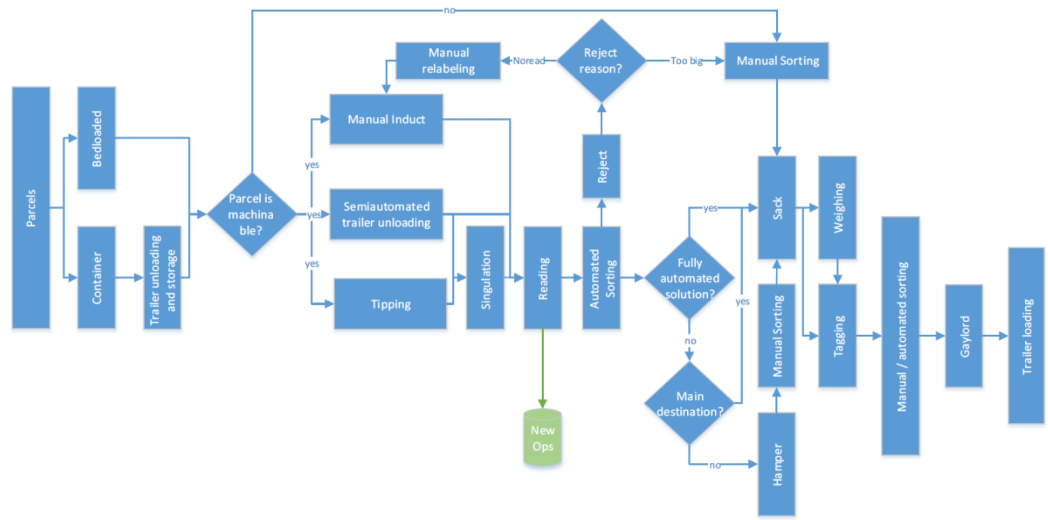


Figure 6-1 - DHL domestic Inbound Mail

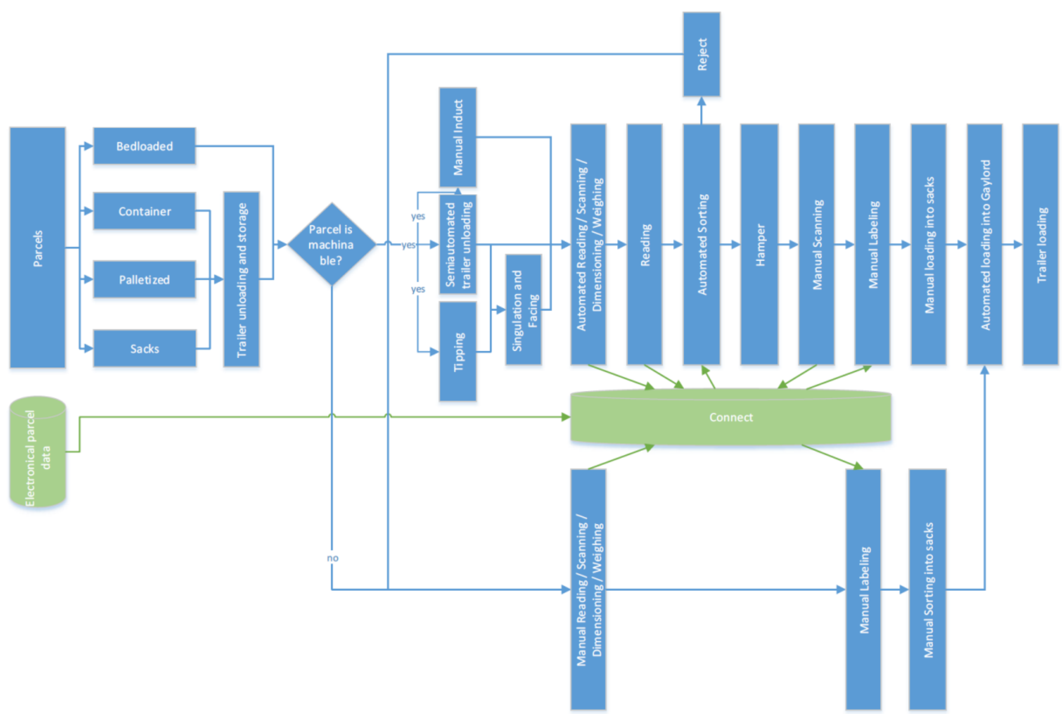


Figure 6-2 - DHL Cross-border Mail

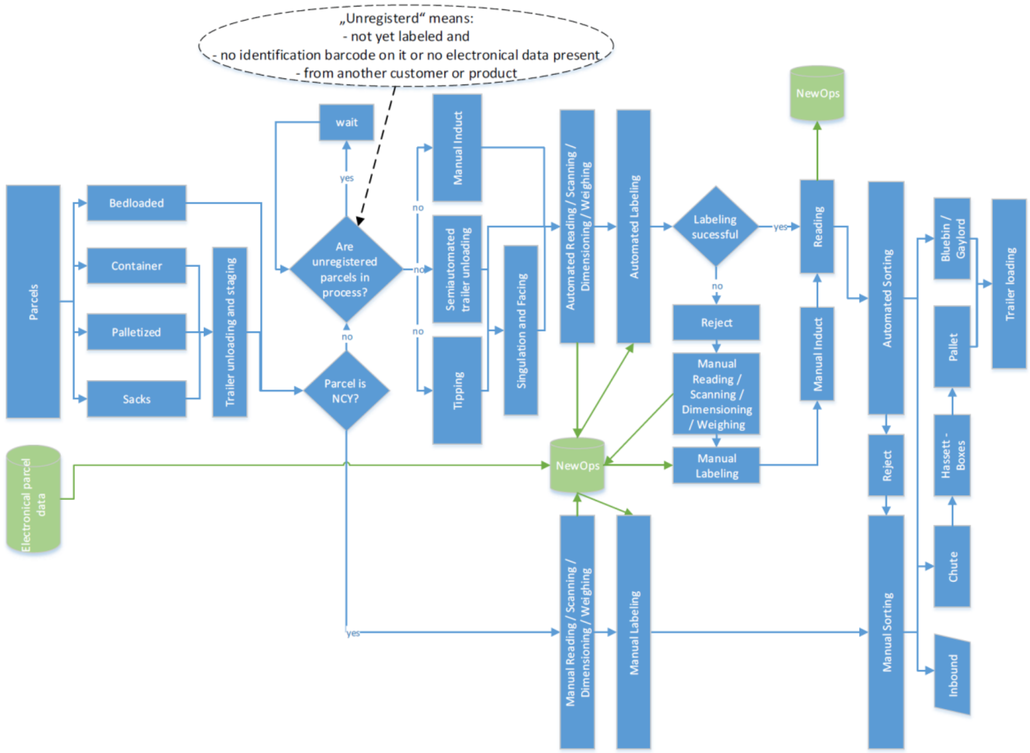


Figure 6-3 - DHL Domestic Inbound Mail